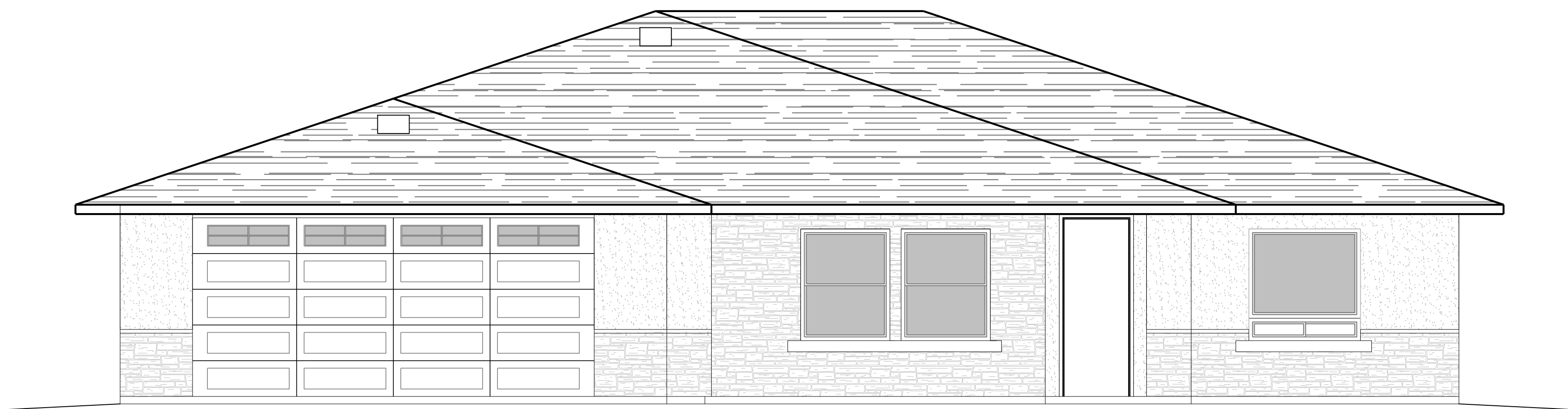


ARCHITECTURAL & STRUCTURAL PLANS FOR
CAPPS CONSTRUCTION
SINGLE FAMILY RESIDENCE



FRONT ELEVATION
N.T.S.

SHEET INDEX

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C-2	GRADING NOTES
C-3	GRADING PLAN
C-4	EROSION CONTROL PLAN
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WSW1	SIMPSON DETAIL
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SP-1	STRUCTURAL SPECIFICATIONS
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PROPERTY INFORMATION SEARCH

Assessment Number:	012-193-028
Owner Name:	Capps Construction
Community Code:	Heritage Ranch
Tax Rate Area:	114-028
Parcel Size:	15,847 SF
Net:	30,000
Structure Type:	Land

PROJECT INFORMATION

OWNER	KIRK AND CARRIE ALLEN
ADDRESS	270 CATALINA PLACE PASO ROBLES, CA 93446
APN	012-193-003
PHONE	805.540.1185

PROJECT STATISTICS

LOT SIZE	15,847 SQ. FT.
OCCUPANCY (CBC 310.1)	R-3, U
CONSTRUCTION TYPE	VB
FIRE SPRINKLERS	YES
BUILDING HEIGHT	±17'-10"
PROPOSED LIVING	2,338 SQ. FT.
PROPOSED GARAGE	708 SQ. FT.
PROPOSED PORCH	710 SQ. FT.

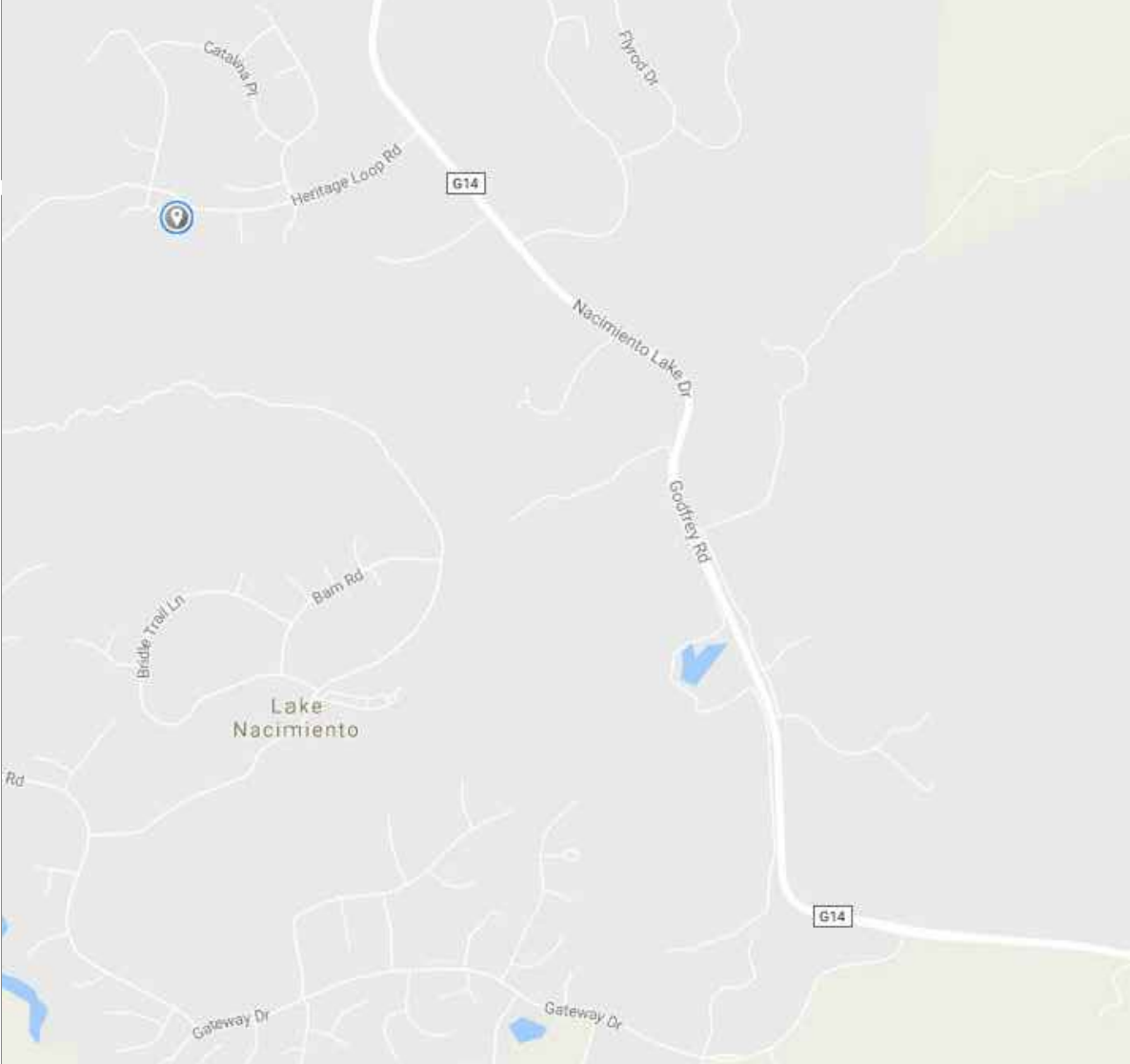
PROJECT DESCRIPTION

PROPOSED SINGLE FAMILY RESIDENCE AS PER PLANS ATTACHED.

PROJECT NOTES

ADDRESSES SHALL BE PLAINLY VISIBLE AND LEGIBLE FROM THE STREET OR ROAD FRONTING THE PROPERTY. ADDRESS NUMBERS SHALL BE 4 INCHES IN HEIGHT, $\frac{3}{8}$ " MINIMUM STROKE WIDTH AND OF CONTRASTING COLOR TO THEIR BACKGROUND. WHERE ADDRESS CAN NOT BE VIEWED FROM PUBLIC WAY, A MONUMENT OR POLE SHALL BE USED. R319.

VICINITY MAP



REVISION LOG

REV.	DESCRIPTION	DATE

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PROJECT NO. —

FILE NAME A-1.1 SITE PLAN.DWG

DRAWN BY JJK

DATE 3/31/2017 7:59 AM

SHEET TITLE:

SITE PLAN

SHEET NUMBER:

A-1.1

GENERAL CONSTRUCTION NOTES

- ALL WORK SHALL CONFORM WITH THE:
2016 CBC (2012 IBC AND CALIFORNIA AMENDMENTS)
2016 CEC (2011 NEC AND CALIFORNIA AMENDMENTS)
2016 CMC (2012 IAPMO UMC AND CALIFORNIA AMENDMENTS)
2016 CPC (2012 IAPMO UPC AND CALIFORNIA AMENDMENTS)
2016 CMC AND T-24
2016 CALIFORNIA GREEN BUILDING CODE
2016 CFC (2012 IFC AND CALIFORNIA AMENDMENTS)
- THESE NOTES SHALL APPLY TO ALL DRAWINGS UNLESS OTHERWISE NOTED OR SHOWN. FEATURES OF CONSTRUCTION SHOWN ARE TYPICAL AND THEY SHALL APPLY GENERALLY THROUGHOUT SIMILAR CONDITIONS. ALL OMISSIONS OR CONFLICTS BETWEEN VARIOUS ELEMENTS OF THE WORKING DRAWINGS AND/OR GENERAL NOTES SHALL BE BROUGHT TO THE ATTENTION OF THE ARCHITECT/ENGINEER BY THE GENERAL CONTRACTOR BEFORE PROCEEDING WITH ANY WORK SO INVOLVED.
- ALL WORK AND CONSTRUCTION METHODS AND MATERIALS SHALL COMPLY WITH ALL PROVISIONS OF THE BUILDING CODES AND OTHER RULES, REGULATIONS AND ORDINANCES GOVERNING THE CONSTRUCTION SITE. BUILDING CODE REQUIREMENTS IN ALL CASES TAKE PRECEDENCE OVER THE DRAWINGS. IT SHALL BE THE RESPONSIBILITY OF ANYONE SUPPLYING LABOR AND/OR MATERIALS TO BRING TO THE ATTENTION OF THE ARCHITECT/ENGINEER ANY DISCREPANCIES OR CONFLICTS BETWEEN THE REQUIREMENTS OF THE CODE AND THE DRAWINGS.
- DO NOT SCALE THE DRAWINGS. DIMENSIONS SHOWN SHALL TAKE PRECEDENCE OVER DRAWING SCALE OR PROPORTION. LARGE SCALE DRAWINGS SHALL TAKE PRECEDENCE OVER SMALLER SCALE DRAWINGS.
- THE CONTRACT DRAWINGS AND SPECIFICATIONS REPRESENT THE FINISHED STRUCTURE. UNLESS OTHERWISE SHOWN, THEY DO NOT INDICATE METHOD OF CONSTRUCTION. CONTRACTOR SHALL SUPERVISE AND DIRECT WORK AND SHALL BE SOLELY RESPONSIBLE FOR ALL CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES AND PROCEDURES. OBSERVATION VISITS TO THE SITE BY FIELD REPRESENTATIVES OF THE ARCHITECT/ENGINEER SHALL NOT INCLUDE INSPECTIONS OF THE PROTECTIVE MEASURES OR THE CONSTRUCTION PROCEDURES REQUIRED FOR SAME, WHICH ARE THE SOLE RESPONSIBILITY OF THE CONTRACTOR. ANY SUPPORT SERVICES PERFORMED BY THE ARCHITECT/ENGINEER DURING CONSTRUCTION SHALL BE DISTINGUISHED FROM CONTINUOUS AND DETAILED INSPECTION SERVICES WHICH ARE FURNISHED BY OTHERS. THESE SUPPORT SERVICES PERFORMED SOLELY FOR THE PURPOSE OF ASSISTING IN QUALITY CONTROL AND IN ACHIEVING CONFORMANCE WITH CONTRACT DRAWINGS AND SPECIFICATIONS, AND THEREFORE THEY DO NOT GUARANTEE CONTRACTOR'S PERFORMANCE AND SHALL NOT BE CONSTRUED AS SUPERVISION OF CONSTRUCTION.
- CONTRACTOR HEREBY GUARANTEES TO THE OWNER AND THE ARCHITECT/ENGINEER THAT ALL MATERIALS, FIXTURES, AND EQUIPMENT FURNISHED TO THE PROJECT ARE NEW UNLESS OTHERWISE SPECIFIED. CONTRACTOR ALSO WARRANTS THAT ALL WORK WILL BE OF GOOD QUALITY AND FREE FROM ANY FAULTS AND DEFECTS FOR A PERIOD OF ONE YEAR AFTER THE DATE OF SUBSTANTIAL COMPLETION, UNLESS A GREATER WARRANTY OR GUARANTEE IS REQUIRED BY THE PROJECT SPECIFICATIONS.
- ANYONE SUPPLYING LABOR AND/OR MATERIALS TO THE PROJECT SHALL CAREFULLY EXAMINE ALL SUBSURFACES TO RECEIVE WORK. ANY CONDITIONS DETRIMENTAL TO WORK SHALL BE REPORTED IN WRITING TO THE CONTRACTOR PRIOR TO BEGINNING WORK. COMMENCEMENT OF WORK SHALL IMPLY ACCEPTANCE OF ALL SUBSURFACES.
- REFER TO ARCHITECTURAL, MECHANICAL, AND ELECTRICAL DRAWINGS FOR DEPRESSED SLABS CURB, FINISHES, TEXTURES, CLIPS, GROUNDS, ETC., NOT SHOWN ON STRUCTURAL DRAWINGS.
- ANY MATERIALS STORED AT THE SITE SHALL BE COMPLETELY SUPPORTED FREE OF THE GROUND, COVERED AND OTHERWISE PROTECTED TO AVOID DAMAGE FROM THE ELEMENTS.
- MORE DETAILED INFORMATION SHALL TAKE PRECEDENCE OVER LESSER DETAILED INFORMATION. SPECIFICATIONS SHALL TAKE PRECEDENCE OVER DRAWINGS.
- GRADING PLANS, DRAINAGE IMPROVEMENTS, ROAD AND ACCESS REQUIREMENTS AND ENVIRONMENTAL HEALTH CONSIDERATIONS SHALL COMPLY WITH ALL APPLICABLE CODES AND LOCAL ORDINANCES.
- THE CONTRACTOR AND ALL SUB-CONTRACTORS WILL BE HELD ACCOUNTABLE TO THE ABOVE GENERAL NOTES FOR THE CONSTRUCTION OF THE PROJECT.
- THE CONTRACTOR SHALL BE RESPONSIBLE TO REMOVE OR DISBURSE ANY EXCESS MATERIAL FROM PROJECT SITE.
- THIS SET OF PLANS TO BE ON JOB SITE AT ALL TIMES DURING CONSTRUCTION. ALL WORK SHALL BE DONE IN ACCORDANCE WITH THE APPROVED PLANS. NO CHANGES OR REVISIONS TO THE APPROVED PLANS OR SPECIFICATIONS SHALL BE PERMITTED UNLESS SUBMITTED TO AND APPROVED BY THE BUILDING OFFICIAL. THE ISSUANCE OF A PERMIT SHALL NOT PREVENT THE BUILDING OFFICIAL FROM REQUIRING THE CORRECTION OF ERRORS OR OMISSIONS FROM THE APPROVED PLANS AND SPECIFICATIONS. [CBC 108]
- ALL CONTRACTORS AND SUB-CONTRACTORS MUST HAVE ON FILE WITH THE BUILDING DEPARTMENT, A LIST OF ALL SUCH CONTRACTORS AND SUB-CONTRACTORS WITH APPROPRIATE CURRENT BUSINESS LICENSE NUMBERS.
- UNLESS NOTED OTHERWISE, ALL VESTIBULES, CLOSETS, COLUMNS, PROJECTIONS, RECESSES, OR OTHER ADJACENT AREAS WITHIN SCHEDULED AREA SHALL HAVE FINISHES AS SCHEDULED FOR THE RESPECTIVE SPACES IN WHICH THEY OCCUR.
- CONTRACTOR SHALL VERIFY ALL SETBACKS, EASEMENTS, CONTOURS, AND BUILDING PAD PRIOR TO CONSTRUCTION.
- TRUSS CALCULATIONS FOR APPROVED PROJECTS ARE REQUIRED TO BE ON THE JOB SITE AT TIME OF FRAMING INSPECTION WITH THE APPROPRIATE REQUIRED SIGNATURES AND STATEMENT AS FOLLOWS: TRUSS CALCULATIONS SHALL INCLUDE THE WET-STAMP AND SIGNATURE OF THE TRUSS DESIGN ENGINEER. IN ADDITION, THEY SHALL INCLUDE ON THE COVER SHEET A WET- SIGNED STATEMENT FROM THE PROJECT'S DESIGN ENGINEER THAT TRUSS CALCULATIONS AND LAYOUTS ARE IN SUBSTANTIAL CONFORMANCE WITH THE STRUCTURAL DESIGN AND INTENT OF THE STRUCTURE. FAILURE TO PROVIDE THEM AS STATED WILL RESULT IN A CORRECTION AND A FAILURE TO PASS FRAMING INSPECTION. [BSP]
- VERIFY LOCATION OF ALL UTILITY TIE-INS AT STREET AND POINT OF CONNECTIONS AT BUILDING PRIOR TO CONSTRUCTION.
- A COPY OF SOILS REPORT SHALL BE ON SITE DURING FOUNDATION INSPECTION.
- ALL PROPERTY CORNERS SHOULD BE ESTABLISHED AT THE TIME OF FOUNDATION INSPECTION WITH THE MARK OF A LICENSED SURVEYOR.

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GENERAL NOTES

1. NO CONSTRUCTION SHALL BE STARTED WITHOUT PLANS APPROVED BY THE COUNTY DEPARTMENT OF PLANNING AND BUILDING. THE COUNTY SHALL BE NOTIFIED AT LEAST 24 HOURS PRIOR TO STARTING CONSTRUCTION AND OF THE TIME LOCATION OF THE PRE CONSTRUCTION CONFERENCE. ANY CONSTRUCTION DONE WITHOUT APPROVED PLANS OR PRIOR NOTIFICATION TO THE COUNTY WILL BE REJECTED AND WILL BE AT THE CONTRACTOR'S AND/OR OWNER'S RISK AND EXPENSE.
2. ALL CONSTRUCTION WORK SHALL CONFORM TO THE COUNTY OF SAN LUIS OBISPO STANDARDS AND SPECIFICATIONS AND ALL WORK SHALL BE SUBJECT TO THE APPROVAL OF THE COUNTY.
3. SOILS TESTS SHALL BE PERFORMED IN ACCORDANCE WITH THE COUNTY STANDARDS AND SPECIFICATIONS SECTION 11-351.1403 AND 11-351.1404. ALL TESTS MUST BE MADE WITHIN 15 DAYS PRIOR TO THE PLACING MATERIAL. THE TEST RESULTS SHALL CLEARLY INDICATE THE LOCATION AND SOURCE OF THE MATERIAL.
4. COMPACTION TESTS SHALL BE MADE ON SUBGRADE MATERIAL AND MATERIAL AS SPECIFIED BY THE ENGINEER. SAID TESTS SHALL BE MADE PRIOR TO THE PLACING OF THE NEXT MATERIAL.
5. SUBGRADE MATERIAL SHALL BE COMPACTED TO A RELATIVE COMPACTION OF 95% IN THE ZONE BETWEEN FINISHED SUBGRADE ELEVATION AND ONE FOOT BELOW. ALL MATERIAL IN FILL SECTIONS BELOW THE ZONE MENTIONED ABOVE SHALL BE COMPACTED TO 90% RELATIVE COMPACTION.
4. AN EFFORT HAS BEEN MADE TO DEFINE THE LOCATION OF UNDERGROUND FACILITIES WITHIN THE JOB SITE. HOWEVER, ALL EXISTING UTILITY AND OTHER UNDERGROUND STRUCTURES MAY NOT BE SHOWN ON THIS PLAN AND THEIR LOCATION WHERE SHOWN IS APPROXIMATE. THE CONSTRUCTION CONTRACTOR AGREES THAT HE SHALL ASSUME SOLE AND COMPLETE RESPONSIBILITY FOR LOCATING OR HAVING LOCATED ALL UNDERGROUND UTILITIES AND OTHER FACILITIES AND FOR PROTECTING THEM DURING CONSTRUCTION.
5. ALL UTILITY COMPANIES MUST BE NOTIFIED PRIOR TO THE START OF CONSTRUCTION. THE CONSTRUCTION CONTRACTOR SHALL CONTACT UNDERGROUND SERVICE ALERT (U.S.A.) AT 1-800-642-2444 2 TO 10 DAYS PRIOR TO THE START OF EXCAVATION AND SHALL VERIFY THE LOCATION OF ANY KNOWN UTILITIES AND WHETHER OR NOT A REPRESENTATIVE OF EACH COMPANY WILL BE PRESENT DURING EXCAVATION.
6. A REGISTERED CIVIL ENGINEER MUST CERTIFY THAT THE IMPROVEMENTS WHEN COMPLETED ARE IN ACCORDANCE WITH THE PLANS PRIOR TO THE REQUEST FOR A FINAL INSPECTION. AS-BUILT PLANS ARE TO BE PREPARED AFTER CONSTRUCTION IS COMPLETED. THE CIVIL ENGINEER CERTIFYING THE IMPROVEMENTS AND PREPARING AS-BUILT PLANS WILL BE PRESENT WHEN THE FINAL INSPECTION IS MADE.
7. AN INSPECTION AGREEMENT IS REQUIRED PRIOR TO THE START OF CONSTRUCTION.
8. ALL UTILITY COMPANIES MUST BE NOTIFIED PRIOR TO THE START OF CONSTRUCTION.
9. A COUNTY ENCROACHMENT PERMIT IS REQUIRED FOR ALL WORK DONE WITHIN THE COUNTY RIGHT-OF-WAY.
10. THE COUNTY INSPECTOR ACTING ON BEHALF OF THE COUNTY DEPARTMENT OF PLANNING AND BUILDING MAY REQUIRE REVISIONS IN THE PLANS TO SOLVE UNFORESEEN PROBLEMS THAT MAY ARISE IN THE FIELD. ALL REVISIONS SHALL BE SUBJECT TO THE APPROVAL OF THE DEVELOPER'S ENGINEER.
11. THE CONTRACTOR SHALL CONFIRM THE STRUCTURAL SECTION WHICH SHALL BE BASED ON SOILS TESTS MADE AT THE TIME OF CONSTRUCTION AND ON A TRAFFIC INDEX OF 4.5 FOR ALL ROADS.
12. HYDROSEEDING SHALL BE PLACED ON ALL DISTURBED SURFACES OTHER THAN PAVED OR GRAVEL SURFACES, PRIOR TO FINAL INSPECTION.
13. FOR ANY PUBLIC IMPROVEMENTS TO BE MAINTAINED BY THE COUNTY, IF ENVIRONMENTAL PERMITS FROM THE U.S. ARMY CORPS OF ENGINEERS, THE REGIONAL WATER QUALITY CONTROL BOARD/STATE WATER RESOURCES CONTROL BOARD, OR THE CALIFORNIA DEPARTMENT OF FISH & GAME ARE REQUIRED, THE DEVELOPER SHALL:
 - A. SUBMIT A COPY OF ALL SUCH COMPLETED PERMITS TO THE COUNTY DEPARTMENT OF PLANNING AND BUILDING
 - OR
 - B. DOCUMENT THAT THE REGULATORY AGENCIES DETERMINED THAT SAID PERMIT IS NOT REQUIRED:

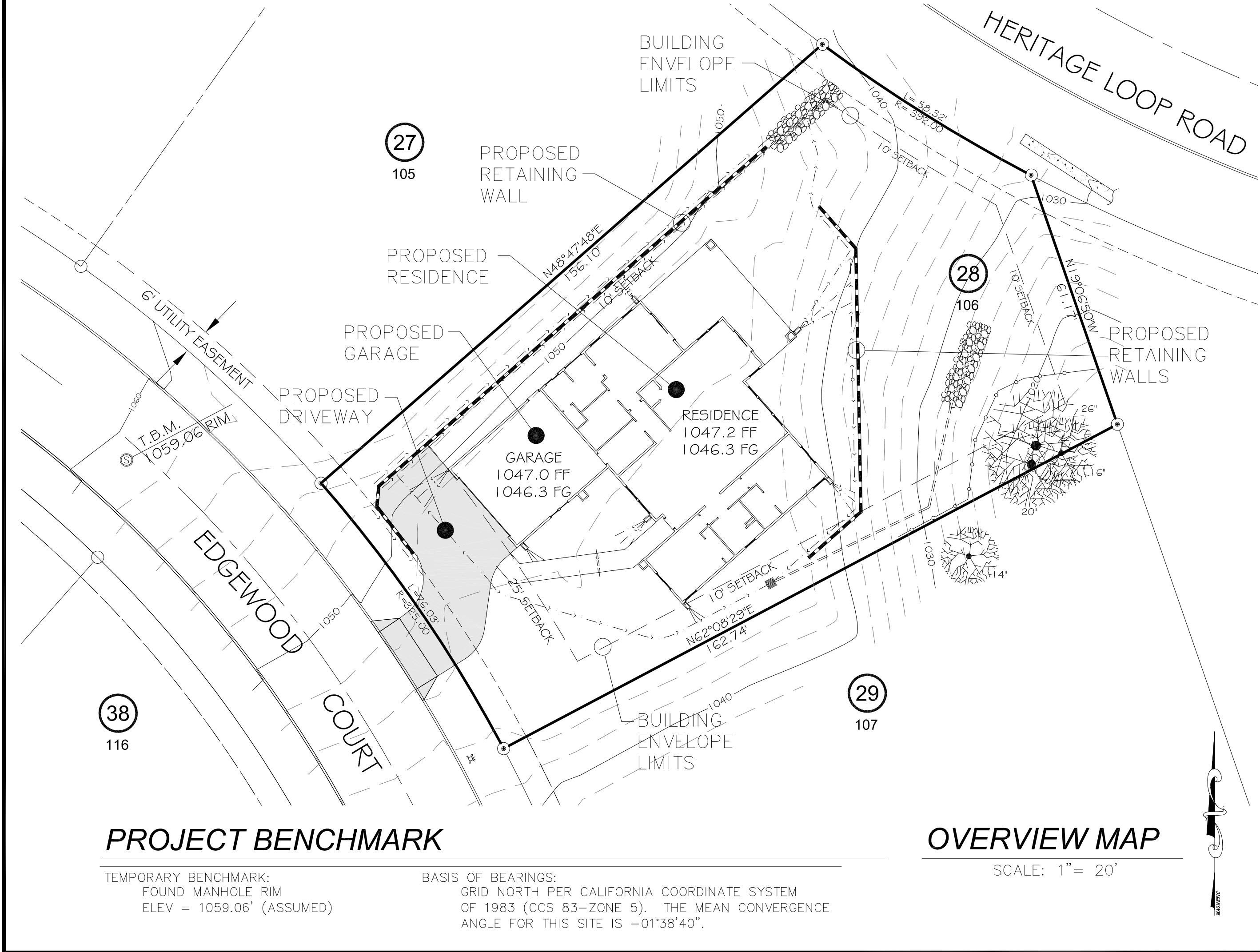
PRIOR TO ACCEPTANCE OF THE COMPLETED IMPROVEMENTS FOR COUNTY MAINTENANCE AND RELEASE OF IMPROVEMENT SECURITY. ANY MITIGATION MONITORING REQUIRED BY SAID PERMITS WILL REMAIN THE RESPONSIBILITY OF THE DEVELOPER.

PROJECT AIR QUALITY

DURING CONSTRUCTION/GROUND DISTURBING ACTIVITIES, THE CONTRACTOR OR BUILDER SHALL DESIGNATE A PERSON OR PERSONS TO MONITOR THE DUST CONTROL PROGRAM AND TO ORDER INCREASED WATERING, AS NECESSARY, TO PREVENT TRANSPORT OF DUST OFF SITE. THEIR DUTIES SHALL INCLUDE HOLIDAY AND WEEKEND PERIODS WHEN WORK MAY NOT BE IN PROGRESS. THE NAME AND TELEPHONE NUMBER OF SUCH PERSONS SHALL BE PROVIDED TO THE APCD PRIOR TO COMMENCEMENT OF CONSTRUCTION.

THE MEASURES FOR DUST CONTROL ARE AS FOLLOWS, BUT NOT LIMITED TO:

- A. REDUCE THE AMOUNT OF DISTURBED AREA WHERE POSSIBLE.
- B. USE OF WATER TRUCKS OR SPRINKLER SYSTEMS IN SUFFICIENT QUANTITIES TO PREVENT AIRBORNE DUST FROM LEAVING THE SITE. INCREASED WATERING FREQUENCY WILL BE REQUIRED WHENEVER WIND SPEEDS EXCEED 15 MPH. RECLAIMED (NON-POTABLE) WATER SHALL BE USED WHENEVER POSSIBLE.
- C. ALL DIRT STOCK PILE AREAS SHALL BE SPRAYED DAILY AS NEEDED.
- D. PERMANENT DUST CONTROL MEASURES IDENTIFIED IN THE APPROVED PROJECT REVEGETATION AND LANDSCAPE PLANS SHALL BE IMPLEMENTED AS SOON AS POSSIBLE FOLLOWING COMPLETION OF ANY SOIL DISTURBING ACTIVITIES.
- E. EXPOSED GROUND AREAS THAT ARE PLANNED TO BE REWORKED AT DATES GREATER THEN ONE MONTH AFTER INITIAL GRADING SHOULD BE SOWN WITH A FAST GERMINATING NATIVE GRASS SEED AND WATERED UNTIL VEGETATION IS ESTABLISHED.
- F. ALL DISTURBED SOIL AREAS NOT SUBJECT TO REVEGETATION MUST BE STABILIZED USING APPROVED CHEMICAL SOIL LINDENS, JUTE NETTING, OR OTHER METHODS APPROVED IN ADVANCE BY APCD.
- G. ALL ROADWAYS, DRIVEWAYS, SIDEWALKS, ETC. TO BE PAVED SHALL BE COMPLETED AS SOON AS POSSIBLE. IN ADDITION, BUILDING PADS SHOULD BE LAID AS SOON AS POSSIBLE AFTER GRADING UNLESS SEEDING OR SOIL BINDERS ARE USED.
- H. VEHICLES SPEED FOR ALL CONSTRUCTION VEHICLES SHALL NOT EXCEED 15 MPH ON ANY UNPAVED SURFACE AT THE CONSTRUCTION SITE.
- I. ALL TRUCKS HAULING DIRT, SAND, SOIL, OR OTHER LOOSE MATERIALS ARE TO BE COVERED OR SHALL MAINTAIN AT LEAST TWO FEET OF FREEBOARD (MINIMUM VERTICAL DISTANCE BETWEEN TOP OF LOAD AND TOP OF TRAILER) IN ACCORDANCE WITH CALIFORNIA VEHICLE CODE SECTION 23114.
- J. INSTALL WHEEL WASHERS WHERE VEHICLES ENTER AND EXIT UNPAVED ROADS ONTO STREETS, OR WASH OFF TRUCKS AND EQUIPMENT LEAVING THE SITE.
- K. SWEEP STREETS AT THE END OF EACH DAY IF VISIBLE SOIL MATERIAL IS CARRIED ONTO ADJACENT PAVED ROADS. WATER SWEEPERS WITH RECLAIMED WATER SHALL BE USED WHERE FEASIBLE.



COUNTY ADOPTED CODE ORDINANCE

THESE PLANS HAVE BEEN DESIGNED AND ENGINEERED TO MEET THE CONDITIONS OF THE COUNTY ADOPTED ORDINANCES LISTED:

- 2016 CALIFORNIA ENERGY CODE
- 2016 CALIFORNIA BUILDINGS CODE (VOLS 1 & 2)
- 2016 CALIFORNIA ELECTRICAL CODE
- 2016 CALIFORNIA FIRE CODE
- 2016 CALIFORNIA GREEN BUILDING CODE
- 2016 CALIFORNIA MECHANICAL CODE
- 2016 CALIFORNIA PLUMBING CODE
- 2016 CALIFORNIA REFERENCE STANDARDS CODE
- 2016 CALIFORNIA RESIDENTIAL CODE
- TITLE 19 COUNTY BUILDING AND CONSTRUCTION ORDINANCE
- TITLE 23 COUNTY COASTAL ZONE LAND USE ORDINANCE
- TITLE 16 COUNTY FIRE CODE ORDINANCE
- TITLE 22 COUNTY LAND USE ORDINANCE
- 2016 INTERNATIONAL BUILDING CODE APPENDIX J GRADING

Table 3-7: PR1 Mandatory Site Design Measures

MANDATORY SITE DESIGN MEASURES (SELECT AT LEAST ONE)	SELECTED	REASON, IF NOT SELECTED	RELEVANT HANDBOOK SECTION
a. Roof runoff directed into cisterns or rain barrels for reuse?	No	No Room for Cisterns	5.2.1
b. Roof runoff directed into vegetated areas (safely away from building foundations and footings)?	No	Not Obtainable	5.2.2
c. Runoff from sidewalks, walkways, and/or patios directed onto vegetated areas (safely away from the building foundation and footings)?	Yes		5.2.3
d. Runoff from driveways and/or uncovered parking lots onto vegetated areas (safely away from the building foundation and footings)?	No	Not Obtainable	5.2.4
e. Construct bike lanes, driveways, uncovered parking lots, sidewalks, walkways, and patios with permeable surfaces?	No	Lot Drains to Rear of Property	5.2.5

SHEET INDEX

GRADING PLAN

- C.1 PROJECT NOTES AND INFORMATION
- C.2 GENERAL NOTES AND INFORMATION
- C.3 SITE AND GRADING PLAN
- C.4 EROSION CONTROL PLAN
- C.5 EROSION CONTROL DEVICES
- R-1.1 SITE RETAINING WALLS
- R-2.1 SITE RETAINING WALLS

SCOPE OF WORK STATEMENT

THE PURPOSE OF THIS PROJECT IS TO CONSTRUCT A SINGLE FAMILY RESIDENCE ON LOT 106, EDGEWOOD COURT TRACT 1990-2 IN HERITAGE RANCH. THIS IS A FINISH GRADED LOT AND SUITABLE FOR THE PROPOSED RESIDENCE.

THIS PROJECT SHALL COMMENCE CONSTRUCTION UPON RECEIPT OF A BUILDING PERMIT AND WILL COMPLETE CONSTRUCTION IN APPROXIMATELY 270 DAYS. THE ESTIMATED START OF CONSTRUCTION IS THE SPRING OF 2017 AND COMPLETION OF CONSTRUCTION IN THE FALL OF 2017.

LOT CERTIFICATION STATUS

LOT 106 IS RECORDED TO HAVE A FINISH PAD ELEVATION OF 1046.0, PER TRACT 1990 - PHASE 2 AS-BUILT PLANS REVIEWED BY WENDELL WILKS, COUNTY OF SAN LUIS OBISPO, DATED 4/25/2011. FIELD SURVEY, BY HORN LAND SURVEYS, PERFORMED ON 9/16/2016, ESTABLISHED FINISH PAD ELEVATION OF APPROXIMATELY 1046.0.

FIRE SAFETY PLAN

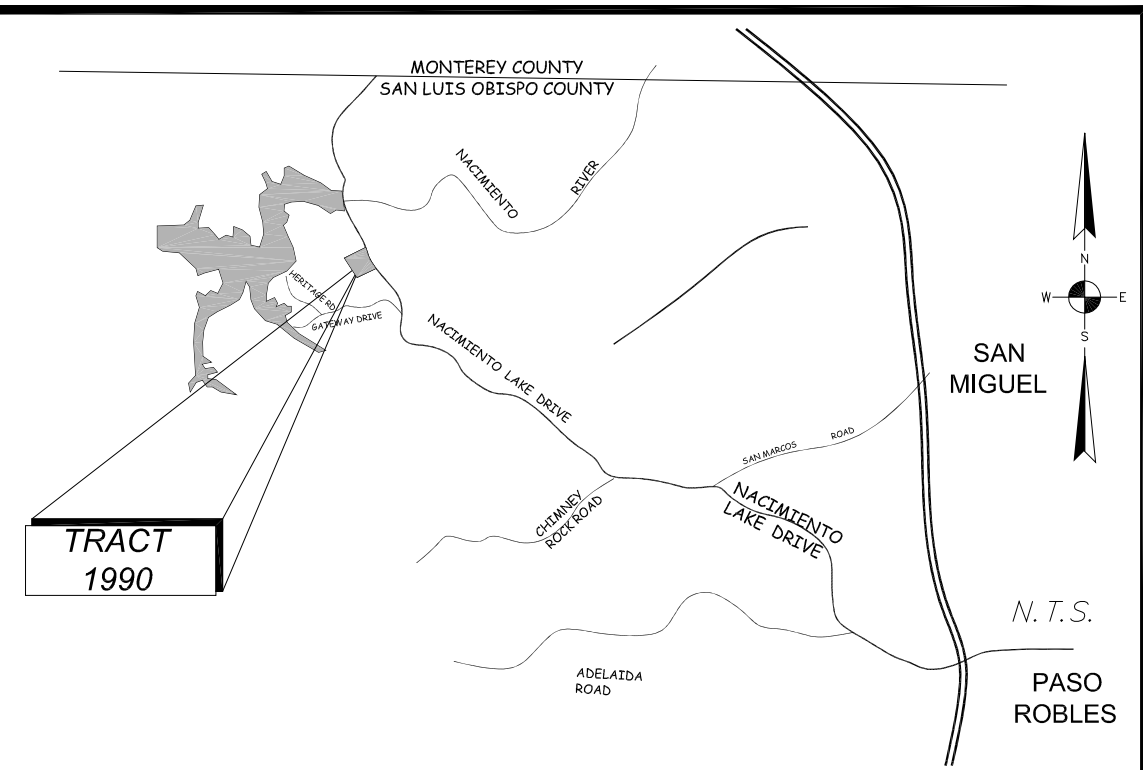
PRIOR TO ISSUANCE OF A BUILDING PERMIT, THE PROPERTY OWNER, OR APPROVED OWNER'S AGENT, SHALL RECEIVE AN APPROVED SIGN OFF FROM THE FIRE DEPARTMENT HAVING JURISDICTION FOR THIS PERMIT AND SHALL SHOW THE DEPARTMENT OF PLANNING AND BUILDING PROOF OF THIS SIGN OFF.

FIRE SPRINKLER PLAN

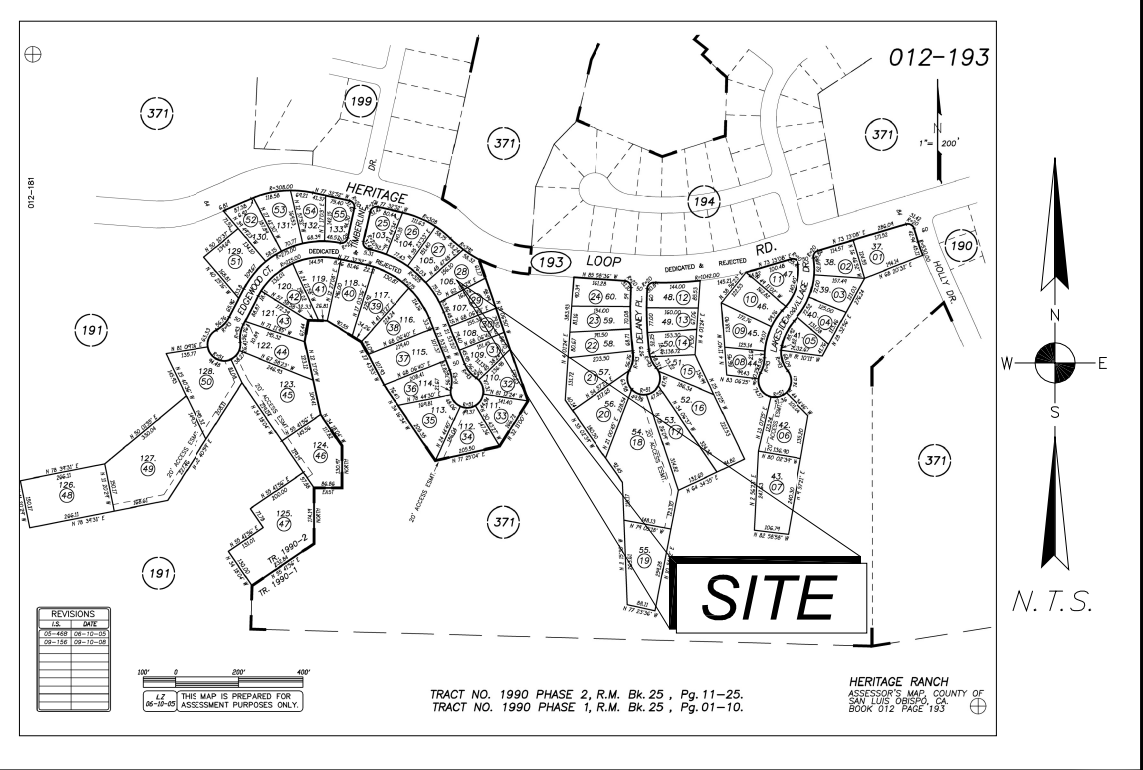
PRIOR TO ISSUANCE OF A BUILDING PERMIT, A FIRE SPRINKLER PERMIT SHALL BE SUBMITTED OR A FIRE SPRINKLER CONTACTOR SHALL CALL THE BUILDING DEPARTMENT STATING THAT A SPRINKLER PLAN IS BEING PREPARED.

GRADING AND DRAINAGE PLAN

LOT 106, TRACT 1990-2
EDGEWOOD COURT
HERITAGE RANCH



VICINITY MAP - HERITAGE RANCH



TRACT 1990-2 - LOT 106

PROJECT INFORMATION

OWNER: KIRK & CARRIE ALLEN
270 CATALINA PLACE
PASO ROBLES, CA 93446

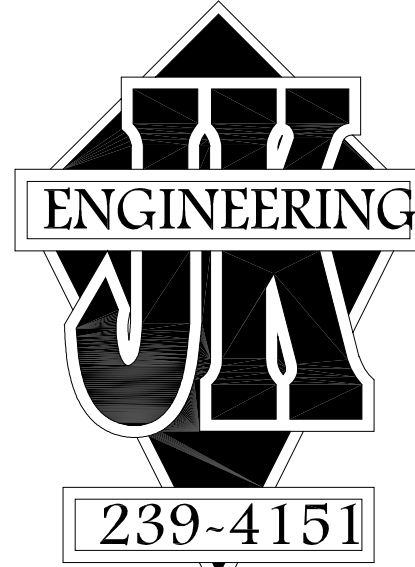
PROJECT: EDGEWOOD COURT
LOT 106 OF TRACT 1990-2,
R.M. BK 25, PG 5A,
IN THE COMMUNITY OF HERITAGE RANCH
IN THE COUNTY OF SAN LUIS OBISPO
IN THE STATE OF CALIFORNIA

APN NO: 012-193-028

EROSION CONTROL MONITOR

THE IMPLEMENTATION AND MONITORING OF THE EROSION AND SEDIMENTATION CONTROL PLAN SHALL BE THE CONTRACTOR OF RECORD, OR HIS APPOINTED REPRESENTATIVE:
CAPPS CONSTRUCTION
DEVIN CAPPS
(805) 540-1185

California Coordinates	E 1145	County Road No.	A5277 LAKESIDE VILLAGE DRIVE
------------------------	--------	-----------------	---------------------------------



John A. Kudla

Civil Engineering &
Structural Design
R.C.E. #50652
610 10th ST. UNIT 'A'
PASO ROBLES, CA.
93446

TITLE SHEET

LIRK & CARRIE ALLEN
LOT 106, TRACT 1990-2
EDGEWOOD COURT
PASO ROBLES, CA 93446



REVISION	LOG
REV.	DESCRIPTION DATE

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SCALE: 1" = 20'
PROJECT: CAPPS
DRAWN BY: FDW
CHECKED BY: JAK
DATE: 3/30/17

SHEET TITLE:

TITLE SHEET

SHEET NUMBER:

C.1

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PROJECT REPORT REQUIREMENTS

- FINAL REPORTS**
FINAL REPORTS SHALL BE REQUIRED IN ACCORDANCE WITH U.B.C. SECTION 3318.1
- PAD CERTIFICATION REQUIREMENT**
A SOIL OR CIVIL ENGINEER SHALL DETERMINE GRADING PERFORMED FOR THIS PROJECT IS IN SUBSTANTIAL CONFORMANCE WITH THE APPROVED PLANS AND IS SUITABLE TO SUPPORT THE INTENDED STRUCTURE(S). THE ENGINEER SHALL SUBMIT A FINAL REPORT TO THE COUNTY.

- SOIL REPORT REQUIREMENTS**
THE SOIL REPORT (ADDENDUM REPORT: H-16897, DATED APRIL15, 2016, BY HALLIN GEOTECHNICAL SERVICES, LLC) OBSERVATIONS AND RECOMMENDATIONS SHALL BE FOLLOWED. A REPRESENTATIVE OF THE GEOTECHNICAL SERVICES SHALL OBSERVE ALL APPLICATIONS REGARDING GENERAL GRADING, PAD GRADING, SLOPE CONSTRUCTION, UTILITY TRENCHES AND FOUNDATION CONSTRUCTION.

PROJECT SURVEY MONUMENTS

EXISTING SURVEY MONUMENTS SHALL BE TIED OUT OR REPLACED IN ACCORDANCE WITH COUNTY STANDARDS AND SPECIFICATIONS.

GRADING APPLICATION REQUIREMENTS:

- A. ALL APPLICATIONS ARE FIELD CHECKED BY THE BUILDING OFFICIAL PRIOR TO APPROVAL.
- B. PROPOSED PROJECT MUST BE COMPLETELY STAKED OUT PRIOR TO FIELD INSPECTION BY THE BUILDING OFFICIAL.
- C. PLACE STAKES AT ALL CULVERT LOCATIONS AND INDICATE SIZE ON STAKES.
- D. THE PERMITTEE OR HIS AGENT SHALL NOTIFY THE BUILDING OFFICIAL TWENTY-FOUR (24) HOURS PRIOR TO THE START OF ANY GRADING WORK.
- E. THE PERMITTEE OR HIS AGENT SHALL CALL THE BUILDING OFFICIAL FOR FINAL INSPECTION WHEN THE PROJECT IS COMPLETED. THE GRADING BOND, IF APPLICABLE, IS RELEASED UPON THE SATISFACTORY COMPLETION OF THE PROJECT.
- F. A FINAL REPORT OF THE GRADING BY THE ENGINEER SHALL BE FILED WHEN REQUIRED BY THE BUILDING OFFICIAL.
- G. NOTICE: ANY GRADING VIOLATIONS MAY RESULT IN A NOTICE OF VIOLATION BEING RECORDED WITH THE MONTEREY COUNTY RECORDER.

GRADING AND VEGETATION REMOVAL

TO CONTROL EROSION, ACTUAL GRADING SHALL BEGIN WITHIN 30 DAYS OF VEGETATION REMOVAL OR THAT AREA SHALL BE PLANTED UNDER THE PROVISIONS OF SECTION 16.08.340. NO VEGETATION REMOVAL OR GRADING WILL BE ALLOWED WHICH WILL RESULT IN SILTATION OF WATER COURSE OR UNCONTROLLABLE EROSION (16.08.300 C.2)

GROUND PREPARATION

PREPARATION OF GROUND FOR FILL. THE GROUND SURFACE SHALL BE PREPARED TO RECEIVE FILL BY THE REMOVAL OF TOPSOIL AND OTHER UNSUITABLE MATERIALS AS DETERMINED BY THE SOILS ENGINEER.

PREPARATION OF THE GROUND. THE GROUND SURFACE SHALL BE PREPARED TO RECEIVE FILL BY REMOVING VEGETATION, NON-COMPLYING FILL, TOPSOIL AND OTHER UNSUITABLE MATERIALS SCARIFYING TO PROVIDE A BOND WITH THE NEW FILL.

FILL MATERIAL PERMITTED. NO ORGANIC MATERIAL SHALL BE PERMITTED IN FILLS EXCEPT AS TOPSOIL USED FOR SURFACE PLANT GROWTH ONLY AND WHICH DOES NOT EXCEED 4 INCHES IN DEPTH. (16.08.310 E)

REGISTERED PORTABLE EQUIPMENT USAGE

PRIOR TO ISSUANCE OF GRADING OR CONSTRUCTION PERMITS, THE APPLICANT SHALL PROVIDE EVIDENCE THEY HAVE CONTACTED APCD ON ANY PROPOSED PORTABLE EQUIPMENT REQUIRING APCD OR CARB REGISTRATION, SUCH AS: PORTABLE GENERATORS AND EQUIPMENT WITH ENGINES THAT ARE 50 HORSEPOWER OR GREATER; CHEMICAL PRODUCT PROCESSING AND/OR MANUFACTURING; THE USE OF A STANDBY GENERATOR BOILERS; IC ENGINES, ETC. SHOULD ANY OF THESE TYPES OF EQUIPMENT BE USED DURING CONSTRUCTION ACTIVITIES CALIFORNIA STATEWIDE PORTABLE EQUIPMENT REGISTRATION (ISSUED BY THE CALIFORNIA AIR RESOURCES BOARD) OR AN APED PERMIT MAY BE REQUIRED.

STRUCTURAL TESTS AND SPECIAL INSTRUCTIONS		
REQUIRED VERIFICATION AND INSPECITON OF SOILS (CBC 1705; TABLE 1705.6)		
VERIFICATION AND INSPECITON TASK	VERIFICATION AND INSPECITON TASK	PERIODICALLY DURING TASK LISTED
1. VERIFY MATERIALS BELOW FOOTING ARE ADEQUATE TO ACHIEVE THE DESIGN BEARING CAPACITY	—	X
2. VERIFY EXCAVATIONS ARE EXTENDED TO PROPER DEPTH AND HAVE REACHED PROPER MATERIAL	—	X
3. PERFORM CLASSIFICATION AND TESTING CONTROLLED FILL MATERIALS	—	X
4. VERIFY USE OF PROPER MATERIALS, DENSITIES AND LIFT THICKNESSES DURING PLACEMENT AND COMPACTION OF CONTROLLED FILL	X	—
5. PRIOR TO PLACEMENT OF CONTROLLED FILL, OBSERVE SUBGRADE AND VERIFY THAT SITE HAS BEEN PREPARED PROPERLY	—	X
SPECIAL INSPECTIONS AND TESTING ARE REQUIRED FOR THIS PROJECT. HALLIN GEOTECHNICAL SERVICES, LLC WILL PERFORM THE REQUIRED INSPECTIONS IN ACCORDANCE TO CBC 1704; TABLE 1704.7 AND WILL PROVIDE INSPECTION REPORTS PRIOR TO THE POURING OF THE FOUNDATION.		

GRADING AND EROSION CONTROL NOTES

1. ALL GRADING CONSTRUCTION SHALL CONFORM TO THE RECOMMENDATIONS OF THE GEOTECHNICAL ENGINEERING REPORTS AND APPLICABLE CODES AS NOTED:
- SOILS ENGINEERING REPORT BY GEOSOLUTIONS DATED OCTOBER 9, 2002
- INTERIM GRADING REPORT FOR TRACT 1990 – PHASES I AND II BY BUENA GEOTECHNICAL SERVICES, LLC
- PAD CERTIFICATION AND EXPANSION INDEX TEST REPORT (H-16897) BY HALLIN GEOTECHNICAL SERVICES, LLC, DATED APRIL 15, 2016
- SOILS ENGINEER FOR THIS PROJECT:
HALLIN GEOTECHNICAL SERVICES, LLC
P.O. BOX 2282
PASO ROBLES, CA 93447-2282
(805) 238-3308
2. ALL GRADING SHALL CONFORM WITH THE COUNTY OF SAN LUIS OBISPO GRADING ORDINANCE AND THE EROSION CONTROL ORDINANCE.
3. ACTUAL GRADING SHALL BEGIN WITHIN 30 DAYS OF VEGETATION REMOVAL OR THE AREA SHALL BE PLANTED TO CONTROL EROSION.
4. DUST CONTROL IS TO BE MAINTAINED AT ALL TIMES DURING CONSTRUCTION.
5. AREAS OF FILL SHALL BE SCARIFIED, KEYED, BENCHED AND RECOMPACTED PER THE BENCH AND KEYWAY RECOMMENDATIONS IN SOILS REPORT, PRIOR TO REPLACING FILL WHILE UNDER OBSERVATION BY A SOIL OR CIVIL ENGINEER.
6. FILL MATERIAL WILL BE RECOMPACTED TO 90% OF MAXIMUM DENSITY.
7. REMOVE ANY DELETERIOUS MATERIAL ENCOUNTERED BEFORE PLACING FILL.
8. NO CUT OR FILL SLOPES WILL BE CONSTRUCTED STEEPER THAN TWO HORIZONTAL TO ONE VERTICAL.
9. ALL DISTRIBUTED AREA SHALL BE HYDRO SEEDED OR PLANTED WITH APPROVED EROSION CONTROL VEGETATION AS SOON AS PRACTICAL AFTER CONSTRUCTION IS COMPLETE.
10. MINIMUM SETBACK TO CREEKS AND BLUFFS SHALL BE MAINTAINED. MINIMUM SETBACK OF TWO FEET FROM ALL PROPERTY LINES WILL BE MAINTAINED FROM ALL GRADING.
11. MINIMUM SLOPE AWAY FROM BUILDINGS SHALL BE 5% FOR THE FIRST TEN FEET AROUND PERIMETER.
12. AN APPROVED EROSION CONTROL PLAN WILL BE REQUIRED TO BE SUBMITTED, APPROVED, IMPLEMENTED AND FUNCTIONAL PRIOR TO THE FIRST INSPECTION.
13. THE SOILS ENGINEER SHALL DETERMINE THE SUITABILITY OF THE SOIL TO SUPPORT THE INTENDED STRUCTURE. A COPY OF ALL COMPACTION TESTS AND FINAL GRADING REPORT SHALL BE SUBMITTED TO THE COUNTY PRIOR TO SCHEDULING ANY INSPECTION.
14. THE SOILS ENGINEER SHALL SUBMIT A LETTER STATING THE GRADING PLANS FOR THIS PROJECT ARE IN CONFORMANCE WITH THE SOILS REPORT OF RECORD.

EROSION CONTROL MEASURES

EROSION AND SEDIMENT CONTROL BEST MANAGEMENT PRACTICES MUST BE IN PLACE AND FUNCTIONAL PRIOR TO THE FIRST INSPECTION. NO INSPECTIONS CAN BE PERFORMED IF THESE DEVICES AREA NOT IN PLACE OR HAVE FAILED TO PROVIDE EROSION CONTROL. FAILURE TO MAINTAIN EROSION CONTROL WILL CAUSE INSPECTIONS TO BE DELAYED UNTIL EROSION CONTROL MEASURES ARE FUNCTIONAL.

PRE-STORM INSPECTIONS SHALL BE CONDUCTED TO ENSURE THAT BEST MANAGEMENT PRACTICES (BMPs) ARE APPROPRIATELY INSTALLED AND MAINTAINED. POST-STORM INSPECTIONS SHALL BE CONDUCTED TO ENSURE THAT BMPs HAVE FUNCTIONED ADEQUATELY. BMPs SHALL BE EVALUATED FOR ADEQUACY AND PROPER IMPLEMENTATION AND WHETHER ADDITIONAL BMPs ARE REQUIRED IN ACCORDANCE WITH THE TERMS OF THE GENERAL PERMIT. QUALIFIED PERSONNEL SHALL CONDUCT INSPECTIONS OF THE SITE:

- PRIOR TO ANTICIPATED STORM EVENTS.
- DURING EXTENDED STORM EVENTS IN 24-HOURS INTERVALS.
- AFTER ACTUAL STORM EVENTS.

FIRE PLAN - SITE PLAN REQUIREMENTS:

- THE FOLLOWING ITEMS ARE REQUIRED TO BE COMPLETED PRIOR TO THE FINAL INSPECTION OF THIS PROJECT:
- A. A FIRE SPRINKLER SYSTEM IS REQUIRED FOR THIS PROJECT.
- B. ALL SETBACKS FROM PROPERTY LINE ARE PER COUNTY PLANNING DEPT.
- C. FIRE HYDRANTS ARE EXISTING AND APPROVED PER TRACT 1990.
- D. DRIVEWAY IS 20'+ WIDE AND CONSTRUCTED OF CONCRETE
- E. 100' DEFENSIVE VEGETATION CLEARANCE IS REQUIRED.
- F. LPG TANK IS TO HAVE MINIMUM SEPARATION FROM BUILDINGS AND PROPERTY LINES.
- G. CLASS 'A' NON-COMBUSTIBLE ROOF IS TO MEET ALL REQUIREMENTS OF CHAPTER 7A OF THE 2010 CALIFORNIA BUILDING CODE.
- H. THIS PROJECT SHALL MEET THE FIRE-RESISTANCE-RATED CONSTRUCTION REQUIREMENTS OF THE 2010 CALIFORNIA BUILDING CODE, CHAPTER 7A.
- I. THE RESIDENCE SHALL HAVE SEPARATE ADDRESS NUMBERS, ASSIGNED BY THE SLO COUNTY PLANNING DEPARTMENT AND SHALL BE HIGHLY VISIBLE PERMANENT ADDRESS NUMBERS. THE ADDRESS NUMBERS SHALL BE PLACED AT THE ENTRANCE OF THE DRIVEWAY AND RESIDENCE. REFLECTIVE NUMBERS ARE HIGHLY RECOMMENDED.
- J. SMOKE DETECTORS ARE REQUIRED IN ALL SLEEPING AREAS AND IN HALLWAYS LEADING TO SLEEPING AREAS.
- WHEN ALL OF THE FIRE SAFETY REQUIREMENTS HAVE BEEN COMPLETED, PLEASE CALL THE FIRE PREVENTION BUREAU AT (805) 543-4244, EXTENSION # 3429, TO ARRANGE FOR THE FINAL INSPECTION. FOR MORE INFORMATION, VISIT THE CAL FIRE WEBSITE AT www.calfireslo.org.

ESTIMATED EARTHWORKS

ESTIMATED TOTAL AREA OF DISTURBANCE: 0.22 ACRES

HOUSE AND DRIVEWAY

ESTIMATED CUT: ±340 CUBIC YARDS
ESTIMATED FILL: ±200 CUBIC YARDS

EXCAVATED CUT: ± 7 FOOT
COMPACTED FILL: ± 7 FOOT

(EXCESS CUT MATERIAL SHALL BE ADVANTAGEOUSLY ADDED TO THE GRADED AREA NOT TO EXCEED 1' LIFTS)

SHRINKAGE, CONSOLIDATION AND SUBSIDENCE FACTORS AND LOSSES DUE TO CLEARING AND DEMOLITION OPERATIONS ARE NOT INCLUDED. ESTIMATED EARTHWORK QUANTITIES ARE BASED ON THE APPROXIMATE DIFFERENCE BETWEEN EXISTING GRADES AND PROPASED ROADWAY SUBGRADES, AS INDICATED ON THE PLANS, AND SHOULD VARY ACCORDING TO THESE FACTORS AND LOSSES.

THE GRADING CONTRACTOR SHALL REVIEW THE SITE AND THE GEOTECHNICAL REPORT(S), SHALL ACCEPT OR CONFIRM EXISTING TOPOGRAPHIC INFORMATION, SHALL PERFORM AN INDEPENDENT EARTHWORK QUANTITY ESTIMATE, AND SHALL BID ACCORDINGLY.

PROJECT TRACT DRAINAGE

THE DRAINAGE SHOWN HAS BEEN VERIFIED TO MEET THE APPROVED DRAINAGE FOR THIS TRACT

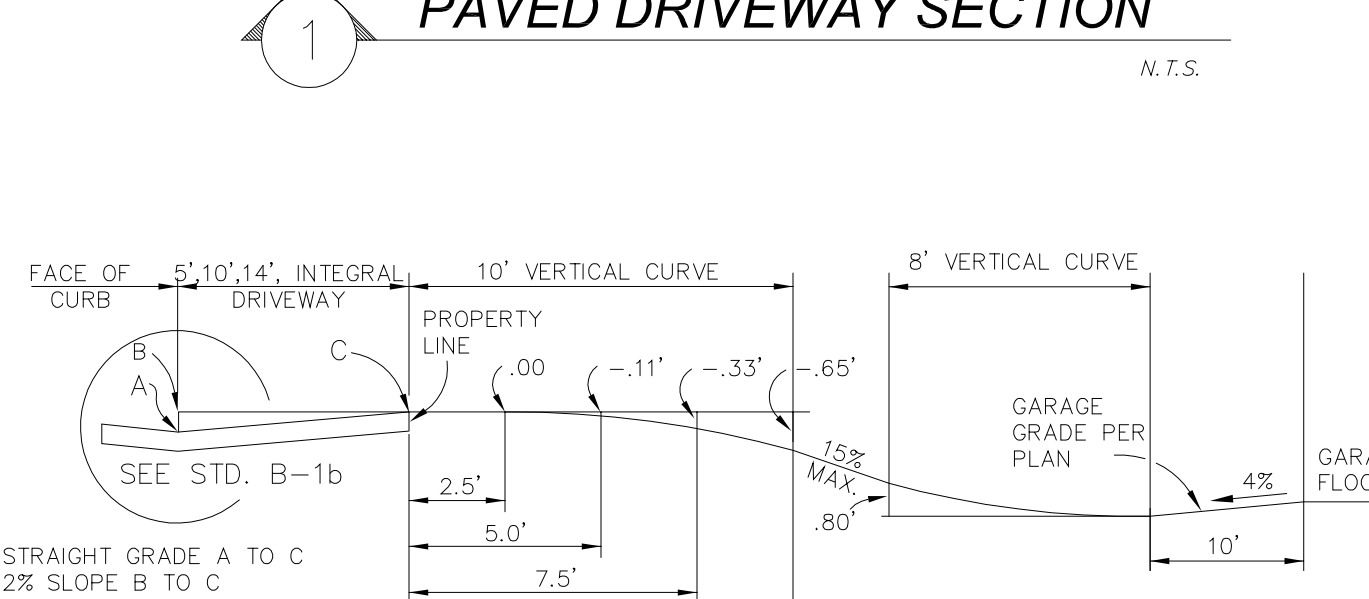
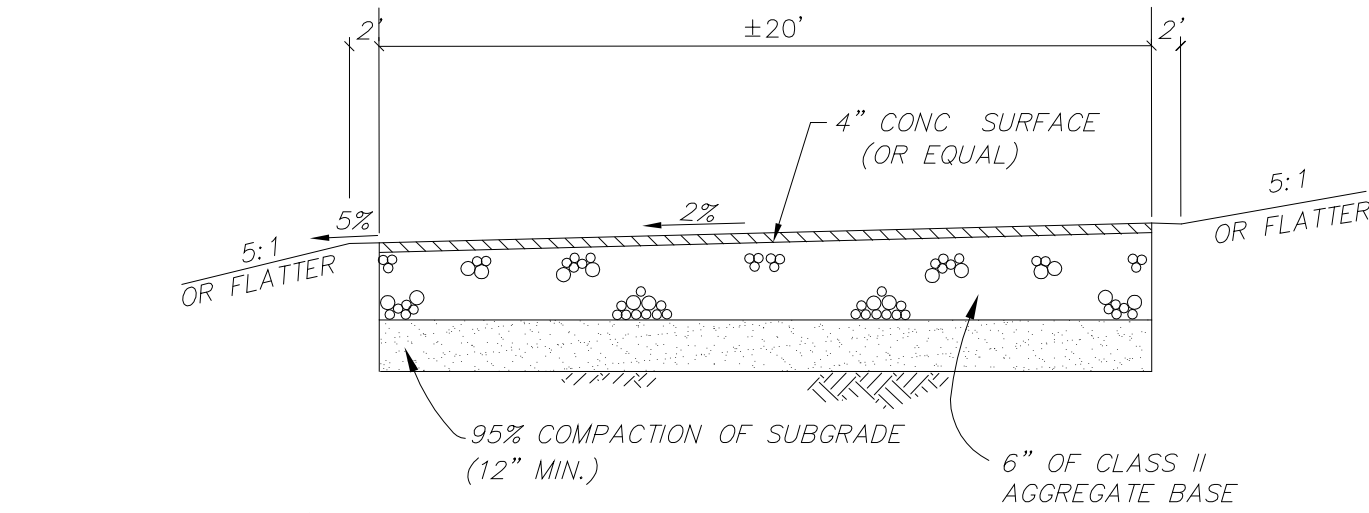
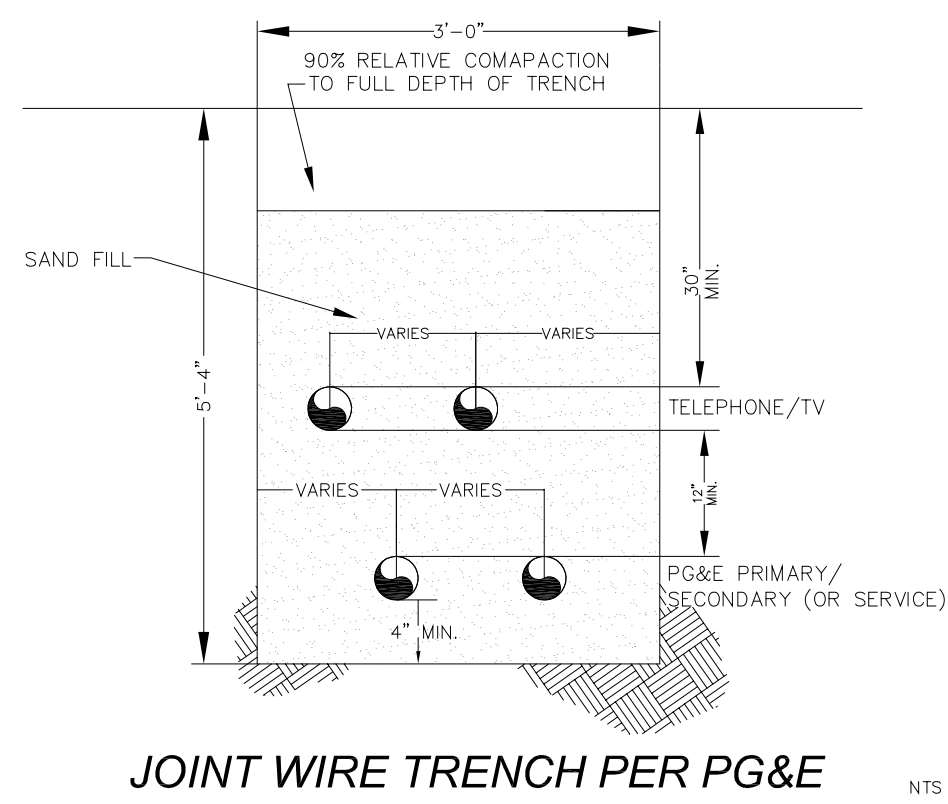
UTILITY SERVICES

1. UNDERGROUND ELECTRICAL SERVICE IS EXISTING AT THE P.U.E. OF EDGEWOOD COURT TO SERVICE THE NEW RESIDENCE.
2. UNDERGROUND TELEPHONE SERVICE IS EXISTING AT THE P.U.E. OF EDGEWOOD COURT TO SERVICE THE NEW RESIDENCE.
3. WATER SERVICE IS EXISTING AT THE P.U.E. OF EDGEWOOD COURT. A 1-1/2" SERVICE LINE SHALL BE INSTALLED TO THE RESIDENCE.
4. PROPANE GAS TANK SHALL BE INSTALLED TO THE NEW RESIDENCE USING 1" POLYETHYLENE PLASTIC PIPE. THE TANK SHALL BE SET BACK 10' FROM PROPERTY LINES AND STRUCTURES. TANK SHALL BE COVERED WITH APPROVED DECORATIVE BLIND.

LEGEND

- W ——— PROPOSED WATER SERVICE LINE
- E T ——— PROPOSED ELECTRICAL, TELEPHONE JOINT TRENCH
- G ——— PROPOSED PROPANE SERVICE LINE
- PROPOSED PROPANE TANK WITH APPROVED BLIND

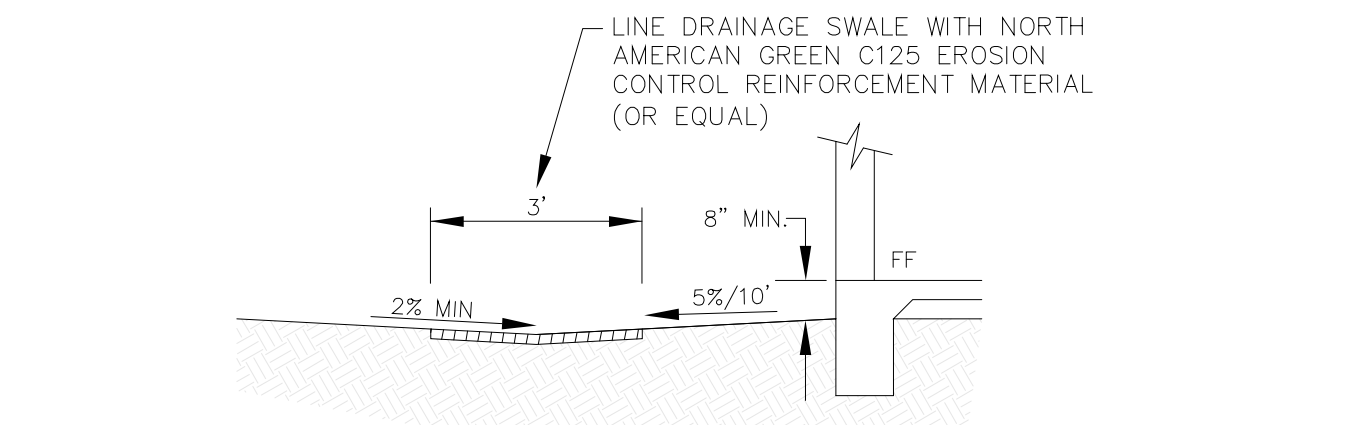
PRIOR TO CONSTRUCTION, THE CONTRACTOR OF RECORD SHALL INSPECT THE LOCATION OF ALL EXISTING UTILITIES, TO VERIFY PROPOSED CONDUIT RUNS FROM THE PROPOSED UTILITY LOCATION TO THE RESIDENCE.



- NOTE:
- MAXIMUM RISE, AND THE RUN, SHALL BE MEASURED FOR THE WORST CONDITION BETWEEN THE BACK OF THE SIDEWALK AND THE FINISHED FLOOR AT THE GARAGE OR CARPORT ENTRANCE.
 - NON-INTEGRAL DRIVEWAY RAMPS MUST BE POURED AS SOON AS PRACTICABLE AFTER CONSTRUCTION OF CURB AND GUTTER.
 - WHENEVER POSSIBLE BUILDING DESIGNERS AND ENGINEERS SHOULD ATTEMPT TO MAXIMIZE A PARKING AREA ADJACENT TO GARAGE WHEN LAYING OUT DWELLINGS AND THEIR DRIVE ACCESS TO THE PUBLIC STREET.

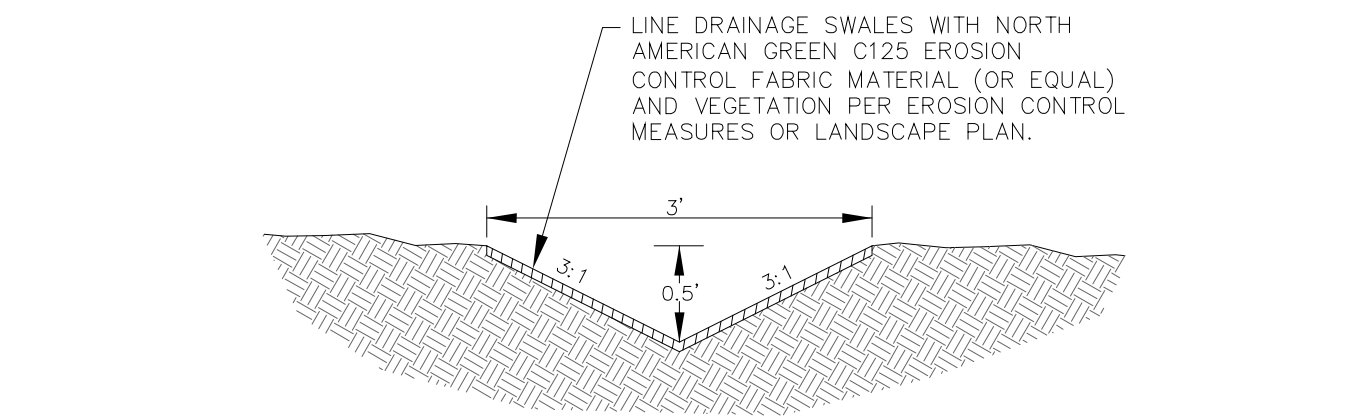
STANDARD DOWNWARD DRIVEWAY

REFER TO COUNTY STD DWG B-3c (SIM) N.T.S.



DRAINAGE SWALE

REFER TO NORTH AMERICAN GREEN CHANNEL SPECIFICATIONS FOR INSTALLATION INFORMATION

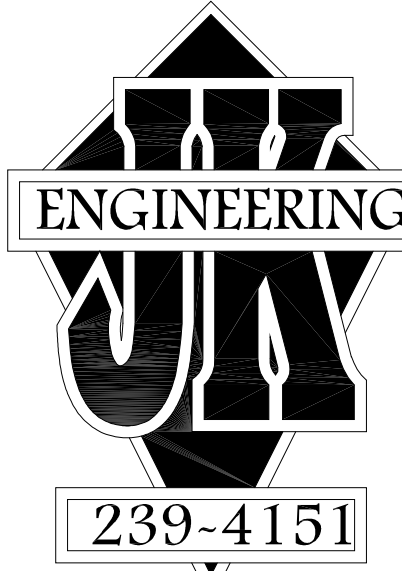
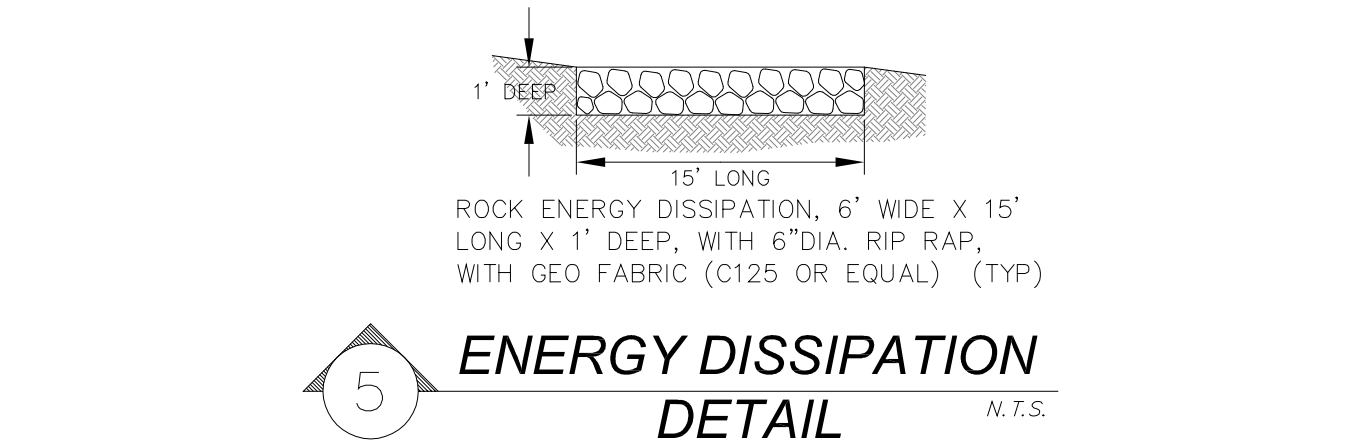


DRAINAGE SWALE

REFER TO NORTH AMERICAN GREEN CHANNEL SPECIFICATIONS FOR INSTALLATION INFORMATION, SEE SHEET C.5

LOW IMPACT DEVELOPMENT (LID) MEASURES

THIS PROJECT IS A TIER ONE PROJECT AND IS REQUIRED TO UTILIZE ONE AGENCY APPROVED STORM-WATER RUNOFF REDUCTION MEASURES. THE LOW IMPACT DEVELOPMENT (LID) MEASURE IS ALL DRAINAGE SWALES SHALL BE VEGETATED.



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GENERAL NOTES
AND INFORMATION
KIRK & CARRIE ALLEN
LOT 106, TRACT 1990-2
EDGEWOOD COURT
PASO ROBLES, CA 93446



REVISION LOG		
REV.	DESCRIPTION	DATE

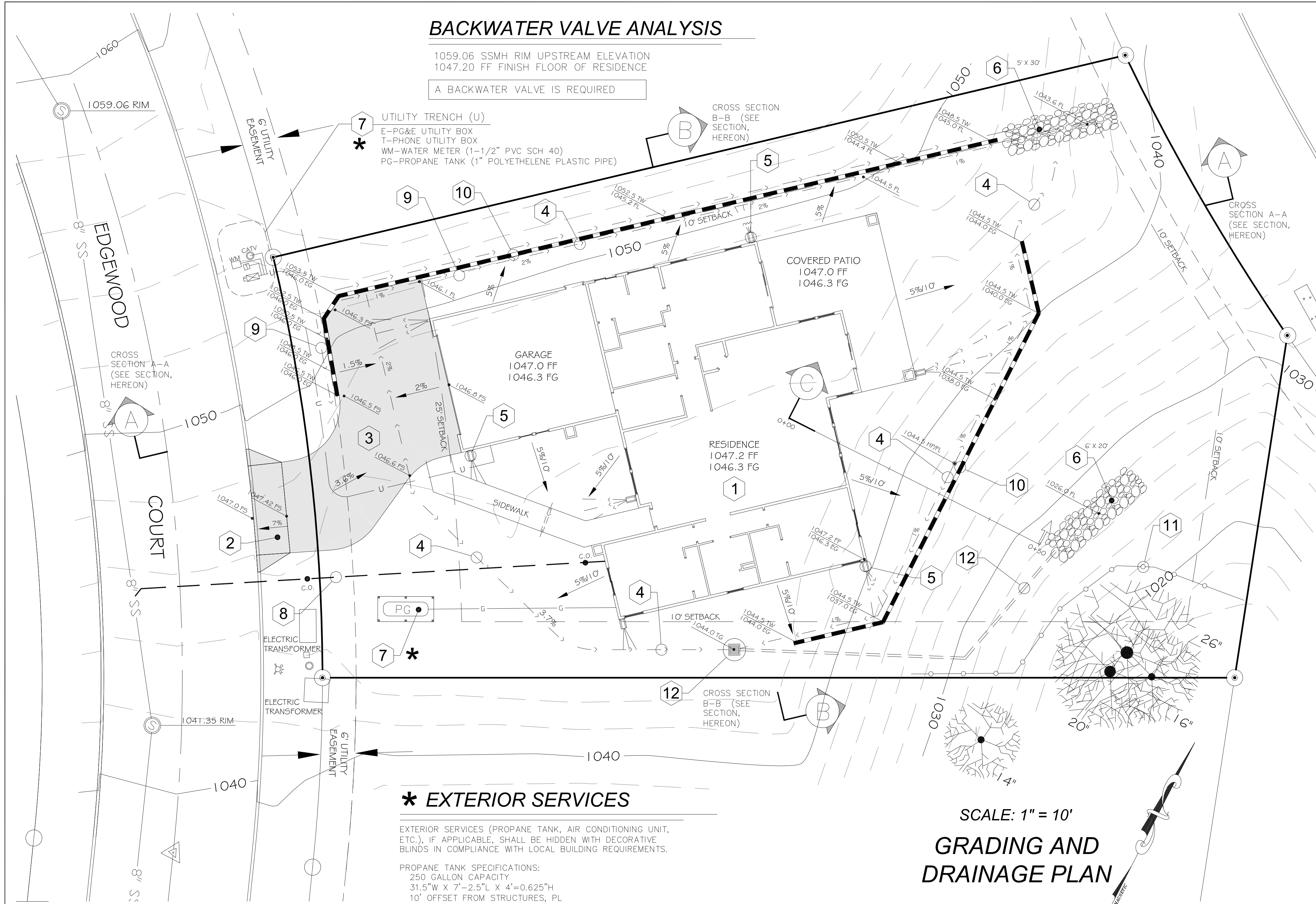
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PROJECT: CAPPS
DRAWN BY: FDW
CHECKED BY: JAK
DATE: 3/28/17

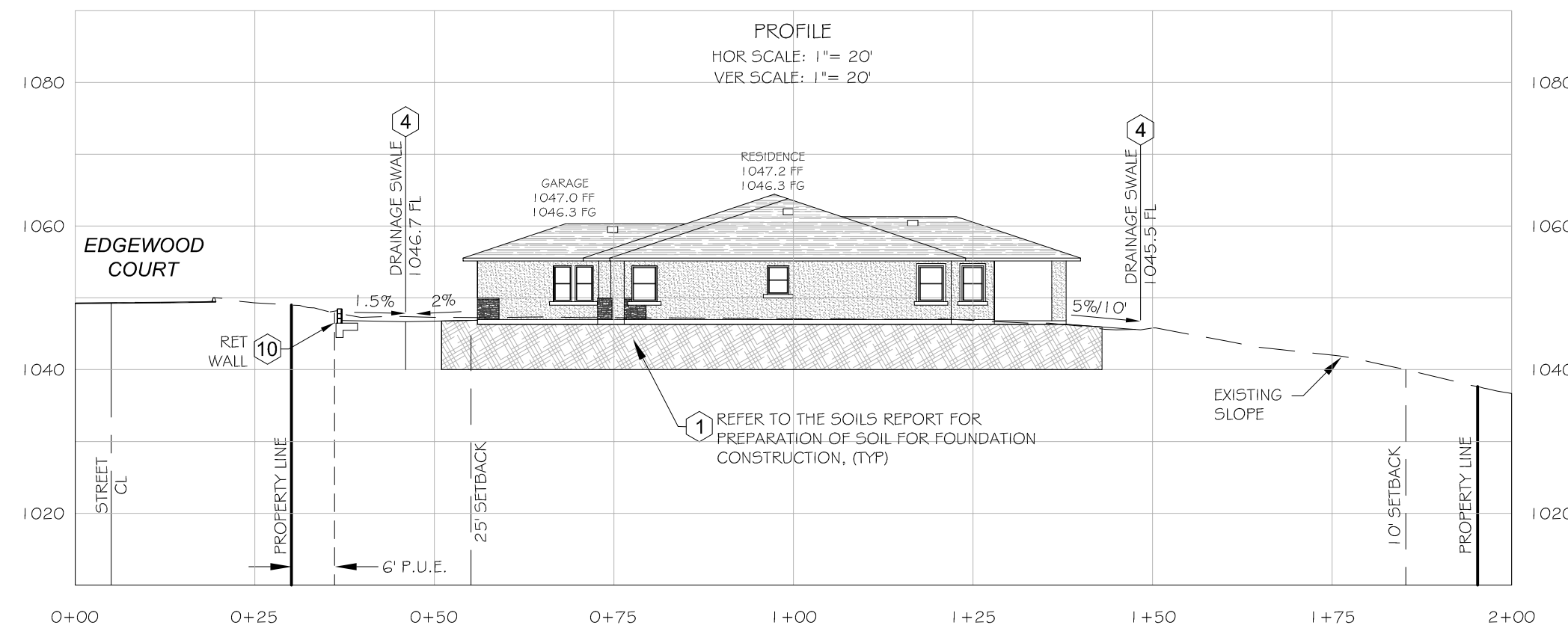
SHEET TITLE:
GENERAL NOTES

SHEET NUMBER:

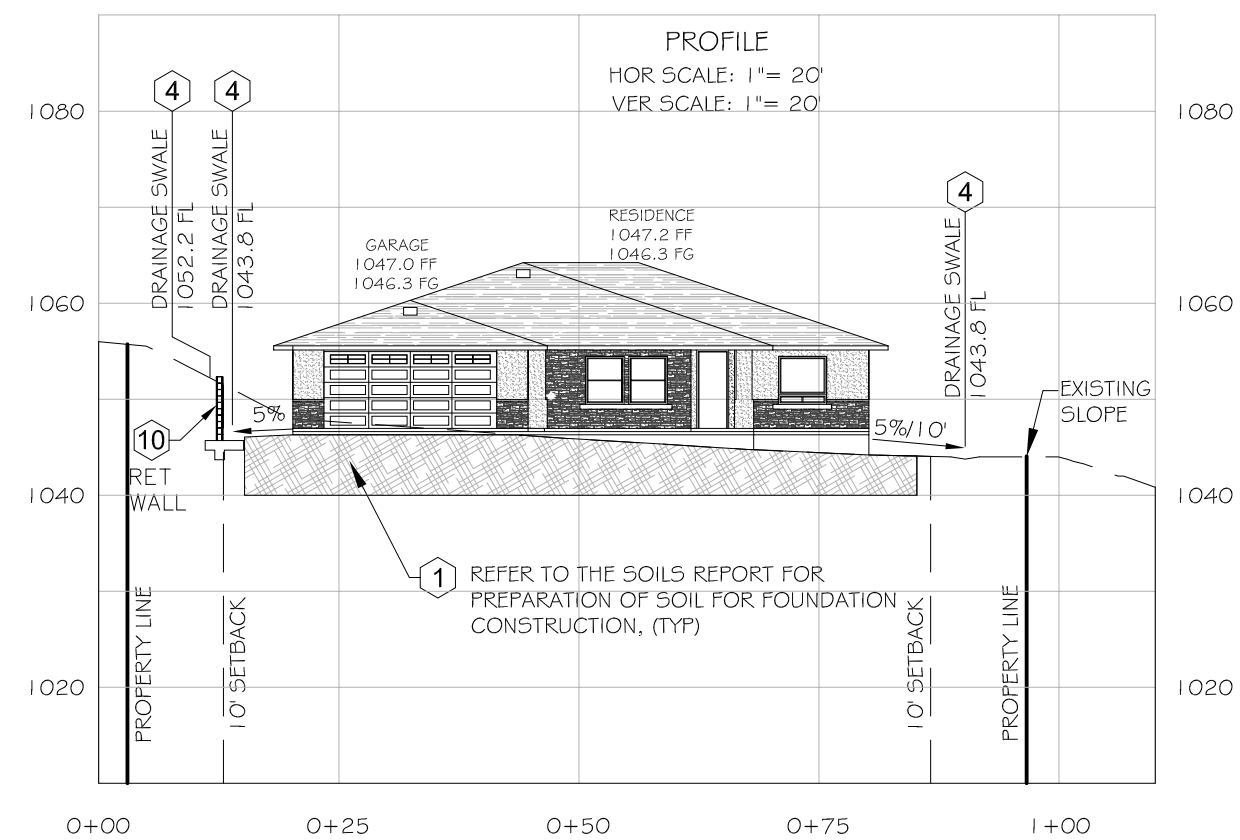
C.2



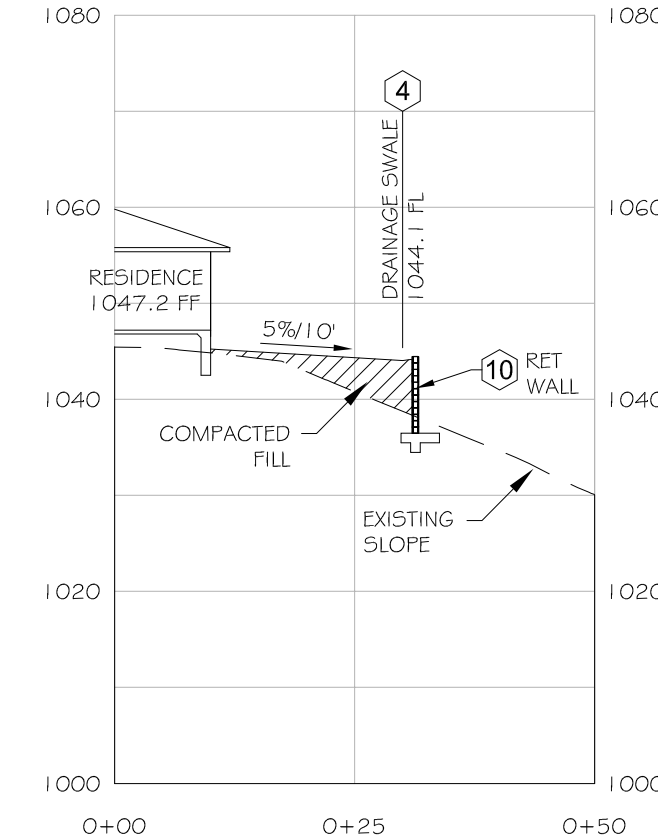
SECTION A-A



SECTION B-B



SECTION C-C

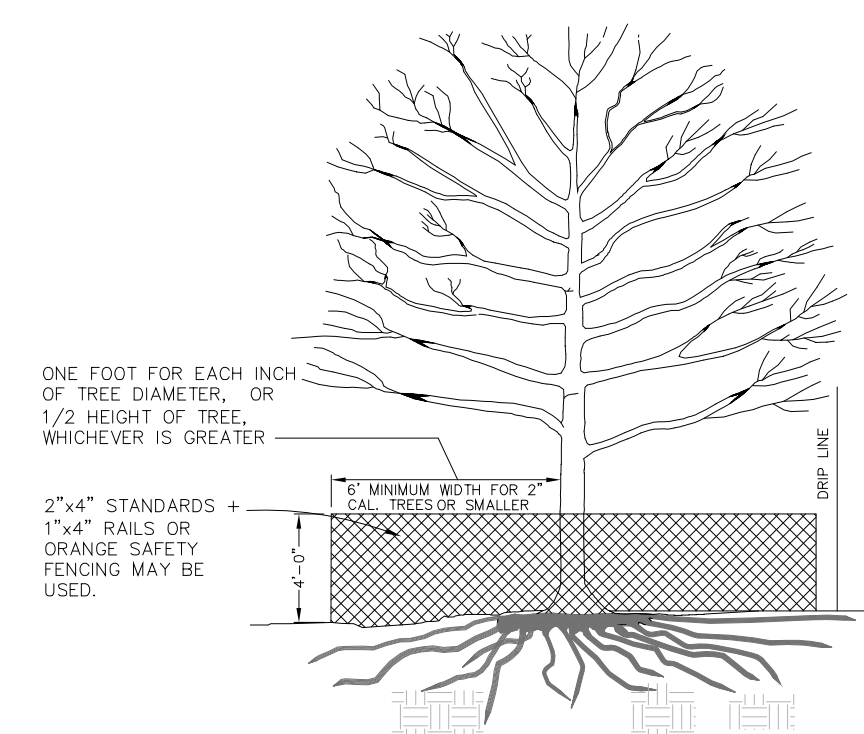


CONSTRUCTION NOTES:

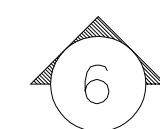
- ① CONSTRUCT PAD ELEVATION TO 1065.5 FINISH GRADE PER LOT CERTIFICATION STATUS DOCUMENT, GEOTECHNICAL SOILS REPORT RECOMMENDATIONS BY HALLIN GEOTECHNICAL (TYP).
- ② CONSTRUCT RURAL DRIVEWAY APPROACH PER CO STD DWG B-2a. AN ENCROACHMENT PERMIT SHALL BE REQUIRED FOR ALL WORK PERFORMED IN THE COUNTY RIGHT OF WAY, (TYP).
- ③ CONSTRUCT PAVED DRIVEWAY PER DETAILS 1 AND 2, SHEET C.2, (TYP).
- ④ CONSTRUCT 3' WIDE DRAINAGE SWALE PER DETAILS 3 AND 4, SHEET C.2, (TYP).
- ⑤ INSTALL DRAIN DOWN SPOUT WITH CONCRETE SPLASH BLOCKS, AND DIRECT DRAINAGE TO DRAINAGE SWALES, (TYP).
- ⑥ INSTALL ROCK SLOPE PROTECTION PER SLO CO DRAWING DETAIL H-5, SEE DETAIL 5, SHEET C.2, AND INSTALL GEOTECH FABRIC UNDER RIPRAP. (TYP)
- * ⑦ INSTALL UTILITIES TO RESIDENCE PER UTILITY SERVICES DETAIL, SHEET C.2, (TYP).
- ⑧ INSTALL SEWER LATERAL PER SLO CO DRAWING DETAIL S-3, WITH CLEAN OUT (C.O.), (TYP).
- ⑨ EXCAVATE CUTS PER RECOMMENDATIONS OF SOILS ENGINEER, (TYP)
- ⑩ CONSTRUCT RETAINING WALL PER DETAILS R-4, R-6, SHEET R1.1, (TYP).
- ⑪ INSTALL TREE PROTECTION FENCE PER DETAIL 6, SHEET C.3, (TYP).
- ⑫ INSTALL 12" CATCH BASIN WITH 4" FLEXIBLE PVC PIPE, DIRECT TO ROCK SLOPE PROTECTION, (TYP)

SURFACE DRAINAGE NOTE

SURFACE DRAINAGE SHALL BE GRADED TO A DRAIN SURFACE WATER AWAY FROM FOUNDATION WALL. THE GRADE SHALL FALL A MINIMUM OF 6" WITH THE FIRST 10'. EXCEPTION: WHERE LOT LINES, WALL, SLOPES, OR OTHER PHYSICAL BARRIERS PROHIBIT 6" OF FALL WITHIN 10', DRAINS OR SWALES SHALL BE CONSTRUCTED TO ENSURE DRAINAGE AWAY FROM STRUCTURE. IMPERVIOUS SURFACES WITHIN 10' OF THE BUILDING FOUNDATION SHALL BE SLOPED A MINIMUM OF 2% AWAY FROM BUILDING (CRC R401.3 DRAINAGE, EXCEPTIONS).



TREE PROTECTION DETAIL



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GRADING PLAN

KIRK & CARRIE ALLEN
LOT 106, TRACT 1990-2
EDGEWOOD COURT
PASO ROBLES, CA 93446

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SCALE:	1" = 10'
PROJECT:	CAPPS
DRAWN BY:	FDW
CHECKED BY:	JAK
DATE:	3/28/17

SHEET TITLE:

*GRADING
PLAN*

SHEET NUMBER

C.3

BEST MANAGEMENT PRACTICES:

1. TEMPORARY SOIL STABILIZATION (EROSION CONTROL)

- SCHEDULING
PRESERVATION OF EXISTING VEGETATION
HYDRAULIC MULCH
HYDROSEEDING
SOIL BINDERS
STRAW MULCH
GEOTEXTILES, PLASTIC COVERS, AND EROSION CONTROL BLANKETS/MATS
WOOD MULCHING
EARTH DIKES/DRAINAGE SWALES AND LINED DITCHES
OUTLET PROTECTION/VELOCITY DISSIPATION DEVICES
SLOPE DRAINS
STREAMBANK STABILIZATION
POLYACRYLAMIDE

2. TEMPORARY SEDIMENT CONTROL

- SILT FENCE
SEDIMENT/DESILTING BASIN
SEDIMENT TRAP
CHECK DAM
FIBER ROLLS
GRAVEL BAG BERM
STREET SWEEPING AND VACUUMING
SANDBAG BARRIER
STRAW BALE BARRIER
STORM DRAIN INLET PROTECTION
CHEMICAL TREATMENT

3. WIND EROSION CONTROL

- WIND EROSION CONTROL

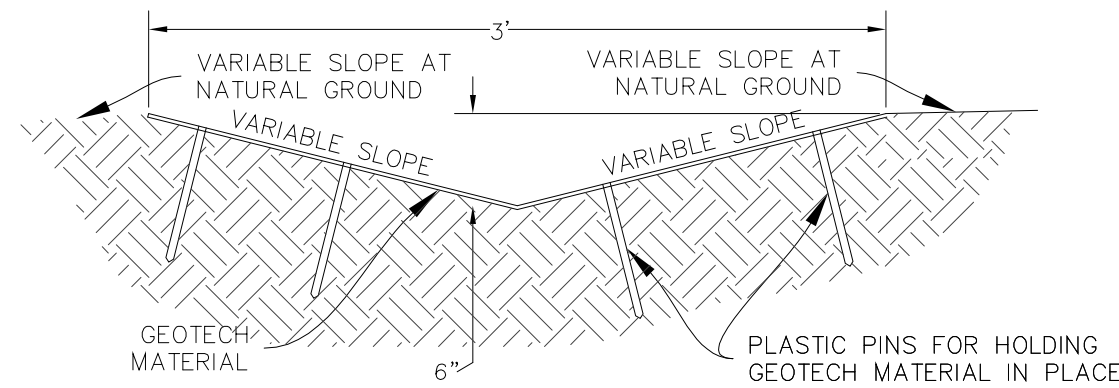
4. TRACKING CONTROL

- STABILIZED CONSTRUCTION ENTRANCE/EXIT
STABILIZED CONSTRUCTION ROADWAY
ENTRANCE/OUTLET TIRE WASH

EROSION CONTROL NOTES:

1. EROSION CONTROL MEASURES SHALL BE IMPLEMENTED ON ALL PROJECTS AND SHALL INCLUDE SOURCE CONTROL, INCLUDING PROTECTION OF STOCKPILES, PROTECTION OF SLOPES, PROTECTION OF ALL DISTURBED AREAS, AND PROTECTION OF ACCESSES. IN ADDITION, PERIMETER CONTAINMENT MEASURES SHALL BE PLACED PRIOR TO THE COMMENCEMENT OF GRADING AND SITE DISTURBANCE ACTIVITIES UNLESS THE PUBLIC WORKS DEPARTMENT TEMPORARY MEASURES TO BE UNNECESSARY BASED UPON LOCATION, SITE CHARACTERISTICS OR TIME OF YEAR. THE INTENT OF EROSION CONTROL MEASURES SHALL BE TO KEEP ALL SEDIMENT FROM ENTERING A SWALE, DRAINAGE WAY, WATERCOURSE, OR ONTO ADJACENT PROPERTIES.
2. SITE INSPECTIONS AND APPROPRIATE MAINTENANCE OF EROSION CONTROL DEVICES SHALL BE CONDUCTED AND DOCUMENTED PRIOR TO, DURING, AND AFTER RAIN EVENTS.
3. THE DEVELOPER SHALL BE RESPONSIBLE FOR THE PLACEMENT AND MAINTENANCE OF ALL EROSION CONTROL DEVICES AS SPECIFIED BY THE APPROVED PLAN UNTIL SUCH TIME THAT THE PROJECT IS ACCEPTED AS COMPLETE BY THE PUBLIC WORKS DEPARTMENT. EROSION CONTROL DEVICES MAY BE RELATED, DELETED OR ADDITIONAL ITEMS MAY BE REQUIRED DEPENDING ON THE ACTUAL SOIL CONDITIONS DISCRETION OF THE ENGINEER OF WORK, COUNTY INSPECTOR, SWPPP MONITOR, OR RWQCB INSPECTOR. GUIDELINES FOR DETERMINING APPROPRIATE EROSION CONTROL DEVICES ARE INCLUDED IN THE APPENDIX OF THE PUBLIC IMPROVEMENT STANDARDS.
4. ALL EROSION CONTROL DEVICES SHALL BE THE FIRST ORDER OF WORK AND SHALL BE IN PLACE PRIOR TO THE START OF CONSTRUCTION AND/OR ANYTIME WHEN THE RAIN PROBABILITY EXCEEDS 30%. THIS WORK SHALL BE INSTALLED OR APPLIED AFTER EACH AREA IS GRADED AND NO LATER THAN FIVE (5) WORKING DAYS AFTER COMPLETION OF EACH AREA.
5. THE ENGINEER OF WORK AND THE PUBLIC WORKS DEPARTMENT SHALL BE NOTIFIED FOR INSPECTION OF INSTALLED EROSION CONTROL DEVICES.
6. A STANDBY CREW FOR EMERGENCY WORK SHALL BE AVAILABLE AT ALL TIMES DURING THE PHASE OF CONSTRUCTION. NECESSARY MATERIALS SHALL BE AVAILABLE AND STOCK PILED AT CONVENIENT LOCATIONS TO FACILITATE RAPID CONSTRUCTION OR MAINTENANCE OF TEMPORARY DEVICES WHEN RAIN IS IMMINENT.
7. PERMANENT EROSION CONTROL SHALL BE PLACED AND ESTABLISHED WITH 90% COVERAGE ON ALL DISTURBED SURFACES OTHER THAN PAVED OR GRAVEL SURFACES, PRIOR TO FINAL INSPECTION. PERMANENT EROSION CONTROL SHALL BE FULLY ESTABLISHED PRIOR TO FINAL ACCEPTANCE. TEMPORARY EROSION CONTROL MEASURES SHALL REMAIN IN PLACE UNTIL PERMANENT MEASURES ARE ESTABLISHED.
8. IN THE EVENT OF A FAILURE, THE DEVELOPER AND/OR HIS REPRESENTATIVE SHALL BE RESPONSIBLE FOR CLEANUP AND ALL ASSOCIATED COSTS OR DAMAGES. IN THE EVENT THAT DAMAGE OCCURS WITHIN THE RIGHT OF WAY AND THE COUNTY IS REQUIRED TO PERFORM CLEANUP, ALL WORK SHALL CEASE ON THE PROJECT UNTIL CLEANUP COSTS ARE FULLY PAID.
9. IF ANY WORK IS NOT IN COMPLIANCE WITH THE PLANS OR PERMITS APPROVED FOR THE PROJECT, THE DEPARTMENT SHALL REVOKE ALL ACTIVE PERMITS AND RECOMMEND THE COUNTY CODE ENFORCEMENT PROVIDE A WRITTEN NOTICE OR STOP WORK ORDER IN ACCORDANCE WITH SECTION 22.52.140 [23.10] OF THE LAND USE ORDINANCE.
10. ALL PROJECTS INVOLVING SITE DISTURBANCE OF ONE ACRE OR GREATER SHALL COMPLY WITH THE REQUIREMENTS OF THE NATIONAL POLLUTANT DISCHARGE ELIMINATION SYSTEM (NPDES). THE DEVELOPER SHALL SUBMIT A NOTICE OF INTENT (NOI) TO COMPLY WITH THE GENERAL PERMIT FOR CONSTRUCTION ACTIVITY WITH THE REGIONAL WATER QUALITY CONTROL BOARD (RWQCB). THE DEVELOPER SHALL PROVIDE THE COUNTY WITH THE WASTE DISCHARGE IDENTIFICATION NUMBER (WQID #) OR WITH VERIFICATION THAT AN EXEMPTION HAS BEEN GRANTED BY RWQCB.
- WQID NUMBER: PROJECT IS EXEMPT FROM WQID REQUIREMENTS
11. PERSON TO CONTACT 24 HOURS A DAY IN THE EVENT THERE IS AN EROSION CONTROL/SEDIMENTATION PROBLEM (STORM WATER COMPLIANCE OFFICER):

NAME: CAPPS CONSTRUCTION
LOCAL PHONE NUMBER: (805) 540-1185



GEOTECH LINED SWALE (EC-9)

N.T.S.

- EC-1
EC-2
EC-3
EC-4
EC-5
EC-6
EC-7
EC-8
EC-9
EC-10
EC-11
EC-12
EC-13

STABILIZED CONSTRUCTION ENTRANCE REQUIREMENTS

THE INGRESS/EGRESS BY CONSTRUCTION VEHICLES SHALL HAVE THE FOLLOWING MEASURE AND REQUIREMENTS IN EFFECT TO MINIMIZE THE TRACKING OF MUD AND DIRT ONTO EXISTING STREETS.

THE FOLLOWING INSTALLATION GUIDELINES SHALL BE IMPLEMENTED:

PROPER GRADE ENTRANCE TO PREVENT RUNOFF FROM THE CONSTRUCTION SITE. THE ENTRANCE ELEVATION OF THE DRIVEWAY SHALL BE LOWER THAN THE CONNECTING STREET.

ROUTE RUNOFF FROM STABILIZED ENTRANCE THROUGH A SEDIMENT TRAPPING DEVICE BEFORE WATER IS DISCHARGED.

DESIGN STABILIZED ENTRANCE TO SUPPORT THE HEAVIEST VEHICLES WHICH WILL USE IT.

SELECT ENTRANCE STABILIZATION (AGGREGATE, ASPHALTIC CONCRETE, CONCRETE) BASED ON LONGEVITY, REQUIRED PERFORMANCE, AND SITE CONDITIONS.

IF AGGREGATE IS SELECTED, PLACE AN 8" THICK COURSE OF AGGREGATE OVER THE GEOTEXTILE FABRIC OR A THICKNESS OF AGGREGATE RECOMMENDED BY A SOILS ENGINEER.

THE FOLLOWING INSPECTION AND MAINTENANCE PROCEDURES SHALL BE FOLLOWED:

INSPECTION OF THE STABILIZED CONSTRUCTION ENTRANCE SHALL BE ROUTINELY PERFORMED FOR DAMAGE AND REPAIR AS NEEDED.

REQUIREMENT THAT ALL EMPLOYEES, SUBCONTRACTORS, AND SUPPLIERS SHALL UTILIZE THE STABILIZED CONSTRUCTION ENTRANCE.

SEDIMENT TRAPPING DEVICES SHALL BE SERVICED ON A REGULAR BASIS.

CONSTRUCTION BMP MAINTENANCE, INSPECTION AND REPAIR

INSPECTIONS WILL BE CONDUCTED AS FOLLOWS:

PRIOR TO A FORECAST STORM

AFTER A RAIN EVENT THAT CAUSES RUNOFF FOR THE CONSTRUCTION SITE

AT 24-HOUR INTERVALS DURING EXTENDED RAIN EVENTS

A FOLLOW-UP PROCEDURE SHALL FOLLOW ANY INSPECTION THAT IDENTIFIES DEFICIENCIES IN THE EROSION CONTROL DEVICES.

POST-CONSTRUCTION BMP CONTROL PRACTICES

THE FOLLOWING POST-CONSTRUCTION BEST MANAGEMENT PRACTICES THAT ARE TO BE USED AT THE CONSTRUCTION SITE AFTER ALL CONSTRUCTION IS COMPLETE:

OUTLET PROTECTION / VELOCITY DISSIPATION DEVICES SHALL BE INSTALLED AT ALL DRAINAGE SWALES AND DRAINAGE OUTLETS

ALL SLOPES WILL BE SEEDED WITH, PLANTED AND PROTECTED WITH STRAW MULCH

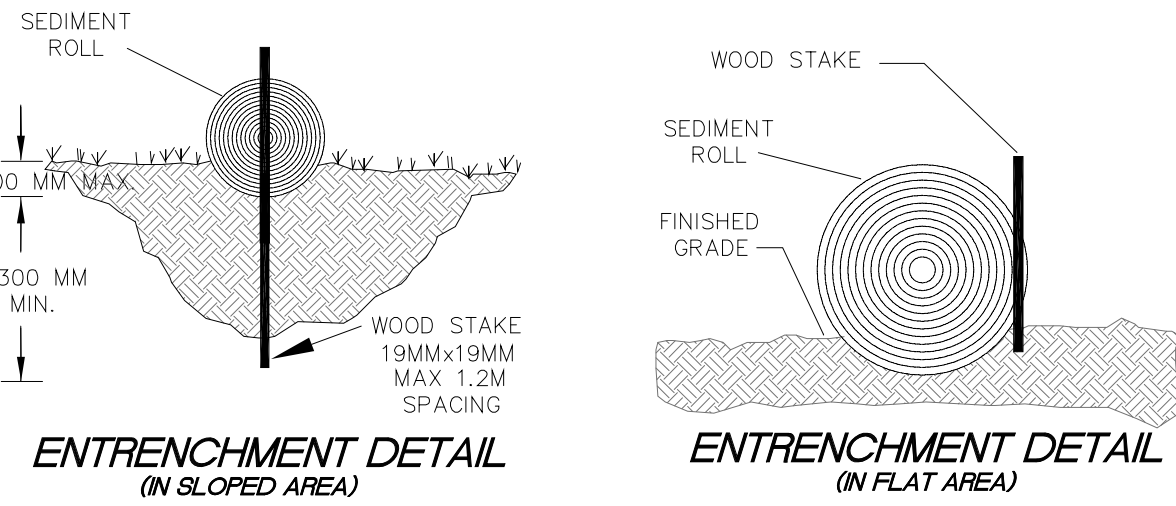
EROSION CONTROL PLANTING NOTES:

A. ALL EROSION CONTROL PLANTING SHALL COMPLY WITH THE COUNTY REQUIREMENTS FOR REVEGETATION AND LANDSCAPE PLANS SHALL BE IMPLEMENTED AS SOON AS POSSIBLE FOLLOWING COMPLETION OF ANY SOIL DISTURBING ACTIVITIES. HYDROSEEDING SHALL BE INSTALLED ACCORDING TO THE FOLLOWING SPECIFICATIONS:

1. THE WORK SHALL CONSIST OF HYDRO-SEEDING EROSION CONTROL MATERIAL CONSISTING OF A MIXTURE OF STABILIZING EMULSION (BINDER), FIBER, SEED, COMMERCIAL FERTILIZER AND WATER TO CUT AND FILL SLOPES ON CONSTRUCTION SITES.
2. THE SEED MIXTURE FOR THE EROSION CONTROL MATERIAL SHALL CONSIST OF THE FOLLOWING APPROXIMATE PROPORTIONS:

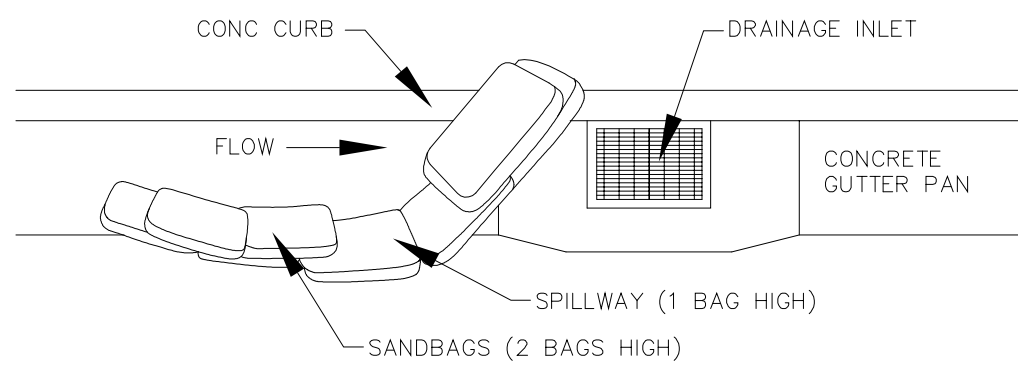
MATERIAL	POUNDS PER ACRE (MEASURED ON SLOPE)
FIBER	1,500 lbs
BARLEY SEED	180 lbs
COMMERCIAL FERTILIZER (16-20-0)	400 lbs
WATER	AS NEEDED FOR APPLICATION
STABILIZING EMULSION	AS RECOMMENDED BY MANUFACTURER

3. MIXING OF MATERIAL FOR APPLICATION WITH HYDRO-SEEDING EQUIPMENT SHALL BE PERFORMED IN A TANK WITH A BUILT-IN CONTINUOUS AGITATION SYSTEM OF SUFFICIENT OPERATING CAPACITY TO PRODUCE A HOMOGENEOUS MIXTURE AND DISCHARGE SYSTEM WHICH SHALL APPLY THE MIXTURE AT A CONTINUOUS AND UNIFORM RATE. THE TANK SHALL HAVE A MINIMUM CAPACITY OF 1,000 GALLONS. THE ENGINEER MAY AUTHORIZE USE OF EQUIPMENT OF SMALLER CAPACITY IF IT DEMONSTRATED SUCH EQUIPMENT IS CAPABLE OF PERFORMING ALL OPERATIONS SATISFACTORILY.



FIBER ROLLS (SE-5)

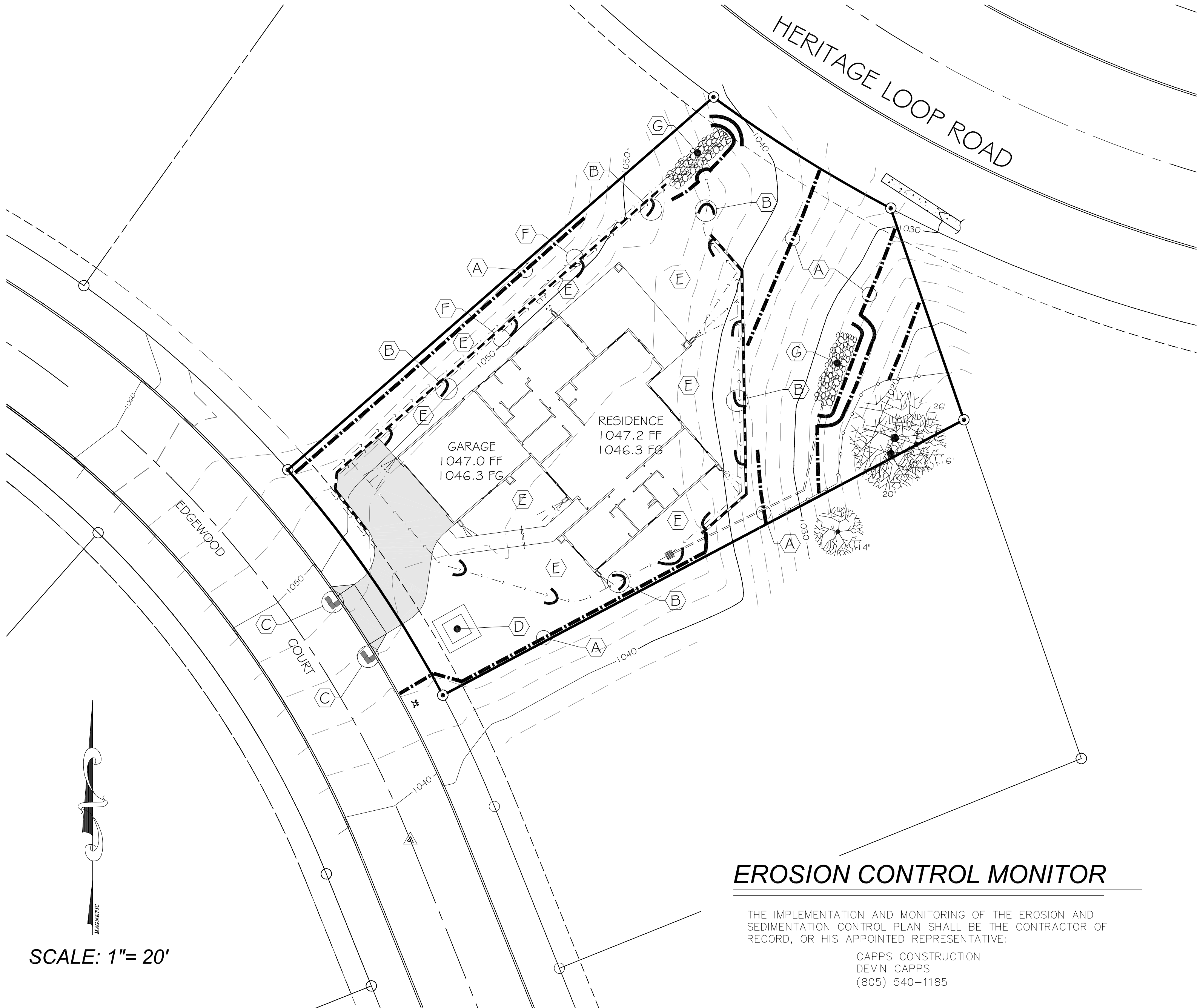
N.T.S.



STORMDRAIN INLET PROTECTION (SE-10)

N.T.S.

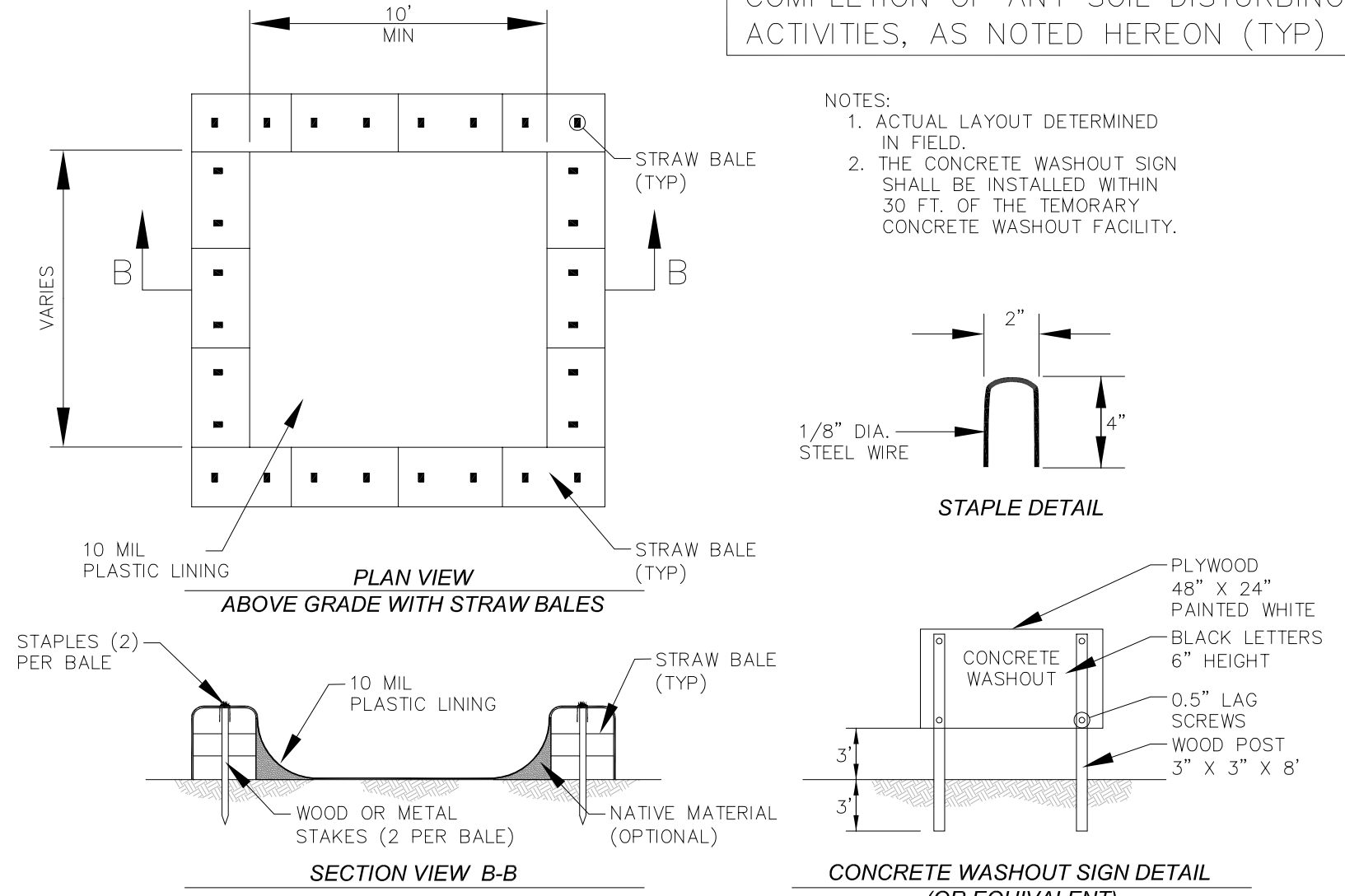
- NOTES:
1. INTENDED FOR SHORT TERM USE.
2. USE TO INHIBIT NON-STORM WATER FLOW.
3. ALLOW THE PROPER MAINTENANCE AND CLEANUP.
4. BAGS MUST BE REMOVED AFTER ADJACENT OPERATION IS COMPLETED.
5. NOT APPLICABLE IN AREAS WITH HIGH SILTS AND CLAYS WITHOUT FIBER FABRIC.



SCALE: 1"= 20'

INSTALL EROSION CONTROL PLANTING, AS SOON AS POSSIBLE, FOLLOWING COMPLETION OF ANY SOIL DISTURBING ACTIVITIES, AS NOTED HEREON (TYP)

- NOTES:
1. ACTUAL LAYOUT DETERMINED IN FIELD.
2. THE CONCRETE WASHOUT SIGN SHALL BE INSTALLED WITHIN 30 FT. OF THE TEMPORARY CONCRETE WASHOUT FACILITY.



CONCRETE WASTE MANAGEMENT (WM-8)

N.T.S.

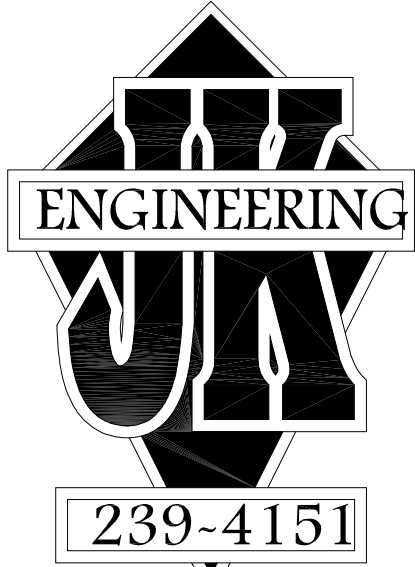
EROSION CONTROL MONITOR

THE IMPLEMENTATION AND MONITORING OF THE EROSION AND SEDIMENTATION CONTROL PLAN SHALL BE THE CONTRACTOR OF RECORD, OR HIS APPOINTED REPRESENTATIVE:

CAPPS CONSTRUCTION
DEVIN CAPPS
(805) 540-1185

EROSION CONTROL NOTES:

- (A) INSTALL FIBER ROLLS (SE-5) PARALLEL TO THE CONTOURS TO PROTECT THE SLOPE BEFORE EROSION CONTROL PLANTING GERMINATION AND TO SLOW DRAINAGE AND TRAP SEDIMENT, REFER TO SHEET C.6 FOR EROSION CONTROL MBP'S (TYP)
- (B) INSTALL FIBER ROLLS (SE-5) IN A CHEVRON SHAPE TO SLOW DRAINAGE AND TRAP SEDIMENT, REFER TO SHEET C.6 FOR EROSION CONTROL MBP'S (TYP)
- (C) INSTALL SANDBAG BARRIER (SE-8, SE-10) IN A CHEVRON SHAPE, AT EACH DRAINAGE OUTLET TO SLOW DRAINAGE AND TRAP SEDIMENT, AND INSTALL SAND BAG BARRIER (SE-10) AT CATCH BASIN. REFER TO SHEET C.6 FOR EROSION CONTROL MBP'S (TYP)
- (D) INSTALL CONCRETE WASTE MANAGEMENT (WM-8) PRIOR TO THE PLACEMENT OF CONCRETE AND STUCCO, REFER TO WM-8 DETAIL, HEREON (TYP)
- (E) INSTALL EROSION CONTROL PLANTING (EC-6, EC-7) OR LANDSCAPING, AS SOON AS POSSIBLE, FOLLOWING COMPLETION OF ANY SOIL DISTURBING ACTIVITIES, AS NOTED HEREON, REFER TO SHEET C.6 FOR EROSION CONTROL MBP'S (TYP)
- (F) INSTALL LINED DRAINAGE SWALES (EC-9), ALL SWALES SHALL BE VEGETATED TO MEET COMPLIANCE OF LOW IMPACT DEVELOPMENT REQUIREMENTS, REFER TO SHEET C.6 FOR EROSION CONTROL MBP'S (TYP)
- (G) INSTALL VELOCITY DISSIPATION DEVICES (EC-10), REFER TO SHEET C.6 FOR EROSION CONTROL MBP'S (TYP).



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EROSION CONTROL PLAN

KIRK & CARRIE ALLEN
LOT 106, TRACT 1980-2
EDGEWOOD COURT
PASO ROBLES, CA 93446



REVISION LOG

REV.	DESCRIPTION	DATE

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SCALE: 1"= 10'
PROJECT: CAPPS
DRAWN BY: FDW
CHECKED BY: JAK
DATE: 3/28/17

SHEET TITLE:

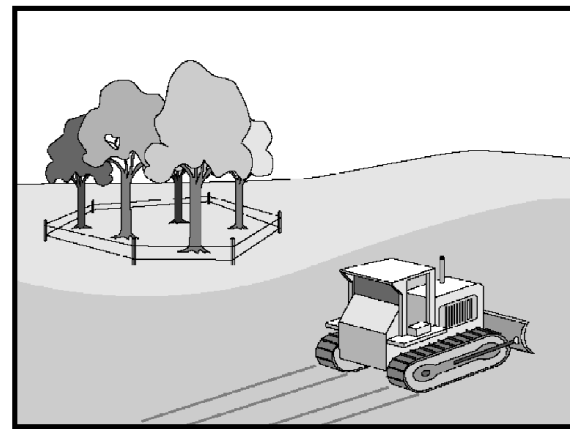
EROSION CONTROL PLAN

SHEET NUMBER:

C.4

EC-2

Preservation Of Existing Vegetation EC-2



Categories	
EC Erosion Control	☒
SE Sediment Control	☐
TC Tracking Control	☐
WE Wind Erosion Control	☐
NS Non-Stormwater Management Control	☐
WM Waste Management and Materials Pollution Control	☐
Legend:	
☒ Primary Objective	
☒ Secondary Objective	

Targeted Constituents	
Sediment	☒
Nutrients	☐
Trash	☐
Metals	☐
Bacteria	☐
Oil and Grease	☐
Organics	☐

Potential Alternatives	
None	

- Areas within the site where no construction activity occurs, or occurs at a later date. This BMP is especially suitable to multi-year projects where grading can be phased.
- Areas where natural vegetation exists and is designated for preservation. Such areas often include steep slopes, watercourse, and building sites in wooded areas.
- Areas where local, state, and federal government require preservation, such as vernal pools, wetlands, marshes, certain oak trees, etc. These areas are usually designated on the plans, or in the specifications, permits, or environmental documents.
- Where vegetation designated for ultimate removal can be temporarily preserved and be utilized for erosion control and sediment control.

Limitations

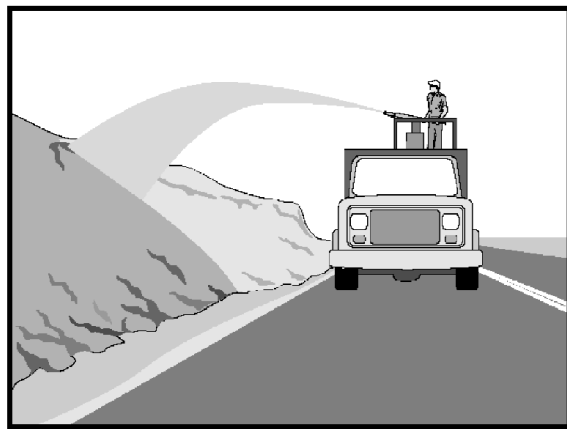
- Requires forward planning by the owner/developer.



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EC-6

Straw Mulch EC-6



Categories	
EC Erosion Control	☒
SE Sediment Control	☐
TC Tracking Control	☐
WE Wind Erosion Control	☒
NS Non-Stormwater Management Control	☐
WM Waste Management and Materials Pollution Control	☐
Legend:	
☒ Primary Category	
☒ Secondary Category	

Targeted Constituents	
Sediment	☒
Nutrients	☐
Trash	☐
Metals	☐
Bacteria	☐
Oil and Grease	☐
Organics	☐

Potential Alternatives	
EC-3 Hydraulic Mulch	
EC-4 Hydroseeding	
EC-5 Soil Binders	
EC-7 Geotextiles and Mats	
EC-8 Wood Mulching	
EC-14 Compost Blanket	

- As a stand-alone BMP on disturbed areas until soils can be prepared for permanent vegetation. The longevity of straw mulch is typically less than six months.
- Applied in combination with temporary seeding strategies to enhance plant establishment and final soil stabilization.
- Applied around containerized plantings to control erosion until the plants become established to provide permanent stabilization.

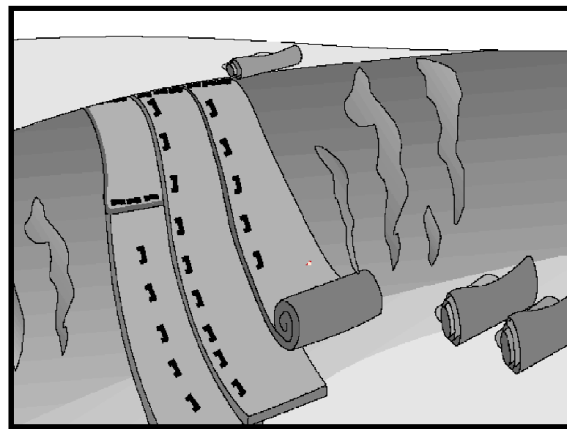
- Limitations**
- Availability of straw and straw blowing equipment may be limited just prior to the rainy season and prior to storms due to high demand.



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EC-7

Geotextiles and Mats EC-7



Categories	
EC Erosion Control	☒
SE Sediment Control	☐
TC Tracking Control	☐
WE Wind Erosion Control	☒
NS Non-Stormwater Management Control	☐
WM Waste Management and Materials Pollution Control	☐
Legend:	
☒ Primary Category	
☒ Secondary Category	

Targeted Constituents	
Sediment	☒
Nutrients	☐
Trash	☐
Metals	☐
Bacteria	☐
Oil and Grease	☐
Organics	☐

Potential Alternatives	
EC-3 Hydraulic Mulch	
EC-4 Hydroseeding	

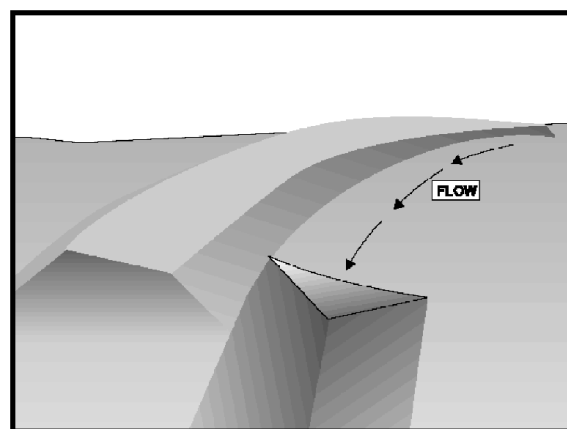
- Description and Purpose**
- Mattings, or Rolled Erosion Control Products (RECPs), can be made of natural or synthetic materials or a combination of the two. RECPs are used to cover the soil surface to reduce erosion from rainfall impact, hold soil in place, and absorb and hold moisture near the soil surface. Additionally, RECPs may be used to stabilize soils until vegetation is established or to reinforce non-woody surface vegetation.
- Suitable Applications**
- RECPs are typically applied on slopes where erosion hazard is high and vegetation will be slow to establish. Mattings are also used on stream banks, swales and other drainage channels where moving water at velocities between 3 ft/s and 6 ft/s are likely to cause scour and wash out new vegetation, and in areas where the soil surface is disturbed and where existing vegetation has been removed. RECPs may also be used when seeding cannot occur (e.g., late season construction and/or the arrival of an early rain season). RECPs should be considered when the soils are fine grained and potentially erosive. RECPs should be considered in the following situations:
- Steep slopes, generally steeper than 3:1 (H:V)
 - Slopes where the erosion potential is high
 - Slopes and disturbed soils where mulch must be anchored
 - Disturbed areas where plants are slow to develop



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EC-9

Earth Dikes and Drainage Swales EC-9



Categories	
EC Erosion Control	☒
SE Sediment Control	☐
TC Tracking Control	☐
WE Wind Erosion Control	☐
NS Non-Stormwater Management Control	☐
WM Waste Management and Materials Pollution Control	☐
Legend:	
☒ Primary Objective	
☒ Secondary Objective	

Targeted Constituents	
Sediment	☒
Nutrients	☐
Trash	☐
Metals	☐
Bacteria	☐
Oil and Grease	☐
Organics	☐

Potential Alternatives	
None	

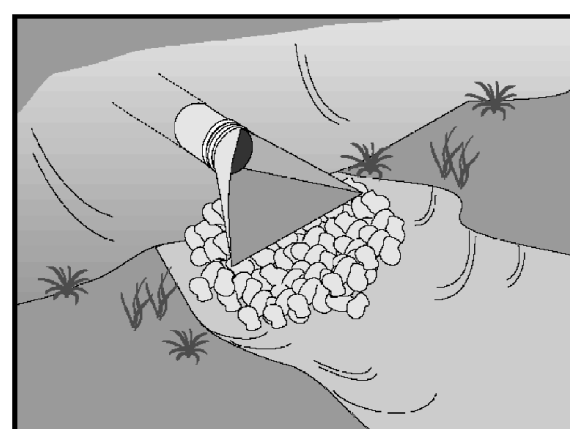
- Description and Purpose**
- An earth dike is a temporary berm or ridge of compacted soil used to divert runoff or channel water to a desired location. A drainage swale is a shaped and sloped depression in the soil surface used to convey runoff to a desired location. Earth dikes and drainage swales are used to divert off-site runoff around the construction site, divert runoff from stabilized areas and disturbed areas, and direct runoff into sediment basins or traps.
- Suitable Applications**
- Earth dikes and drainage swales are suitable for use, individually or together, where runoff needs to be diverted from one area and conveyed to another.
- Earth dikes and drainage swales may be used:
 - To convey surface runoff down sloping land
 - To intercept and divert runoff to avoid sheet flow over sloped surfaces
 - To divert and direct runoff towards a stabilized watercourse, drainage pipe or channel
 - To intercept runoff from paved surfaces
 - Below steep grades where runoff begins to concentrate
- Along roadways and facility improvements subject to flood drainage



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EC-10

Velocity Dissipation Devices EC-10



Categories	
EC Erosion Control	☒
SE Sediment Control	☐
TC Tracking Control	☐
WE Wind Erosion Control	☐
NS Non-Stormwater Management Control	☐
WM Waste Management and Materials Pollution Control	☐
Legend:	
☒ Primary Objective	
☒ Secondary Objective	

Targeted Constituents	
Sediment	☒
Nutrients	☐
Trash	☐
Metals	☐
Bacteria	☐
Oil and Grease	☐
Organics	☐

Potential Alternatives	
None	

- Description and Purpose**
- Outlet protection is a physical device composed of rock, grouted riprap, or concrete rubble, which is placed at the outlet of a pipe or channel to prevent scour of the soil caused by concentrated, high velocity flows.
- Suitable Applications**
- Whenever discharge velocities and energies at the outlets of culverts, conduits, or channels are sufficient to erode the next downstream reach. This includes temporary diversion structures to divert runoff during construction.
- These devices may be used at the following locations:
 - Outlets of pipes, drains, culverts, slope drains, diversion ditches, swales, conduits, or channels.
 - Outlets located at the bottom of mild to steep slopes.
 - Discharge outlets that carry continuous flows of water.
 - Outlets subject to short, intense flows of water, such as flash floods.
 - Points where lined conveyances discharge to unlined conveyances

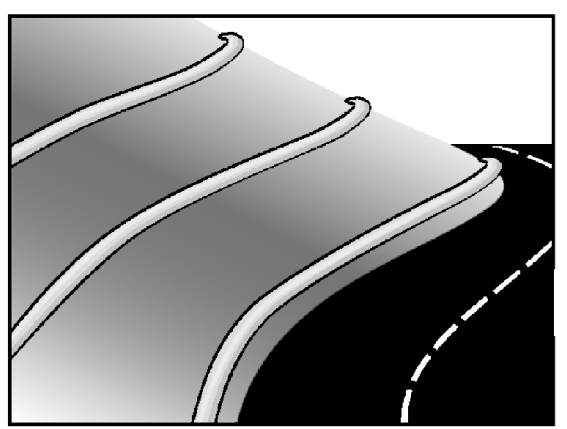
- Limitations**
- Large storms or high flows can wash away the rock outlet protection and leave the area susceptible to erosion.



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SE-5

Fiber Rolls SE-5



Categories	
EC Erosion Control	☒
SE Sediment Control	☒
TC Tracking Control	☐
WE Wind Erosion Control	☐
NS Non-Stormwater Management Control	☐
WM Waste Management and Materials Pollution Control	☐
Legend:	
☒ Primary Category	
☒ Secondary Category	

Targeted Constituents	
Sediment	☒
Nutrients	☐
Trash	☐
Metals	☐
Bacteria	☐
Oil and Grease	☐
Organics	☐

Potential Alternatives	
SE-1 Silt Fence	
SE-6 Gravel Bag Berm	
SE-8 Sandbag Barrier	
SE-14 Biofilter Bags	

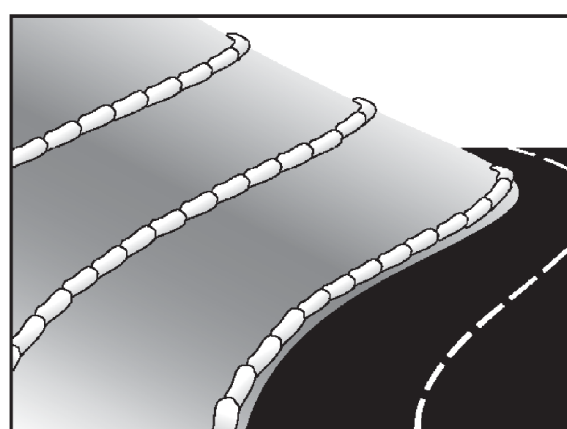
- Description and Purpose**
- A fiber roll consists of straw, coir, or other biodegradable materials bound into a tight tubular roll wrapped by netting, which can be photodegradable or natural. Additionally, gravel core fiber rolls are available, which contain an unbedded ballast material such as gravel or sand for additional weight when staking the rolls over non-forests (such as use as inlet protection). When fiber rolls are placed at the toe and on the face of slopes along the contours, they intercept runoff, reduce its flow velocity, release the runoff as sheet flow, and provide removal of sediment from the runoff (through sedimentation). By interrupting the length of a slope, fiber rolls can also reduce sheet and rill erosion until vegetation is established.
- Suitable Applications**
- Fiber rolls may be suitable:
- Along the toe, top, face, and at grade breaks of exposed and erodible slopes to shorten slope length and spread runoff as sheet flow.
 - At the end of a downward slope where it transitions to a steeper slope.
 - Along the perimeter of a project.
 - As check dams in unlined ditches with minimal grade.
 - Down-slope of exposed soil areas.
 - At operational storm drains as a form of inlet protection.



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SE-6

Gravel Bag Berm SE-6



Categories	
EC Erosion Control	☒
SE Sediment Control	☒
TC Tracking Control	☐
WE Wind Erosion Control	☐
NS Non-Stormwater Management Control	☐
WM Waste Management and Materials Pollution Control	☐
Legend:	
☒ Primary Category	
☒ Secondary Category	

Targeted Constituents	
Sediment	☒
Nutrients	☐
Trash	☐
Metals	☐
Bacteria	☐
Oil and Grease	☐
Organics	☐

Potential Alternatives	
SE-1 Silt Fence	
SE-5 Fiber Roll	
SE-8 Sandbag Barrier	
SE-14 Biofilter Bags	

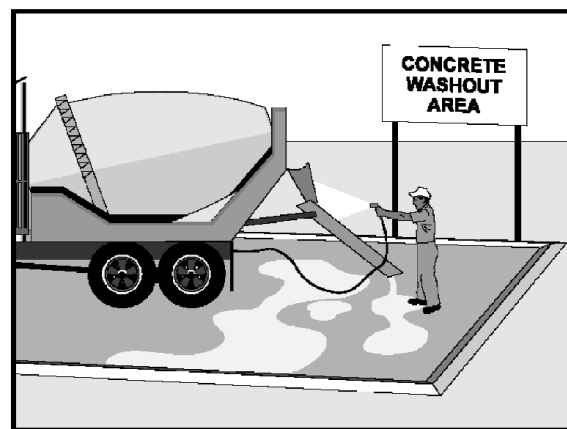
- Description and Purpose**
- A gravel bag berm is a series of gravel-filled bags placed on a level contour to intercept sheet flows. Gravel bags pond sheet flow runoff, allowing sediment to settle out, and release runoff slowly as sheet flow, preventing erosion.
- Suitable Applications**
- Gravel bag berms may be suitable:
- As a linear sediment control measure:
 - Below the toe of slopes and erodible slopes
 - As sediment traps at culvert/pipe outlets
 - Below other small cleared areas
 - Along the perimeter of a site
 - Down slope of exposed soil areas
 - Around temporary stockpiles and spoil areas
 - Parallel to a roadway to keep sediment off paved areas
 - Along streams and channels
 - As a linear erosion control measure:
 - Along the face and at grade breaks of exposed and erodible slopes to shorten slope length and spread runoff as sheet flow



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WM-8

Concrete Waste Management WM-8



Categories	
EC Erosion Control	☐
SE Sediment Control	☐
TC Tracking Control	☐
WE Wind Erosion Control	☐
NS Non-Stormwater Management Control	☒
WM Waste Management and Materials Pollution Control	☒
Legend:	
☒ Primary Category	
☒ Secondary Category	

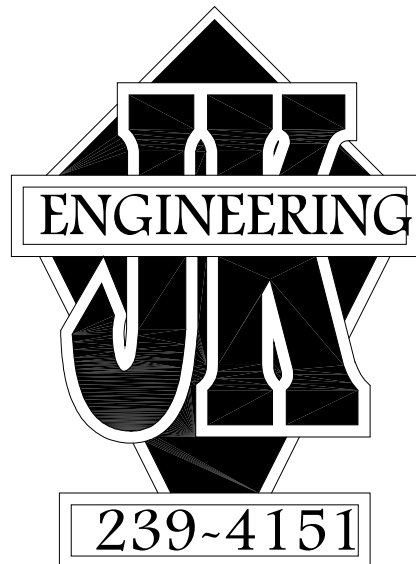
Targeted Constituents	
Sediment	☒
Nutrients	☐
Trash	☐
Metals	☒
Bacteria	☐
Oil and Grease	☐
Organics	☐

Potential Alternatives	
None	

- Description and Purpose**
- Prevent the discharge of pollutants to stormwater from concrete waste by conducting washout onsite or offsite in a designated area, and by employee and subcontractor training.
- The General Permit incorporates Numeric Effluent Limits (NEL) and Numeric Action Levels (NAL) for pH (see Section 2 of this handbook to determine your project's risk level and if you are subject to these requirements).
- Many types of construction materials, including mortar, concrete, stucco, cement and block and their associated wastes have basic chemical properties that can raise pH levels outside of the permitted range. Additional care should be taken when managing these materials to prevent them from coming into contact with stormwater flows and raising pH to levels outside the accepted range.
- Suitable Applications**
- Concrete waste management procedures and practices are implemented on construction projects where:
- Concrete is used as a construction material or where concrete dust and debris result from demolition activities.
 - Shingles containing portland cement concrete (PCC) are generated, such as from saw cutting, coring, grinding, grinding, and hydro-concrete demolition



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239-4151
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EROSION CONTROL
DEVICES
KIRK & CARRIE ALLEN
LOT 106, TRACT 1990-2
EDGEWOOD COURT
PASO ROBLES, CA 93446



REVISION LOG

REV.	DESCRIPTION	DATE

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SCALE: N.T.S.
PROJECT: CAPPS
DRAWN BY: FDW
CHECKED BY: JAK
DATE: 3/30/17

SHEET TITLE:
EROSION
CONTROL
DEVICES
SHEET NUMBER:

C.5



PLAN PREPARED FOR:
KIRK & CARRIE ALLEN
LOT 106 TRACT 1990-2
PASO ROBLES, CA 93446

5/6 TYPE "P" GYPSUM BOARD ON GARAGE SIDE OF COMMON WALL AND CEILING OF GARAGE AND HOUSE. DRYWALL GARAGE COMPLETE (CRC 408.1.4).

WHEN THE INTERIOR OF THE GARAGE IS BEING CONSTRUCTED, THE CEILING SHALL BE PROTECTED, THE WALLS AND /OR BEAMS SUPPORTING THE CEILING SHALL BE PROTECTED WITH AN EQUIVARIANT MATERIAL (CRC 408.1.4).

CONSTRUCTION (CRC 174)

INSTANT TANKLESS GAS WATER HEATER SHALL BE NATIONALLY LISTED AND BE INSTALLED TO COMPLY WITH THE ANTI-SCALDING REQUIREMENTS OF THE CODE AS PART OF THEIR LISTING. FUEL-BURNING WATER HEATERS MAY BE INSTALLED IN A CLOSET LOCATED IN THE GARAGE PROVIDED THE CLOSET MEETS THE REQUIREMENTS OF THE CODE FOR CLOSET DOOR ASSEMBLY AND A LISTED SELF-CLOSING DEVICE. THE CLOSET ASSEMBLY SHALL BE INSTALLED WITH A THRESHOLD AND BOTTOM SEAL.

2/6 GARAGE AND CLOSET DOORS SHALL BE INSTALLED IN FIXED PANELS ADJACENT TO A DOOR WHERE NEAREST EXPOSED EDGE OF THE GLAZING IS WITHIN A 24" ARC OF EITHER SIDE OF THE DOOR. GLAZING SHALL BE INSTALLED TO MEET THE REQUIREMENTS OF THE CODE. IF THE GLAZING IS LESS THAN 60" ABOVE WALKING SURFACE, CRC SECTION 260.4.1 EITHER WITHIN 38" OF FLOORS WITH TUB - SHOWER ENCLOSURES, WITHIN HOT - TUB WHIRLPOOL, SAUNA AND HOT TUB ENCLOSURES, OR WITHIN 38" OF THE BOTTOM OF THE GLAZING IS LESS THAN 60" ABOVE THE WALKING SURFACE, THE GLAZING SHALL BE INSTALLED TO MEET THE REQUIREMENTS OF THE CODE.

3/6 F.A.U. IN AT-FLAT. PROVIDE SWITCH, LIGHT, AND OUTLET NEAR ACCESS AND UNIT. PROVIDE 3" X 3" ACCESS TO MECHANICAL UNIT. A 22" X 3" ACCESS OPENING CAN BE USED TO PROVIDE ACCESS TO MECHANICAL UNITS. PROVIDE ACCESS TO MECHANICAL UNITS THROUGH AN OPENING OF THAT SIZE. ACCESS TO BE WITHIN 20" F.A.U. AND HAVE A VENT TO THE OUTSIDE WALKING SURFACE. ALSO PROVIDE 3" CLEAR UNOBSTRUCTED WORKING SPACE IN FRONT OF F.A.U.

4/6 KITCHEN COUNTER ON CONCRETE PAD. PROVIDE 5" MINIMUM CLEAR PASSAGE AROUND COUNTER.

5/6 KITCHEN COUNTER, SINK, STOVE, REFRIG. AND DISHWASHER SHALL BE INSTALLED IN ROOMS CONTAINING BATHTUBS, SHOWERS, SPAS AND SIMILAR BATHING FIXTURES SHALL BE INSTALLED TO MEET THE REQUIREMENTS OF THE CODE.

6/6 APPROVED MULTIPLE PURPOSE CARBON MONOXIDE / SMOKE DETECTOR INSTALLED AS REQUIRED AND AS INDICATED. CARBON MONOXIDE / SMOKE DETECTOR COMBINED ALARMS SHALL BE INSTALLED IN THE IMMEDIATE VICINITY OF THE DETECTED ALARMS SHALL BE LOCATED OUTSIDE OF EACH SEPARATE DWELLING UNIT SLEEPING AREA IN THE IMMEDIATE VICINITY OF EACH SEPARATE DWELLING UNIT. DETECTED ALARMS SHALL BE INSTALLED WHEN MORE THAN ONE CARBON MONOXIDE ALARM IS REQUIRED TO BE INSTALLED, THE ALARMS SHALL BE INTERCONNECTED IN A MANNER THAT ACTIVATION OF ONE UNIT WILL ACTIVATE ALL OTHERS. DETECTED ALARMS SHALL BE INSTALLED TO MEET THE REQUIREMENTS OF THE CODE.

7/6 CARBON MONOXIDE DETECTORS SHALL BE LISTED PER UL 2075. FOR ALTERATIONS, REPAIRS, OR REPLACEMENTS EXISTING CARBON MONOXIDE DETECTORS SHALL BE REPLACED WITH CARBON MONOXIDE DETECTORS OR GAS-FUEL BURNING APPLIANCES SHALL BE PROVIDED WITH A CARBON MONOXIDE ALARM.

8/6 MINIMUM 100,000 BPH INCHEN IN DOOR OF A LAUNDRY CLOSET OR PROVIDE OTHER APPROVED MEANS TO MAKE UP AIR PER CRC 504.3.1.

9/6 PROVIDE MEANS TO MAKE UP AIR PER CRC 504.3.1.

10/6 INDIVIDUAL VALVES OF THE PRESSURE BALANCE OR THE THERMOSTATIC MIXING VALVE TYPE ARE REQUIRED AT THE SHOWERS AND TUB-SHOWER COMBINATION PER CRC 420.

11/6 PROVIDE MEANS TO MAKE UP AIR PER CRC 504.3.1.

12/6 PROVIDE MEANS TO MAKE UP AIR PER CRC 504.3.1.

13/6 SHOWERS AND WALLS ABOVE BATHTUBS WITH SHOWER HEADS SHALL BE FINISHED WITH A SMOOTH, NONABRASIVE SURFACE TO A HEIGHT NOT LESS THAN 70" ABOVE THE DRAIN INLET. CRC 408.1.4.

14/6 DETECTED ALARMS HARDWIRED AND INTERCONNECTED TO ONE ANOTHER. PROVIDE BATTERY BACKUP FOR DETECTED ALARMS. DETECTED ALARMS SHALL BE INSTALLED TO MEET THE REQUIREMENTS OF THE CODE.

15/6 DETECTED ALARMS SHALL BE CLEARLY AUDIBLE. CRC 807.2.10.3.

THE FOLLOWING FIXTURES SHALL BE OF WATER CONSERVATION:

THE FOLLOWING FIXTURES SHALL BE OF WATER CONSERVATION:

RESIDENTIAL:

WATER CLOSETS:	1.28 GALLON PER FLUSH MAXIMUM
SHOWER HEAD FLOW:	2.0 GALLON PER MINUTE AT 80 PSI
LAVATORY / SINK FIXTURE:	1.5 GALLON PER MINUTE AT 60 PSI
KITCHEN FAUCETS:	2.2 GALLON PER MINUTE AT 60 PSI

NON COMPLIANT FIXTURES MEANS ANY OF THE FOLLOWING:

RESIDENTIAL:
ANY TOILET MANUFACTURED TO USE MORE THAN 1.6 GAL OF WATER PER FLUSH.
ANY SHOWERHEAD MANUFACTURED WITH 2.5 GALLONS PER MINUTE.
ANY INTERIOR FAUCET THAT EMITS MORE THAN 2.2 GALLONS PER MINUTE.

SYM.	QTY.	SIZE	NOTES
1	1	4000	SH WINDOW AT 7'-6"
2	8	3500	SH WINDOW AT 7'-6"
3	2	5010	SH WINDOW AT 3'-6"
4	2	5040	FX WINDOW AT 7'-6"
5	1	50" x 50"	GLASS BLK WINDOW AT 7'-6"
6	2	4050	SH WINDOW AT 7'-6"
7	1	4010	GLASS BLOCK WINDOW AT 7'-6"
8	2	3550	FX WINDOW AT 7'-6"
9	1	5050	FX WINDOW AT 7'-6"
10	2	3640	SH WINDOW AT 7'-6"

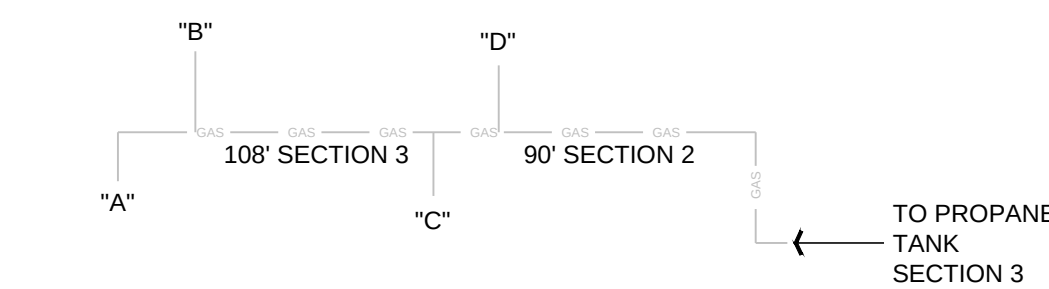
EXTERIOR WINDOWS, WINDOW WALLS, GLAZED DOORS AND GLAZED OPENING WITHIN EXTERIOR DOORS SHALL BE INSULATING-GLASS UNITS WITH A MINIMUM OF ONE TEMPERED PANE OR GLASS BLOCK UNITS OR HAVE A FIRE-RESISTANCE RATING OF NOT LESS THAN 20 MINUTES

SYM.	QTY.	SIZE	NOTES	
A	1	2480	INT	
B	3	2680	INT	
C	4	3080	INT	
D	1	3080	FIRE RATED SELF CLOSING	
E	2	3080	EXT	
F	1	4080	BIFOLD	
G	1	6080	BIFOLD	
H	1	60710	SL GL	
I	1	80710	SL GL	
J	1	18'-0" x 9'-0"	GARAGE	

GAS LINE SIZES FOR THIS PROJECT PER CPC 2010 CHAPTER 12 TABLE 1216.2(19) ARE AS FOLLOWS:

GAS LINE SIZES FOR THIS PROJECT PER CPC 2010 CHAPTER 12 TABLE 1216.2(19) ARE AS FOLLOWS:

- FOLLOWS:
- | | |
|---|-------------|
| • OUTLET A- OVEN (65,000 BTU)= | 108" X 3/2" |
| • OUTLET B- FAU (40,000 BTU)= | 90" X 3/2" |
| • SECTION 1 = 108" X 3/2" FROM OUTLET B TO P.T. (105,000 BTU) | 100" X 3/2" |
| • OUTLET C- DRYER (35,000 BTU)= | 90" X 5/2" |
| • OUTLET D - W/H (200,000 BTU)= | 85" X 1" |
| • SECTION 2 = 90" X 1" FROM OUTLET E TO P.T. (235,000 BTU) | |
| • TOTAL BTU =240,000 BTU | |
| • SECTION 3 = 60" X 1" FROM P.T. (240,000 BTU) | |



REVISION LOG

These drawings are the exclusive property of J.B. drafting and design and shall be used solely for the purpose of this project on this site. Any use other than the project upon which it is intended for without the written consent of J.B. drafting and design and John M Butler II is prohibited.

PROJECT NO. —

FILE NAME A/2.1 FLOOR PLAN.DWG

DRAWN BY JJK

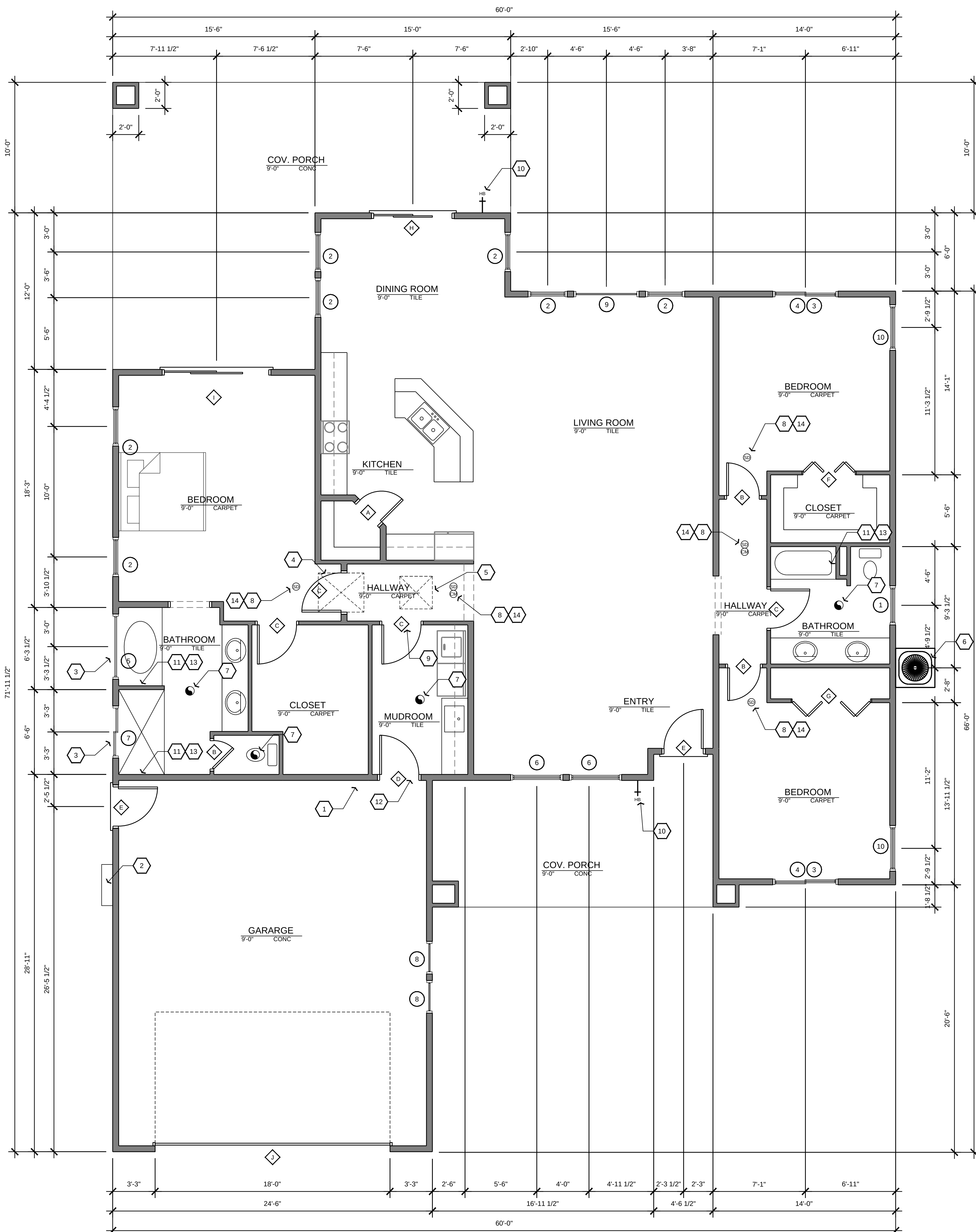
DATE 3/31/2017:

SHEET TITLE:

FLOOR PLAN

SHEET NUMBER

A-2.1



$$1/4'' = 1'$$



KIRK & CARRIE ALLEN
LOT 106 TRACT 1990-2
PASO ROBLES, CA 93446

1. COMPOSITION SINGLE ROOFING OF MIN OF CLASS A OVER 30 LB MINIMUM ROOFING FELT
2. 2" X 8" HEM FIRM FASCIA (TYP)
3. 7/8" CEMENT PLASTER @ 3/4" RIBBED LATH AND OVER BUILDING PAPER
4. 3/4" WYCOND SHINGLING WITH 60-62 OF 12" ON HOR. SURFACE OF EXTERIOR (TYP)
5. STONE SLIDING OVER APPROVED MOISTURE BARRIER
6. NONCOMBUSTIBLE OR IGNITION RESISTANT MATERIAL ON EXPOSED UNDERSIDE.
7. EXPOSED VENTING TO BE PROTECTED BY 1/2" MINIMUM THICKNESS OF 1/4" MINIMUM
8. 2X TRIM AROUND ALL DOORS AND WINDOWS AND AT ALL CORNERS (TYP)
9. ATTIC VENTILATION CALCULATIONS: TOTAL ATTIC AREA = 3760 SQ FT
REQUIRED ATTIC VENTILATION = 3760 SQ FT / 150 = 25.07 SQ FT
USE 10" L4" X 24" LUMINOX DORMER (NVA = 90) = 900 SQ. IN.
USE 22" X 33" UNDER EAVE VENTS (NVA = 41) = 943 SQ. IN.
REQUIRED VENTS APPROVED EQUAL TO REQUIRED AREA OF FLAME & EMBER INTO ATTIC AREA.
10. 24 GA. WEEP SCREED FLASHING AT BASE OF CEMENT PLASTER AND INSTALLED PER MANUFACTURER'S INSTRUCTIONS. WEEP SCREED RESISTANT WITH A MINIMUM VERTICAL ATTACHMENT FLANGE OF 3-1/2" AND SHALL BE PROVIDED AT OR BELOW THE FOUNDATION PLATE LINE. THE SCREED SHALL BE PLACED A MINIMUM OF 4" ABOVE THE EARTH SURFACE.
11. SLOPE AWAY FROM BUILDING 5% FOR 10'-0" MINIMUM (TYP)

BUILDING INSULATION:		
EXTERIOR WALL:		R-21 MINIMUM(TYP)
CEILING:		R-38 MINIMUM(TYP)
INTERIOR FINISH MATERIAL:		
WALLS:		1/2" GYPSUM BOARD
CEILING:		5/8" GYPSUM BOARD (GARAGE WALLS & CEILING USE 5/8" TYPE "X" BOARD AND FOR USABLE AREA UNDER STAIRS)
WALL FRAMING:		
EXTERIOR WALLS:		2X6 STUD WALLS @ 16" O/C
INTERIOR WALLS:		2X4 STUD WALLS @ 16" O/C
HEADERS UNLESS OTHERWISE NOTED:		
EXTERIOR BEARING:		6X12 D.F. #1 (U.O.N.)
EXTERIOR NON-BEARING:		6X12 D.F. #1 (U.O.N.)
INTERIOR BEARING:		4X12 D.F. #2 (U.O.N.)
INTERIOR NON-BEARING:		4X8 D.F. #2 (U.O.N.)



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PROJECT NO. ---

FILE NAME: A.D.A.F.D.(ACTION)ONG

DRAWN BY JJK

SHEET TITLE:

ELEVATIONS

SHEET NUMBER:

A-3.1



E-1.1

CERTIFICATE OF COMPLIANCE - RESIDENTIAL PERFORMANCE COMPARISON METHOD										CFR-609-61	
Project Name: Capex Conversion Corp										Page 6 of 6	
Calculation Date/Time: 1/30, Tue, Feb 07, 2017											
Input File Name: Capex-16101-0101.rpt											
PERFORATION / SLABING											
E1	E2	E3	E4	E5	E6	E7	E8	E9	E10	E11	E12
Name	Type	Surface Orientation	Width (ft)	Height (ft)	Material	Area (sq ft)	Factor	SHC	U-factor	SHC	U-factor
Vincent	Vinwood	North Wall (Sloped)	—	—	—	11.92	0.25	0.25	—	—	—
Vincent	Vinwood	North Wall (Sloped)	—	—	—	14.7	0.25	0.25	—	—	—
Vincent	Vinwood	North Wall (Sloped)	—	—	—	11.92	0.25	0.25	—	—	—
Vincent	Vinwood	North Wall (Sloped)	—	—	—	14.7	0.25	0.25	—	—	—
Vincent	Vinwood	South Wall (Sloped)	—	—	—	11.92	0.25	0.25	—	—	—
Vincent	Vinwood	South Wall (Sloped)	—	—	—	14.7	0.25	0.25	—	—	—
Vincent	Vinwood	South Wall (Sloped)	—	—	—	11.92	0.25	0.25	—	—	—
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Vincent	Vinwood	South Wall (Sloped)	—	—	—	14.7	0.25	0.25	—	—	—
Vincent	Vinwood	South Wall (Sloped)	—	—	—	11.92	0.25	0.25	—	—	—
Vincent	Vinwood	South Wall (Sloped)	—	—	—	14.7	0.25	0.25	—	—	—
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Vincent	Vinwood	South Wall (Sloped)	—	—	—	14.7	0.25	0.25	—	—	—
Vincent											

CEMPATE PLANT COMPLIANCE - RESIDENTIAL PERFORMANCE COMPLIANCE STATE										CFR-PRF-01	
Project Name: Cempate Constaration Corp					Calculation Date/Time: 10:06 AM, Tue, Sep 07, 2017					Page 3 of 6	
Calculation Description: Title 24 Analysis					Input File Name: Cempate 10/10/16.rvt						
ZONE INFORMATION											
Zone Name	Zone Type	HVAC System Type	Zone Floor Area (m ²)	Zone Floor Area (ft ²)	Avg. Ceiling Height (m)	Water Heating System 1		Water Heating System 2			
Living area	Condo Type	Split System	2478.0	0	2.70	DHW Sys 1					
OPaque SURFACES											
SI	SI	SI	SI	SI	SI	SI	SI	SI	SI	SI	
North Wall	Zone	Construction	SI	SI	SI	SI	SI	SI	SI	SI	
North Wall	Living Area	9.25 Wall	100	1076	0	0	0	0	0	0	
South Wall	Living Area	9.25 Wall	180	1944	0	0	0	0	0	0	
East Wall	Living Area	9.25 Wall	90	972	0	0	0	0	0	0	
West Wall	Living Area	9.25 Wall	270	2916	0	0	0	0	0	0	
Interior Surface	Living Area - Garage	9.25 Wall	0	0	0	0	0	0	0	0	
North Wall - Garage	Garage	9.25 Wall	0	0	0	0	0	0	0	0	
South Wall - Garage	Garage	9.25 Wall	0	0	0	0	0	0	0	0	
Interior Surface 2	Garage	9.25 Wall	270	2916	0	0	0	0	0	0	
Interior Surface 3	Garage - Unlabeled area	9.25 Wall	0	0	0	0	0	0	0	0	
Roof 1	Garage	Roof 16.00 Flat w/6.00 Slope	0	0	0	0	0	0	0	0	
ATTIC											
SI	SI	SI	SI	SI	SI	SI	SI	SI	SI	SI	
Attic - North	Construction	Type	Roof Rise	Roof Reflectance	Roof Emittance	Radiant Barrier		Cool Roof			
Attic - North	Attic Garage Ceiling	Unlabeled	0	0.7	0.05						
Attic - South	Attic Garage Ceiling	Unlabeled	0	0.7	0.05						

[illegible]

CERTIFICATE OF COMPLIANCE - RESIDENTIAL PERFORMANCE COMPLIANCE METHOD				CPR-RPF-01	
Project Name: Capasa Construction Corp.				Page 1 of 8	
Calculation Details: The 24 Analysis				Calculation Date: May 10/24, 2017	
Input File Name: Capasa 10/24/2016.rvt				Input File Name: Capasa 10/24/2016.rvt	
GENERAL INFORMATION					
01	Project Name	Capasa Construction Corp.			
02	Calculation Description	The 24 Analysis			
03	Project Location	10/24			
04	City	Windsor, Ontario			
05	Address	10/24			
06	City	Windsor, Ontario			
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08	City	Windsor, Ontario			
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100	City	Windsor, Ontario			
COMPLIANCE RESULTS					
01	Building Compliance with Computer Performance				
02	This building incorporates measures that require full third-party verification by a certified HERS rater under the supervision of a CEC-registered HERS provider.				
03	This building incorporates one or more special provisions known below:				
ENERGY USE SUMMARY					
	Energy Use (kWh/m ² /yr)	Equivalent Demand	Proposed Design	HT	SB
01	Base Heating	15.22	20.29	3.28	11.4%
02	Base Cooling	24.32	1.01	-23.31	-95.8%
03	Mech Ventilation	1.39	1.39	0.00	0.0%
04	Base Heating	14.83	20.29	5.46	36.8%
05	Photovoltaic Offset	0.00	0.00	0.00	0.0%
06	Compliance Energy Total	44.83	63.01	23.82	8.5%
REGISTERED INFORMATION					
CPR Rating Energy Efficiency Standard - 2014 Residential Compliance			HERS Provider		
Registration Date: 03/17/2017			Report Generated: 2017-05-07 16:58		
Report Version: 03/17/2022-07-04-05					

CERTIFICATE OF COMPLIANCE - RESIDENTIAL PERFORMANCE COMPLIANCE FORM		CFR-PBP-3
Project Name: Capco Construction Corp.		Calculator Software Title: LSA, Tm, Top Pro 2017
Calculation Description: Title 24 Analysis		Input File Name: Capco 10101-101616
DOCUMENTATION AUTHOR'S DECLARATION STATEMENT		
I certify that this Certificate of Compliance documentation is accurate and complete.		
Documentation Author Name: Jim Reed Cad Service		Documentation Author Signature:
Company: Jim Reed	Date: 2/7/2017	
Address: 510 10th St, Suite C Paso Robles, CA 93446	CEA/RES Declaration Verification (if applicable): Phone: 805 239 9158	
RESPONSIBLE PERSON'S DECLARATION STATEMENT		
I certify the following under penalty of perjury, under the laws of the State of California:		
1. That I am the owner or authorized agent of the Responder and Participants listed in stated requirements. 2. That I have the energy source and performance specifications identified on the Certificate of Compliance conform to the requirements of Title 24, Part 6 and Part 7 of the California Code of Regulations. 3. The building design features or system design features identified on the Certificate of Compliance are analyzed with the information provided in other applicable compliance documents, whenever calculations, plans and specifications submitted to the enforcement agency to approve such existing project applications.		
Responder Name: CDS Building	Responsible Designer Signature:	
Company: CDS Drafting & Design	Date Signed:	
Address: 510 10th St Paso Robles, CA 93446	Phone: 805 237 0850	

CERTIFICATE OF COMPLIANCE - RESIDENTIAL PERFORMANCE COMPLIANCE METHOD										CFR-PRF-01	
Project Name: Cappa Construction Div. 24 Analysis						Calculation Date/Time: 10:04 Tue Feb 07, 2017				Page 7 of 7	
Input File Name: Cappa 10150.rvtbfile											
HVAC - DISTRIBUTION SYSTEMS											
01	02	03	04	05	06	07	08	09	10	11	12
Name	Type	Duct Leakage	Insulation R-value	Duct Location	System Size	HERS Verification					
Air Handler 10150-1	Ductwork	Insulated and Sealed	5	Att.	None	None					
HVAC DISTRIBUTION - HERS DISTRIBUTION											
01	02	03	04	05	06	07	08	09	10	11	12
Name	Duct Leakage	Duct Location	Verified Duct Type	Verified Duct R-value	Battled Ducts	Dealt Battled Ducts	Air Handler	Low Leakage			
Air Distribution System 10150-1	Required	E.O.	Not Required	Not Required	Not Required	Not Required	Not Required	---			
HVAC - FAN SYSTEMS											
01	02	03	04	05	06	07	08	09	10	11	12
Name	Speed	Typical	Fan Power (Watt/CAPG)	HERS Verification							
HVAC Fan 10150-1	Single Speed	Typical	0.58	HVAC Fan 10150-1							
HVAC FAN SYSTEMS - HERS VERIFICATION											
01	02	03	04	05	06	07	08	09	10	11	12
Name	Speed	Typical	Fan Power (Watt/CAPG)	HERS Verification							
HVAC Fan 10150-1	Single Speed	Typical	0.58	HVAC Fan 10150-1							
HVAC FAN SYSTEMS - HERS VERIFICATION											
01	02	03	04	05	06	07	08	09	10	11	12
Name	Speed	Typical	Fan Power (Watt/CAPG)	HERS Verification							
HVAC Fan 10150-1	Single Speed	Typical	0.58	HVAC Fan 10150-1							
IAQ (Indoor Air Quality) FANS											
01	02	03	04	05	06	07	08	09	10	11	12
Ducting Unit	IAQ CMR	IAQ Recovery (Efficiency/Sec)	IAQ Fan	HERS Verification							
None	None	None	None	None							

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RESIDENTIAL MEASURES SUMMARY										RMS-1
Project Name		Building Type		B3 Single Family		B3 Addition Area		03/05/21		
Contractor/Construction Corp		B3 Multi-Family		B3 Construction Addition/Alteration				03/05/21		
Project Address		Calumet Energy Service Zone Dr		Total Gross Floor Area		Addition		# of Units		
L70156 Heritage Ranch/ Lakeview Village		CA Climatic Zone Dri		2,478		600				
INSULATION										
Construction Type		Cavity		Area		Special Features		Status		
Wall	Horizontal Insulation	R-13	12.00					None		
Door	Double Door	R-10	10.00					None		
Roof	Unvented Dbl-on-Deck	R-49	10.00	24" EPS, Plastic w/ F				None		
Attic	Unvented Dbl-on-Deck	R-38	12.00	24" EPS, Plastic w/ F				None		
Basement	Horizontal Insulation	R-13	12.00					None		
Demolish	Wood Framed	R-13	12.00					None		
FENESTRATION										
Orientation		Total Area		Area		Shading Percentage		15.4%		
Orientation Area(ft²)		U-Fac		SHG		Overhang		Slofinas		
North (N)		81.4	0.250	0.25	none			Status		
East (E)		88.0	0.250	0.25	none			None		
South (S)		66.0	0.250	0.25	none			None		
West (W)		67.0	0.250	0.25	none			None		
HVAC SYSTEMS										
Qty.		Heating		Cooling		Min. Eff.		Thermostat		
Boiler/ Furnace		80% EFF.		100% EFF.		100% EFF.		Status		
HVAC DISTRIBUTION										
Qty.		Heating		Cooling		Duct Location		Duct R-Value		
Split System		Ducted		Ducted		Attic		R-8		
WATER HEATING										
Qty.		Type		Gallons		Min. Eff.		Distribution		
Water Heater		Electric		100.0		100.0		Status		
Worksheet: 7.12.2020 User: Noreen - 4/31/21										
Page 11 of 12										

These drawings are the exclusive property of DUCT DYNASTY and shall be used solely for the purpose of this project on this site. Any use other than the project upon which it is intended for without the written consent of DUCT DYNASTY AND Jim Reed is prohibited.

PROJECT NO. ---
FILE NAME T-24 ENERGY COMPLIANCE.DWG
DRAWN BY JJK

SHEET TITLE:

ENERGY

CONCLUSION

SHEET NUMBER:

[illegible]

	Duct System Summary Entire House JB REED CAD Service			Joh: 04/04/2017 Jim: 04/05/2017 Jim: 04/07/2017							
				Print							
B35 1/2" Black, Pvc, 1/2" Steel Pvc, 1/2" Steel Pvc, 1/2" Steel Pvc, 1/2" Steel Pvc, 1/2" Steel Pvc											
Project Information											
For:		Cassa, Cassa Construction 12710 Lakeside Wy, Heritage Ranchy Pva Robins, Ca 92460									
External static pressure											
		Heating		Cooling							
		0.80 in H ₂ O		0.60 in H ₂ O							
		2 in H ₂ O		2 in H ₂ O							
Available static pressure											
		0.453 / 0.414 in H ₂ O		0.453 / 0.414 in H ₂ O							
Supply return available pressure											
		0.453 / 0.414 in H ₂ O		0.453 / 0.414 in H ₂ O							
Losses from trunk											
		0.87 in H ₂ O		1.113 in H ₂ O							
		276 ft									
Supply Branch Detail Table											
Name	Design (Btu/h)	Hg (in)	Qig (cfm)	Design (Btu/h)	Qm (cfm)	H _v W/ (in)	Max (in)	Actual L in W/ (in)	Fig W/ L in W/ (in)	Trunk	
air-flow	C 1345	63	152	62	0.250	0.60	0.60	0.60	27.7	1739	a2a
Pressure losses	C 1326	87	33	0.250	0.250	0.60	0.60	0.60	11.5	11.5	a1
air-flow	C 3469	85	159	129	0.70	0.60	0.60	0.60	27.7	190	a2a
Pressure losses	C 3446	117	145	0.70	0.70	0.60	0.60	0.60	14.0	14.0	a1a
air-flow	C 3051	102	166	127	0.80	0.60	0.60	0.60	33.8	1500	a1a
Pressure losses	C 3028	117	145	0.80	0.80	0.60	0.60	0.60	14.0	14.0	a1a
COURT	E 238	6	6	0.238	0.40	0.60	0.60	0.60	13.3	10.0	a2
Pressure losses	E 238	6	6	0.238	0.40	0.60	0.60	0.60	13.3	10.0	a2
air-flow	C 3245	148	148	148	0.70	0.60	0.60	0.60	33.8	1500	a1a
Pressure losses	C 3245	148	148	0.70	0.70	0.60	0.60	0.60	14.0	14.0	a1a
air-flow	C 3245	148	148	0.70	0.70	0.60	0.60	0.60	24.5	1150	a1a
Pressure losses	C 3245	148	148	0.70	0.70	0.60	0.60	0.60	14.0	14.0	a1a
air-flow	E 423	11	9	0.907	0.40	0.60	0.60	0.60	4.3	8.5	a2
Pressure losses	E 423	11	9	0.907	0.40	0.60	0.60	0.60	4.3	8.5	a2
Supply Trunk Detail Table											
Name	Trunk Type	Hg (in)	Qig (cfm)	Design Veloc (ft/min)	Design Veloc (ft/min)	H _v W/ (in)	Qm (cfm)	Actual L in W/ (in)	Fig W/ L in W/ (in)	Trunk	
a1a	Pvc A/V	219	312	0.227	0.27	10.0	0	0	0	a1	
a2a	Pvc A/V	247	169	0.227	0.27	10.0	0	0	0	a1	
a2b	Pvc A/V	267	593	0.219	0.26	19.0	0	0	0	a1	
a2c	Pvc A/V	155	221	0.219	0.26	500	0	0	0	a2	
wrightsoft High-Speed 3D/4D 2017 3.1.1 15/04/2018						2016-04-07 10:00:00					

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Enure House	d	2331	26324	29100	637	1113
Other equip funds			0	0		
Funds @	1.00 RSM			29100		
Latent coding				1394		
TOTALS		2331	26324	30493	637	1113

w/ripsoft:		Right-16 Worksheet Entire House Jim Reed C&D Service				Job: Date: Feb. 05, 2017 Time: 2:50	
615 NW 18th, West Palatka, Ca 92061 Phone 805 233 1072 Email: jimreed@jimreed.com							
4. Rooms Name Occupancy Room Number		5. Max. Load 30.0 A @ 120 V 15.0 A @ 240 V		6. Max. Load 12.5 A @ 120 V 6.25 A @ 240 V		7. LAMPING 30.0 A @ 120 V 15.0 A @ 240 V	
8. Description		9. Demand Factor	10. Demand Factor	11. Demand Factor	12. Demand Factor	13. Demand Factor	14. Demand Factor
15. Description		16. Demand Factor	17. Demand Factor	18. Demand Factor	19. Demand Factor	20. Demand Factor	21. Demand Factor
22. Description		23. Demand Factor	24. Demand Factor	25. Demand Factor	26. Demand Factor	27. Demand Factor	28. Demand Factor
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225. Description		226. Demand Factor	227. Demand Factor	228. Demand Factor	229. Demand Factor	230. Demand Factor	231. Demand Factor
232. Description		233. Demand Factor	234. Demand Factor	235. Demand Factor	236. Demand Factor	237. Demand Factor	238. Demand Factor

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 Residential Plans Examiner Review Form for HVAC System Design (Ducts, Equipment, Ducts)										Form #RES-1 19-Mar-20	
Contractor: Jim Reed CAD Service			San Luis Obispo County Housing								
Medical Review:			REQUIRED ATTACHMENTS 1. ☐ Load Calculations (Manual J) 2. ☐ Equipment Sizing (Manual S) 3. ☐ Ductwork Sizing (Manual D) 4. ☐ Equipment Selection (Manual E) 5. ☐ Equipment Installation (Manual I) 6. ☐ Equipment Maintenance (Manual M) 7. ☐ Equipment Replacement (Manual R)								
Building class:			Form #RES-1 19-Mar-20								
How Address (Circuit or Load, Substation):			How Address (Circuit or Load, Substation):								
Lot: L50			Lot: L50								
HVAC EQUIPMENT SELECTION (RAC M1001-5)											
Design Conditions											
Winter Design Conditions		Summer Design Conditions		Building Construction Information		Roofing		Attached		Attached	
Winter temperature: 20 °F Winter humidity: 65 % Winter wind speed: 10 mph		Summer temperature: 90 °F Summer humidity: 72 % Summer wind speed: 10 mph		Building construction: 2013 Number of stories: 1 Number of units: 1 Number of bedrooms: 3 Number of bathrooms: 2		Roofing: 2013 Number of stories: 1 Number of units: 1 Number of bedrooms: 3 Number of bathrooms: 2		Attached: 2013 Number of stories: 1 Number of units: 1 Number of bedrooms: 3 Number of bathrooms: 2		Attached: 2013 Number of stories: 1 Number of units: 1 Number of bedrooms: 3 Number of bathrooms: 2	
Severe Design Conditions		Winter		Summer		Roofing		Attached		Attached	
Severe winter temperature: 27 °F Severe winter humidity: 65 % Severe winter wind speed: 10 mph		Winter temperature: 20 °F Winter humidity: 65 % Winter wind speed: 10 mph		Summer temperature: 90 °F Summer humidity: 72 % Summer wind speed: 10 mph		Building construction: 2013 Number of stories: 1 Number of units: 1 Number of bedrooms: 3 Number of bathrooms: 2		Roofing: 2013 Number of stories: 1 Number of units: 1 Number of bedrooms: 3 Number of bathrooms: 2		Attached: 2013 Number of stories: 1 Number of units: 1 Number of bedrooms: 3 Number of bathrooms: 2	
How Address (Circuit or Load, Substation):		How Address (Circuit or Load, Substation):		How Address (Circuit or Load, Substation):		How Address (Circuit or Load, Substation):		How Address (Circuit or Load, Substation):		How Address (Circuit or Load, Substation):	
Lot: L50		Lot: L50		Lot: L50		Lot: L50		Lot: L50		Lot: L50	
HVAC EQUIPMENT SELECTION (RAC M1001-5)											
Modeling Equipment Data											
Modeling Equipment Data		Equipment Data		Equipment Data		Equipment Data		Equipment Data		Equipment Data	
Model: 2013 Manufacturer: 2013 Model: 2013 Manufacturer: 2013		Model: 2013 Manufacturer: 2013 Model: 2013 Manufacturer: 2013		Model: 2013 Manufacturer: 2013 Model: 2013 Manufacturer: 2013		Model: 2013 Manufacturer: 2013 Model: 2013 Manufacturer: 2013		Model: 2013 Manufacturer: 2013 Model: 2013 Manufacturer: 2013		Model: 2013 Manufacturer: 2013 Model: 2013 Manufacturer: 2013	
How Address (Circuit or Load, Substation):		How Address (Circuit or Load, Substation):		How Address (Circuit or Load, Substation):		How Address (Circuit or Load, Substation):		How Address (Circuit or Load, Substation):		How Address (Circuit or Load, Substation):	
Lot: L50		Lot: L50		Lot: L50		Lot: L50		Lot: L50		Lot: L50	
HVAC EQUIPMENT SELECTION (RAC M1001-5)											
Design Conditions											
Winter Design Conditions		Summer Design Conditions		Building Construction Information		Roofing		Attached		Attached	
Winter temperature: 20 °F Winter humidity: 65 % Winter wind speed: 10 mph		Summer temperature: 90 °F Summer humidity: 72 % Summer wind speed: 10 mph		Building construction: 2013 Number of stories: 1 Number of units: 1 Number of bedrooms: 3 Number of bathrooms: 2		Roofing: 2013 Number of stories: 1 Number of units: 1 Number of bedrooms: 3 Number of bathrooms: 2		Attached: 2013 Number of stories: 1 Number of units: 1 Number of bedrooms: 3 Number of bathrooms: 2		Attached: 2013 Number of stories: 1 Number of units: 1 Number of bedrooms: 3 Number of bathrooms: 2	
Severe Design Conditions		Winter		Summer		Roofing		Attached		Attached	
Severe winter temperature: 27 °F Severe winter humidity: 65 % Severe winter wind speed: 10 mph		Winter temperature: 20 °F Winter humidity: 65 % Winter wind speed: 10 mph		Summer temperature: 90 °F Summer humidity: 72 % Summer wind speed: 10 mph		Building construction: 2013 Number of stories: 1 Number of units: 1 Number of bedrooms: 3 Number of bathrooms: 2		Roofing: 2013 Number of stories: 1 Number of units: 1 Number of bedrooms: 3 Number of bathrooms: 2		Attached: 2013 Number of stories: 1 Number of units: 1 Number of bedrooms: 3 Number of bathrooms: 2	
How Address (Circuit or Load, Substation):		How Address (Circuit or Load, Substation):		How Address (Circuit or Load, Substation):		How Address (Circuit or Load, Substation):		How Address (Circuit or Load, Substation):		How Address (Circuit or Load, Substation):	
Lot: L50		Lot: L50		Lot: L50		Lot: L50		Lot: L50		Lot: L50	
HVAC EQUIPMENT SELECTION (RAC M1001-5)											
Modeling Equipment Data											
Modeling Equipment Data		Equipment Data		Equipment Data		Equipment Data					

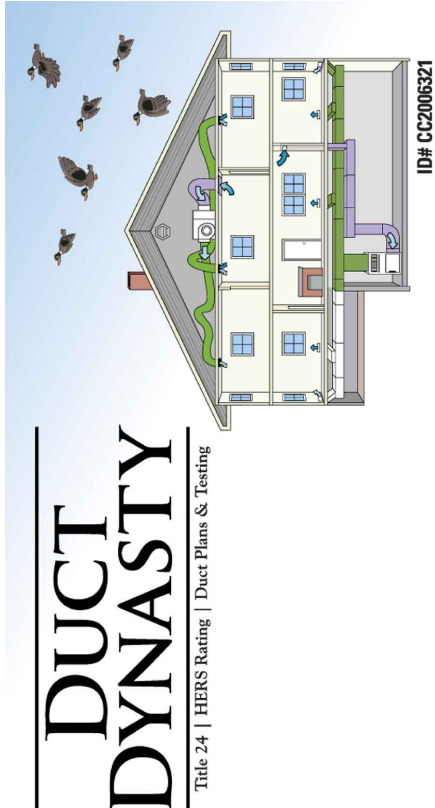
[illegible]

	Manual S Compliance Report Entire House JRM Reed CAD Service			Job: _____ By: Jim Reed
	005 10th Street, Palm Point, CA 92058 Phone: 650 238-8511 Email: reed@wrightcad.com			
Project Information				
For: Cape, Cape Construction Lot 100 Lakeside Village, Heritage Ranches Phase Robles, CA 93046				
Coaling Equipment				
Outdoor design DB: 58.8°F Outdoor design WB: 68.8°F Indoor design DB: 75.0°F Indoor design RH: 77%	Sensitive gas: 29100 Btu/h Sensitive mass: 10480 Btu/h Total gas: 32480 Btu/h Estimated airflow: 1113 cfm	Entering coil DB: 78.8°F Entering coil WDB: 52.8°F		
Manufacturer's Performance Data Actual Conditions				
Equipment type: Split AC Manufacturer: Carrier Capacity: 11.0 cfm Sensible capacity: 0 Btu/h Latent capacity: 0 Btu/h Total capacity: 0 Btu/h	Model: 38BC0R038-314LPH28120PFA8-B7D 0 Btu/h 0 Btu/h 0 Btu/h 0 Btu/h			
Heating Equipment				
Outdoor design DB: 58.8°F Indoor design DB: 65.0°F	Heat loss: 29524 Btu/h	Entering coil DB: 55.8°F		
Manufacturer's Performance Data Actual Conditions				
Equipment type: Manufacture Actual output: 1837 Btu/h Capacity: 3450 Btu/h	Model: 38ST72J45-08 116% of load 0 Btu/h	Temp. rise: 50 °F		

Meets all requirements of ACCA Manual S.

*w/whrptst:		Right-39 Worksheet Enire House Jim Reed CAD Service				Job: Date: Feb. 04. 2017		Job Ref: JMS 008			
101 010 Street, Pease Heights, Ca 92061 Phone: 650 338 8873 Email: jim@jimreed.com											
		BATHROOM				CLIMBER					
Notes		12.0		12.0		6.0		6.0			
Dimensions		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'			
Room Name		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'			
Room Number		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'			
Room Area		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'			
Room Volume		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'			
Room Height		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'			
Room Width		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'			
Room Depth		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'			
Room Length		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'			
Room Area		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'			
Room Volume		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'			
Room Height		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'			
Room Width		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'			
Room Depth		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'			
Room Length		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'			
Room Area		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'			
Room Volume		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'			
Room Height		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'			
Room Width		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'			
Room Depth		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'			
Room Length		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'		12.0' x 12.0'			

Name	Grille Size (in)	Htg (cfm)	Cp (cfm)	TEL (ft)	Design FR	Veloc (fpm)	Diam (in)	H x W (in)	Stud/Uolet Opening (in)	Duct Matl	Trunk
rb1	0x0	346	654	46.2	0.318	490	16.0	0x 0		VPx	
rb2	0x0	418	682	67.2	0.219	487	16.0	0x 0		VPx	



PLAN PREPARED FOR:
KIRK & CARRIE ALLEN
LOT 106 TRACT 1990-2
PASO ROBLES CA 93446

[illegible]

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PROJECT NO. ---
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DRAWN BY JJK
DATE 3/31/2017 8:00 AM
SHEET TITLE:
**MECHANICAL
PLAN**

SHEET NUMBER:

MP-1

RESIDENTIAL MANDATORY MEASURES

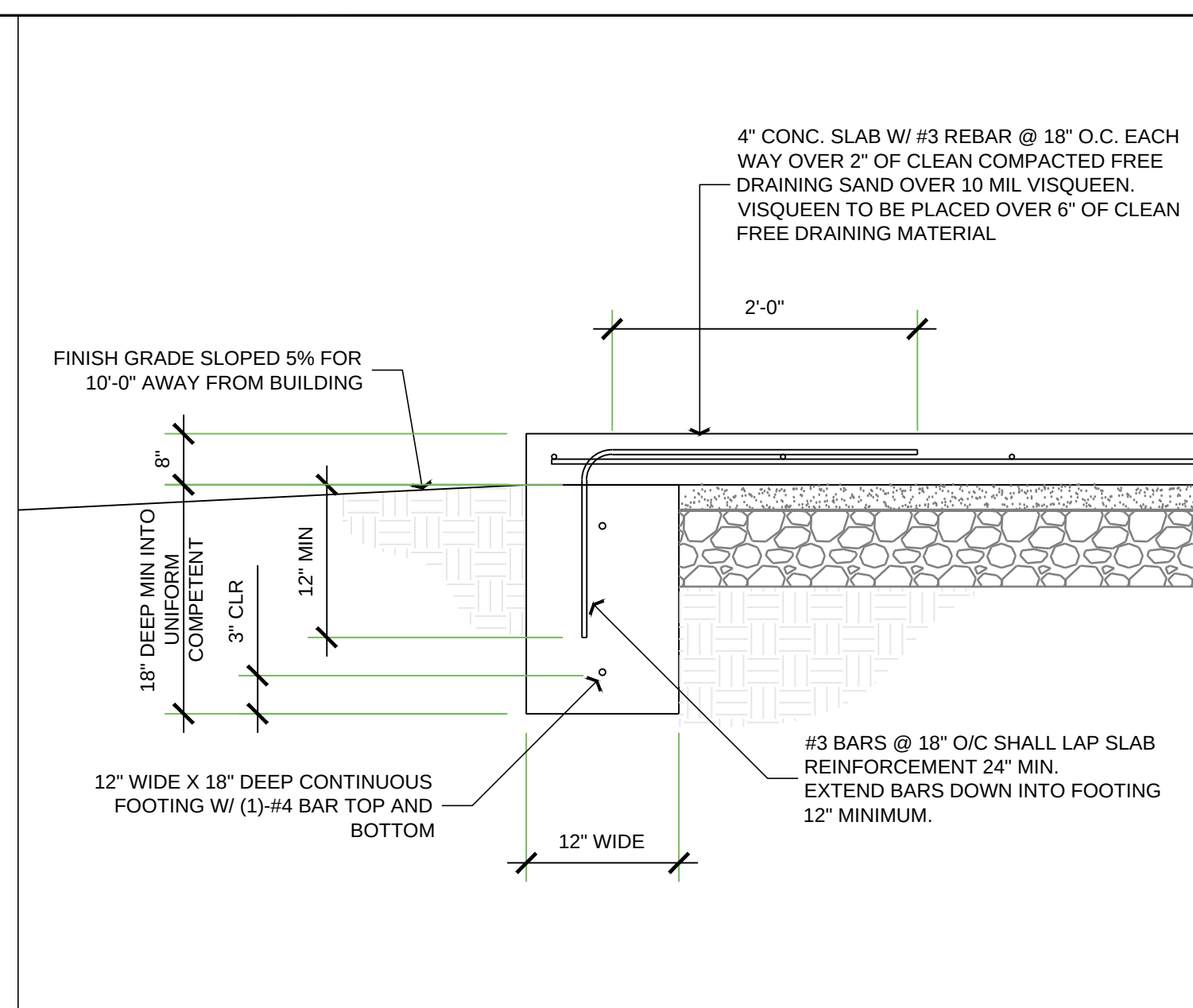
28

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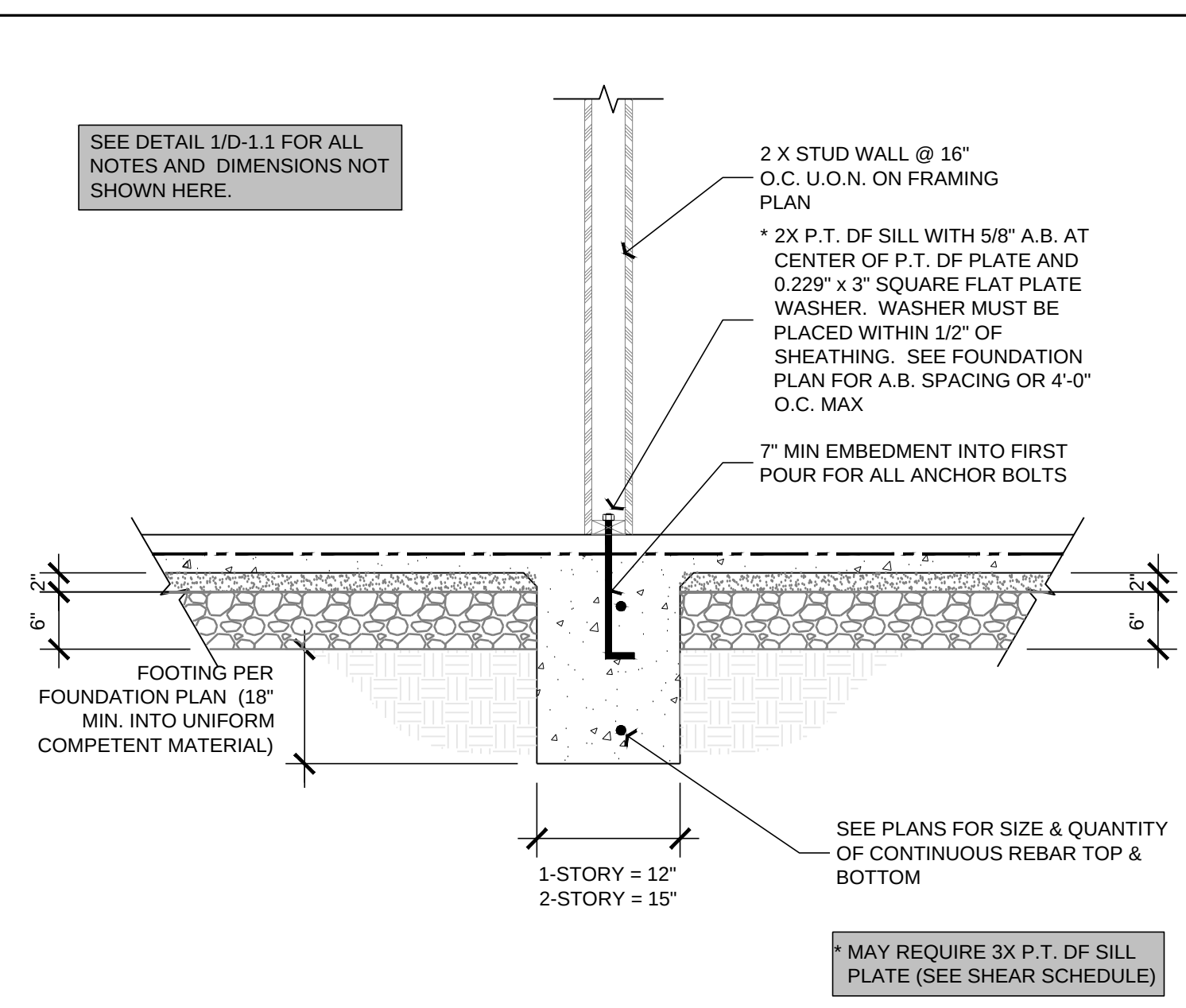
Coating Category and Less Exempt Compositions	Effective	
	1/19/2015	1/1/2022
Flat coatings	150	
Nonflat coatings	100	
Nonflat high gloss coatings	50	
SPECIALTY COATINGS		
Aluminum roof coatings	400	
Fluorocarbon specialty coatings	400	
Bituminous roof coatings	50	
Bituminous roof primers	350	
Isol sealers	350	
Concrete/masonry compounds	250	
Concrete/masonry sealers	100	
Driveway sealers	50	
Deck finishing	150	
Deck finishing coatings	350	
Fire retarding coatings	350	
Floor coatings	100	
Fluorocarbon coatings	250	
Gloss (spec. on coating edge paint)	400	
High temperature coatings	420	
Industrial maintenance coatings	250	
Lead solubility coatings	120	
Machinability coatings	400	
Mastic resins coatings	100	
Metallic pigmented coatings	500	
Multifunctional coatings	250	
Penetrating sealers	420	
Primers, sealers, and undercoats	350	
Reactive penetrating sealers	350	
Recoated coatings	250	
Roof coatings	50	
Rust penetrative coatings	400	250
Stethics		
Clear	730	
Opaque	550	
Specialty primers, sealers and undercoats	350	100
Slane	250	
Stone consolidants	450	
Swimming pool coatings	340	
Tracking marking coatings	100	
Tie and weld coatings	250	
Waterproofing membranes	250	
Wood coatings	275	
Wood preservatives	350	
Zinc-rich coatings	250	



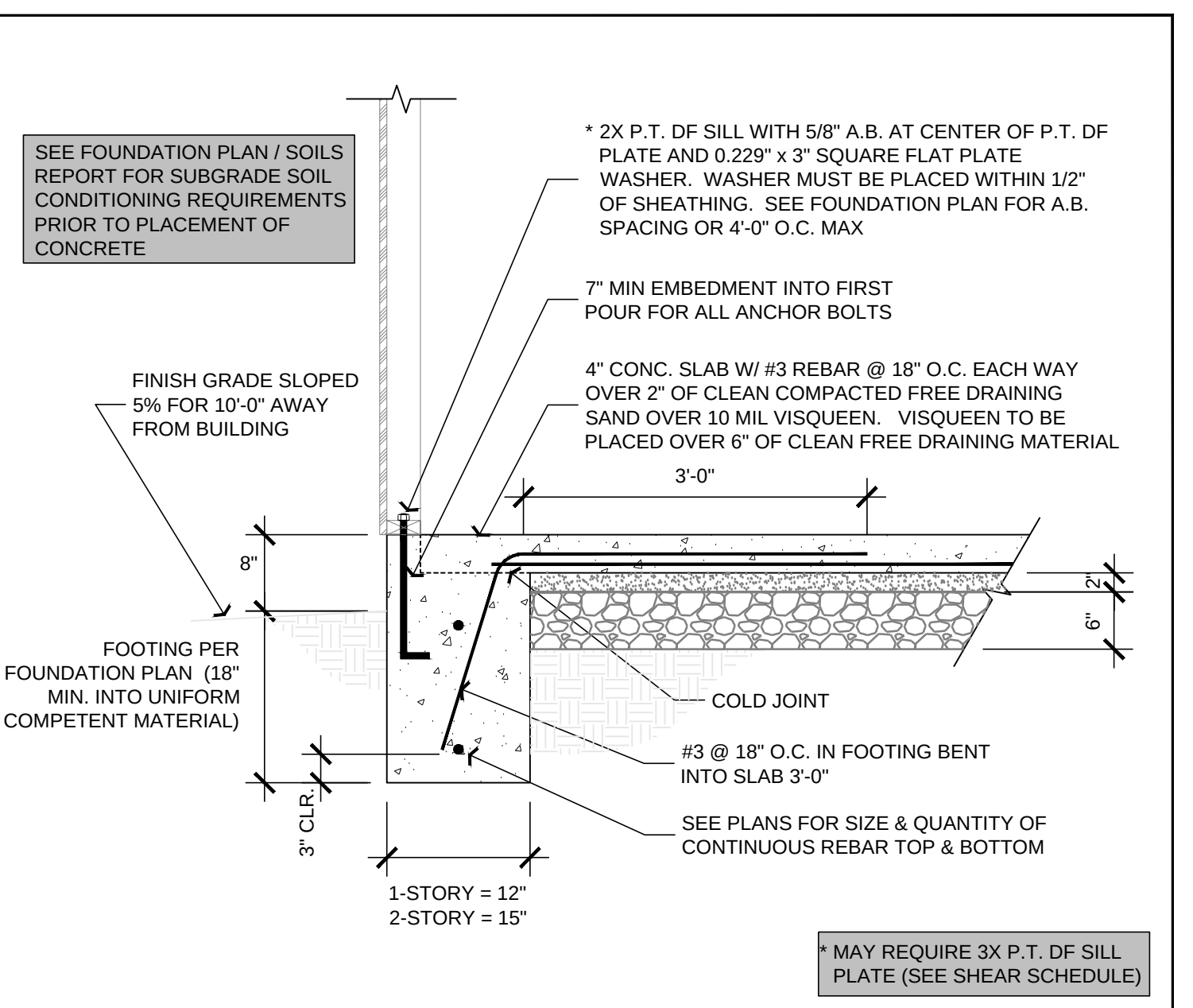
S-1.1



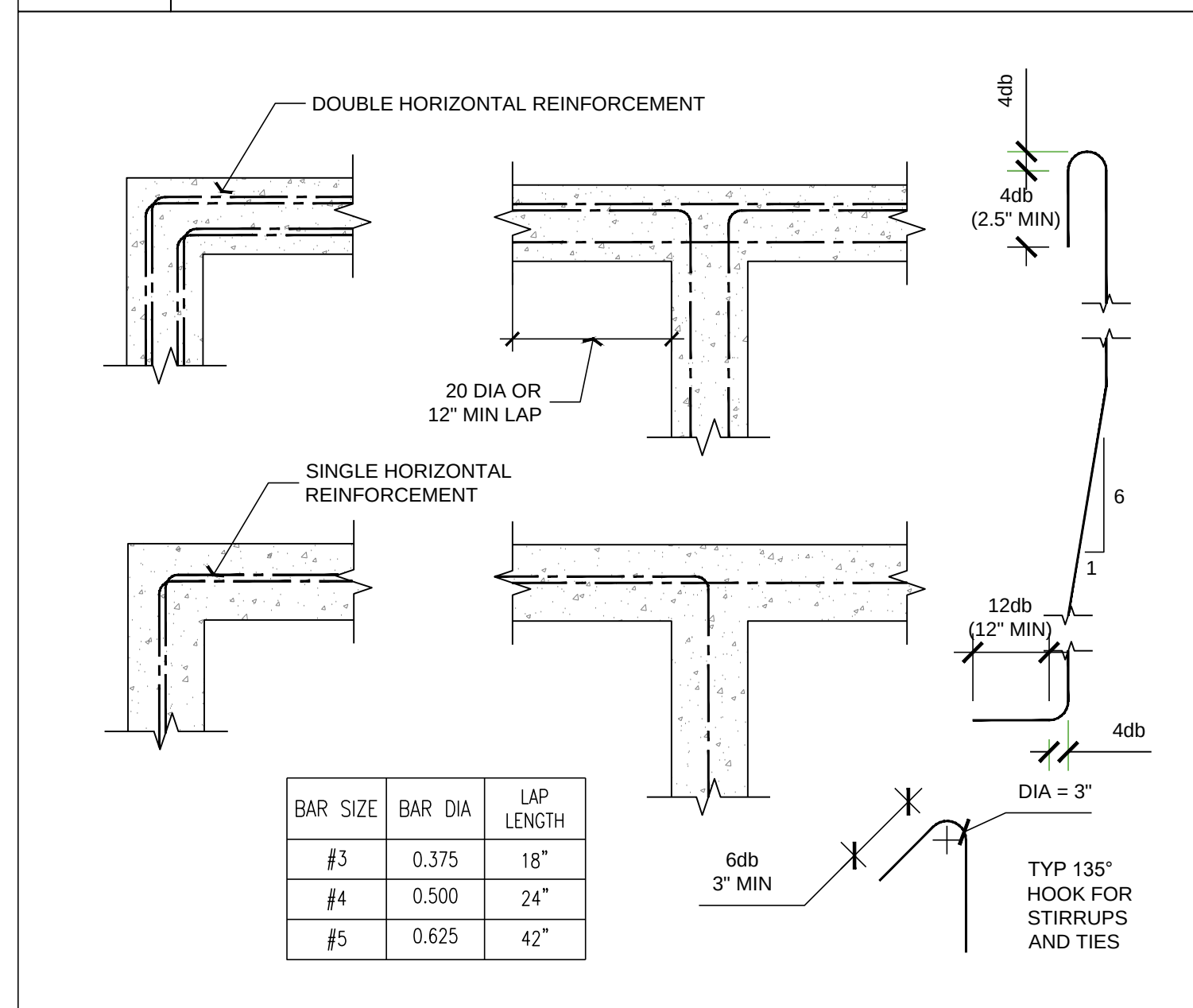
7	PERIMETER FOOTING
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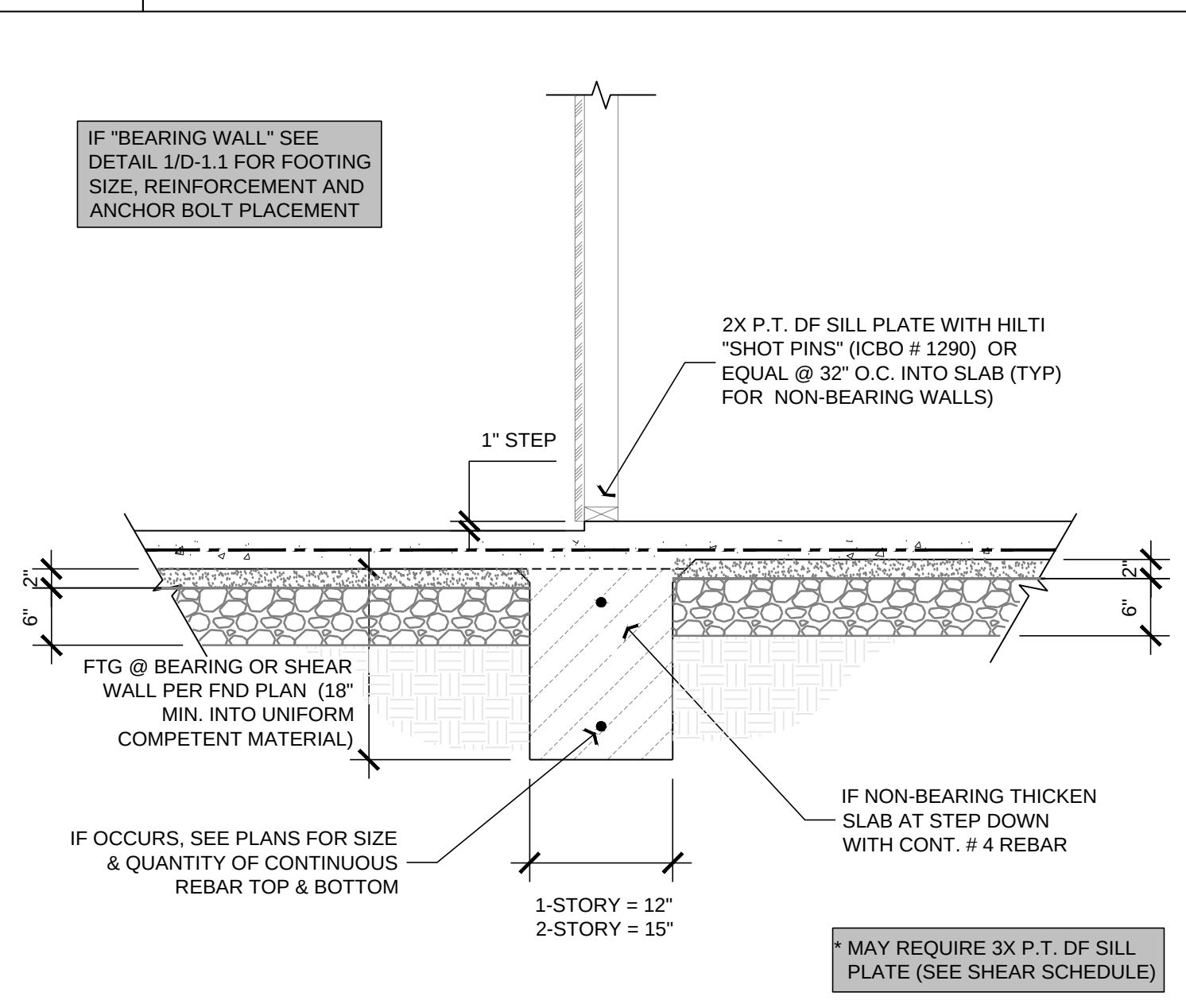
4	INTERIOR FOOTING
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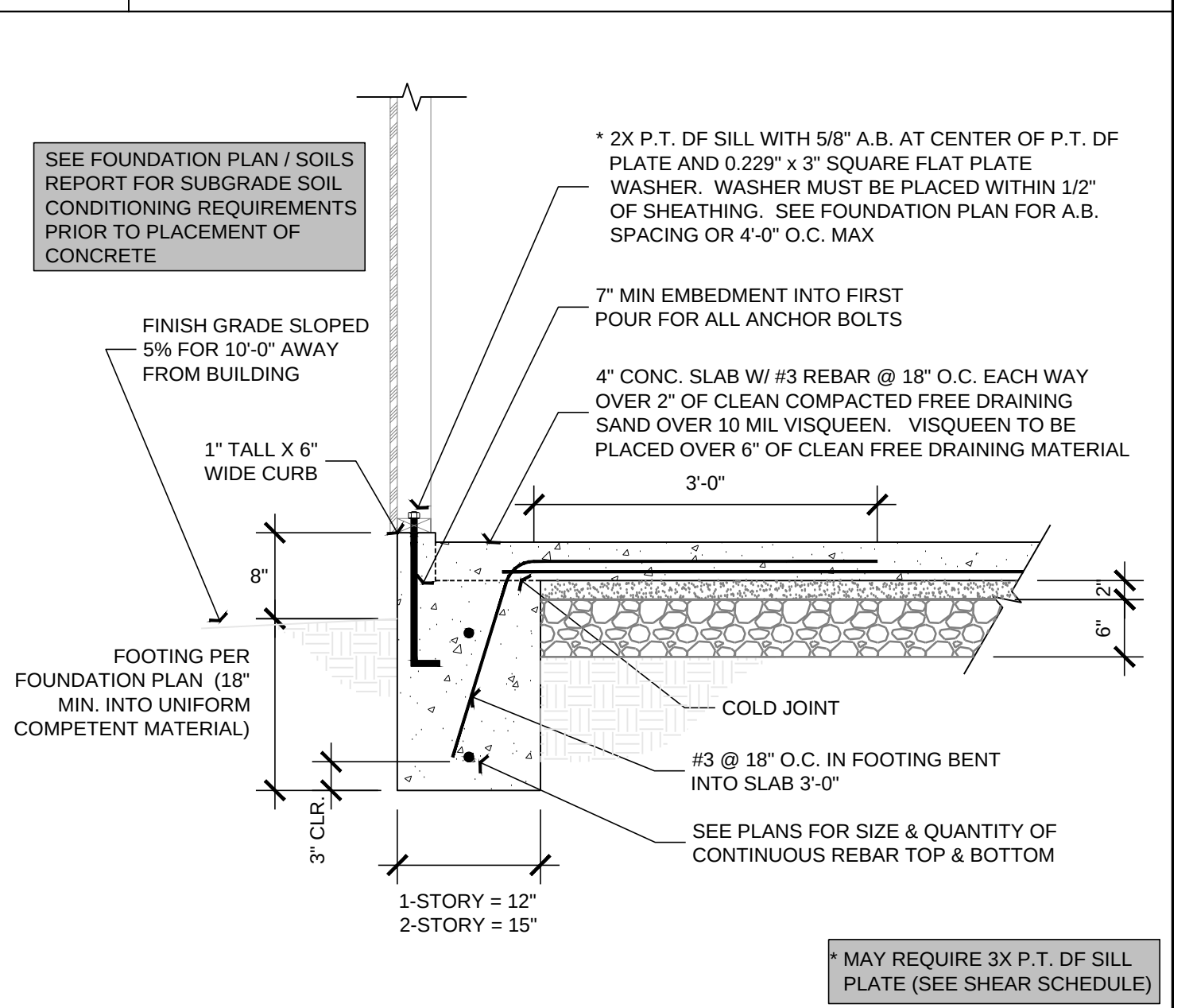
1	EXTERIOR FOOTING
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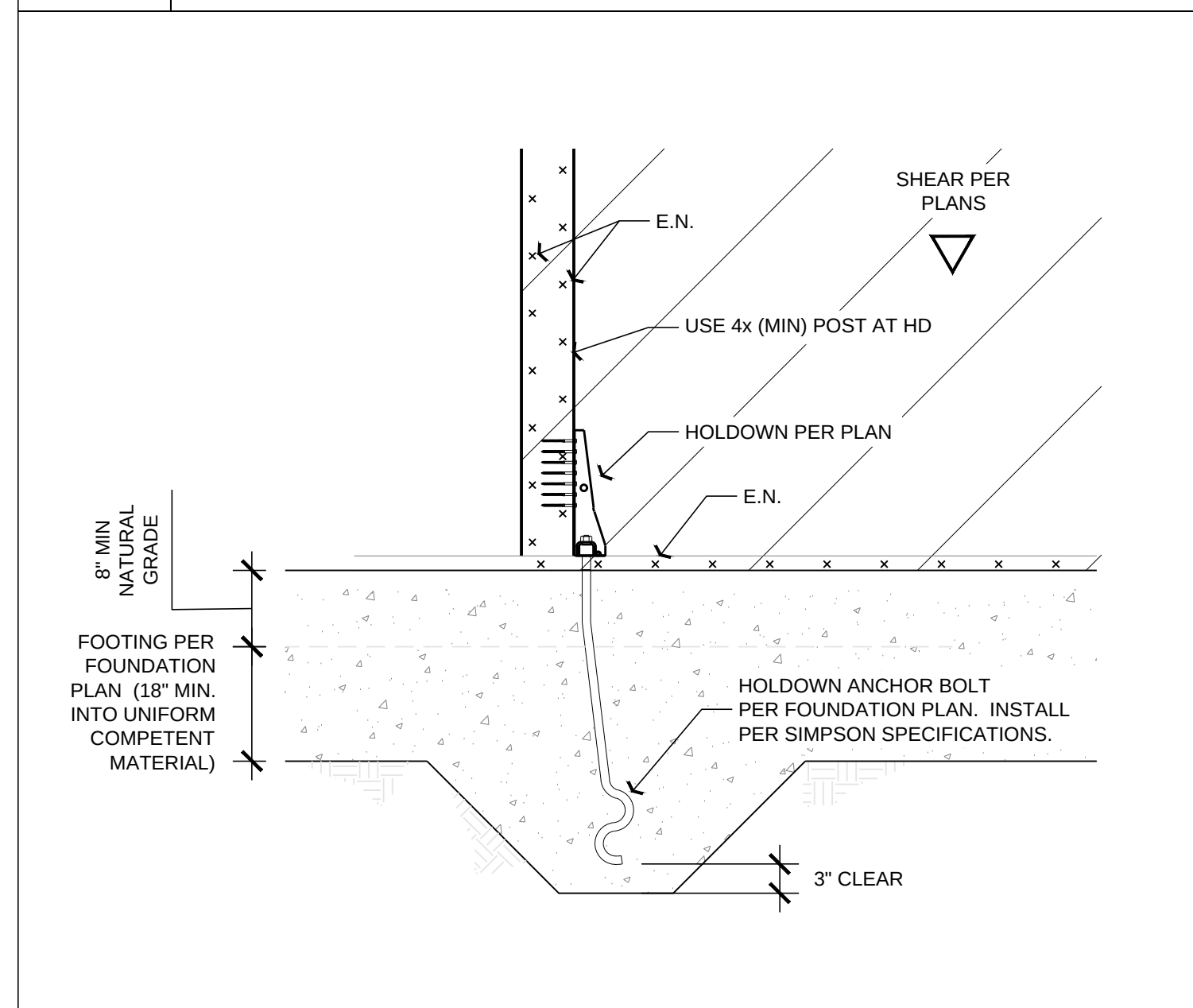
8 TYPICAL HORIZONTAL REINFORCING



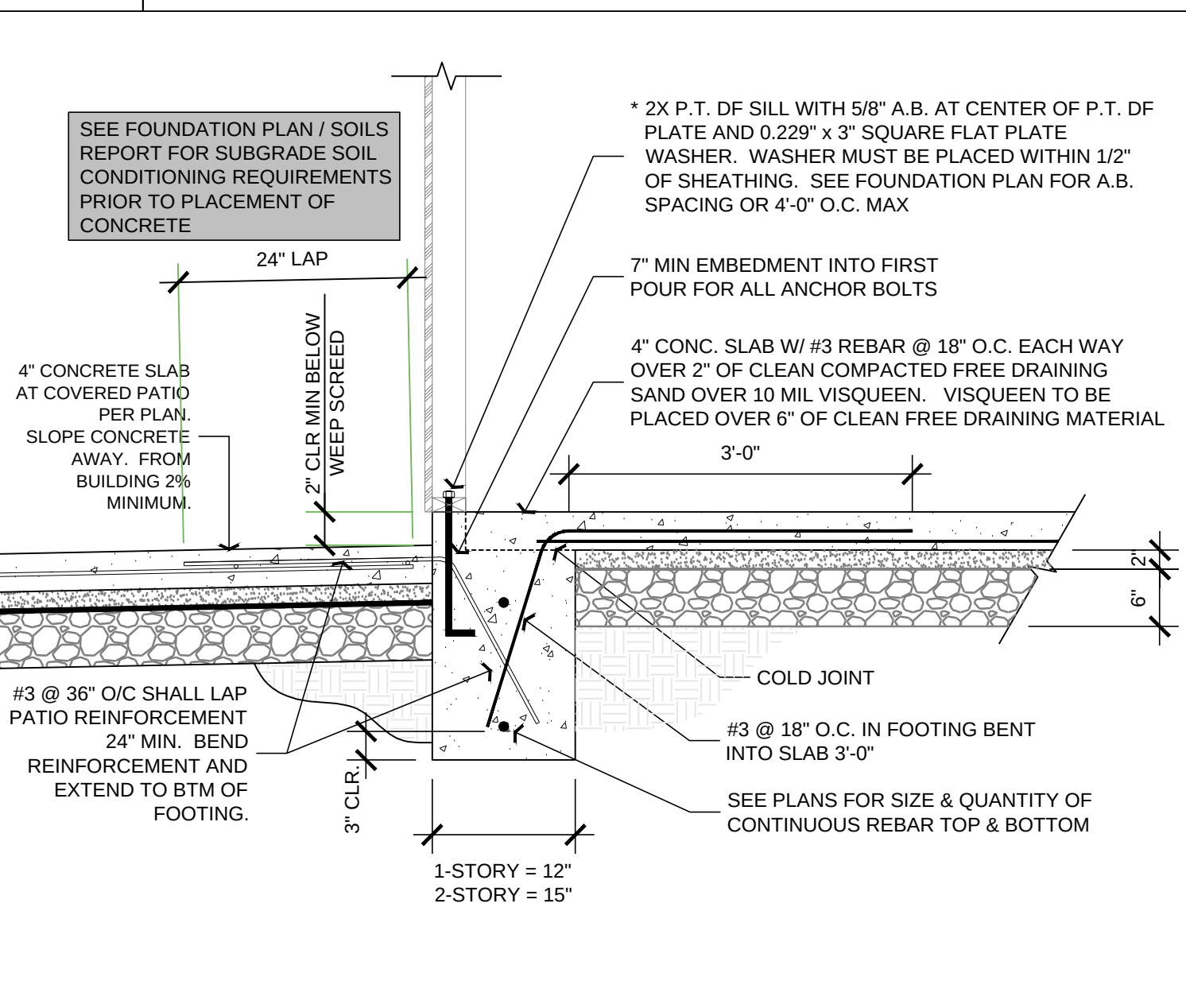
5	SLAB @ GARAGE
---	---------------



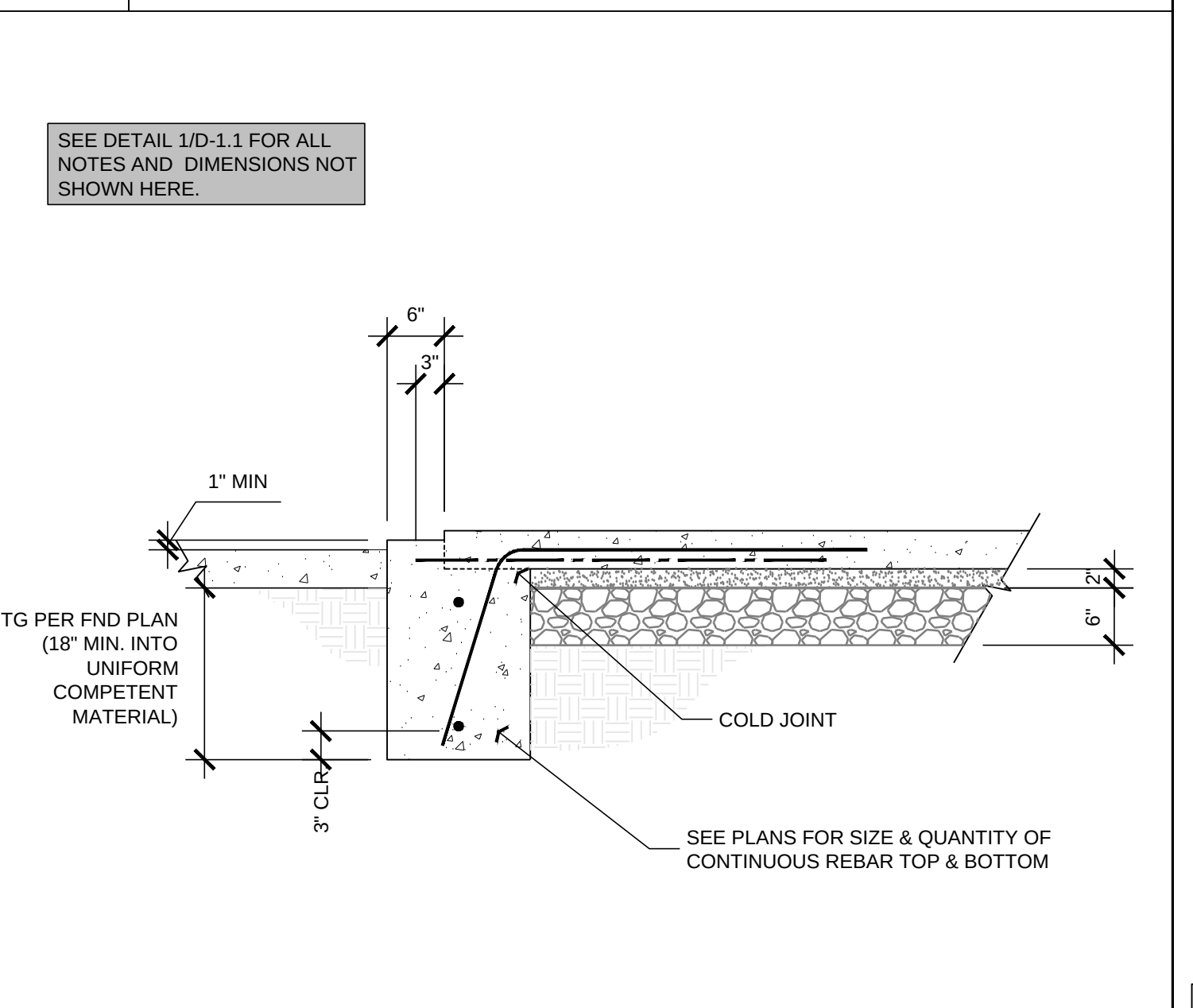
2 EXTERIOR FOOTING AT GARAGE



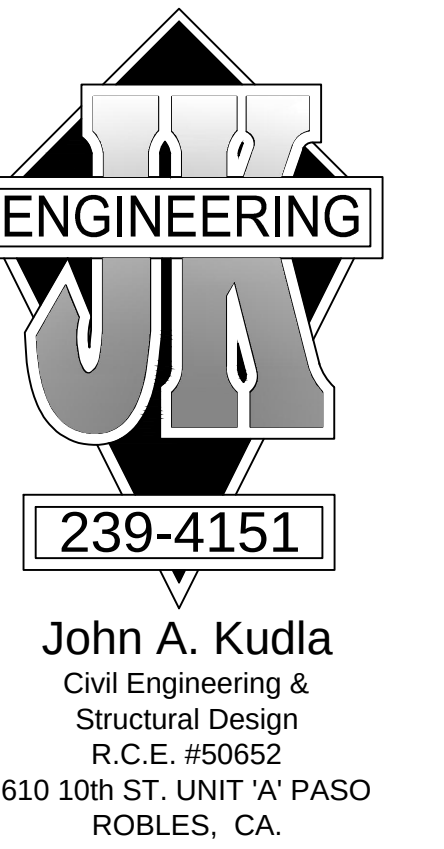
9	HOLDOWN DETAIL
---	----------------



6 EXTERIOR FOOTING AT PATIO SLAB



3	FOOTING AT GARAGE DOOR
---	------------------------



PLAN PREPARED FOR:
KIRK & CARRIE ALLEN
LOT 106 TRACT 1990-2
PASO ROBLES, CA 93446

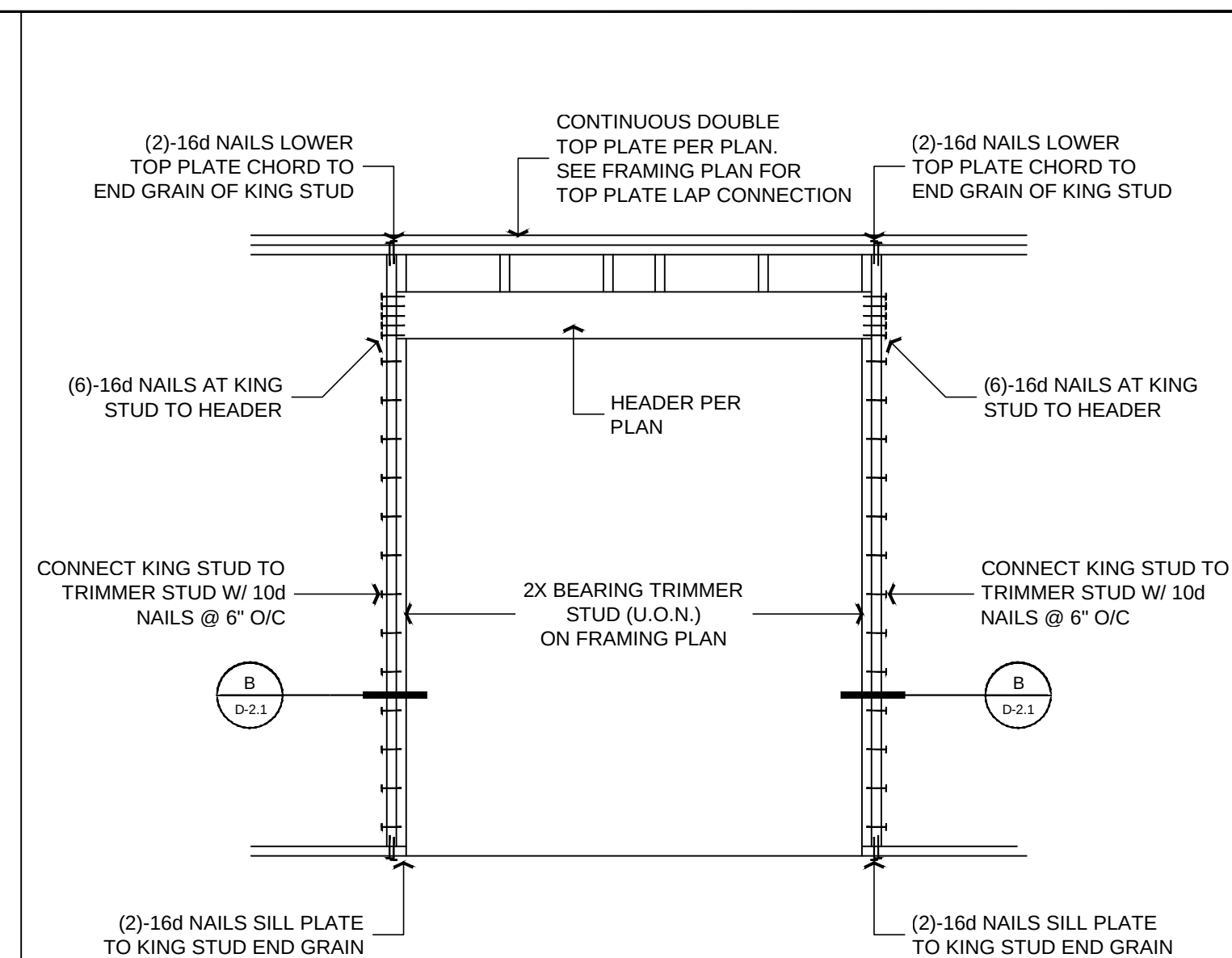
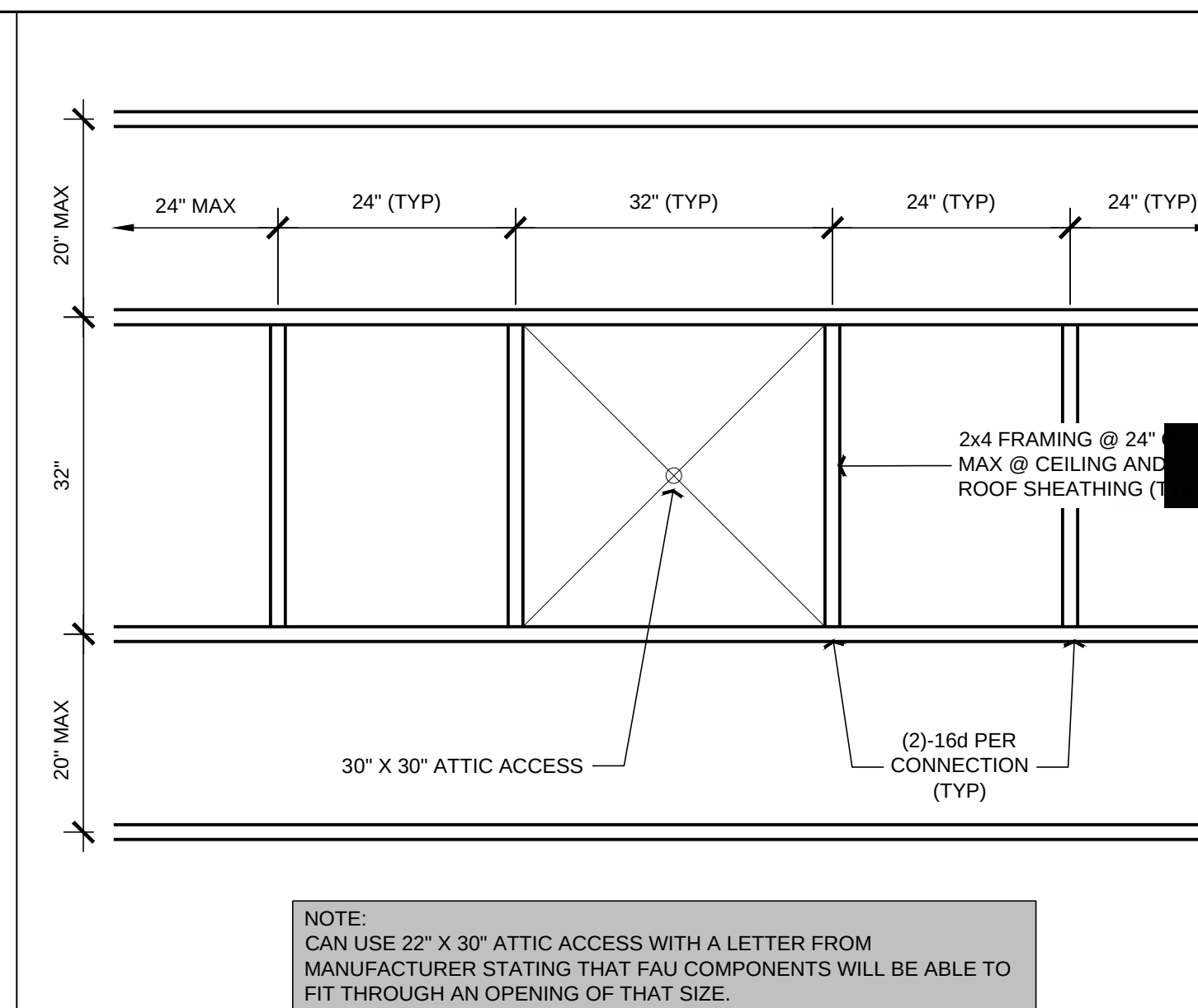
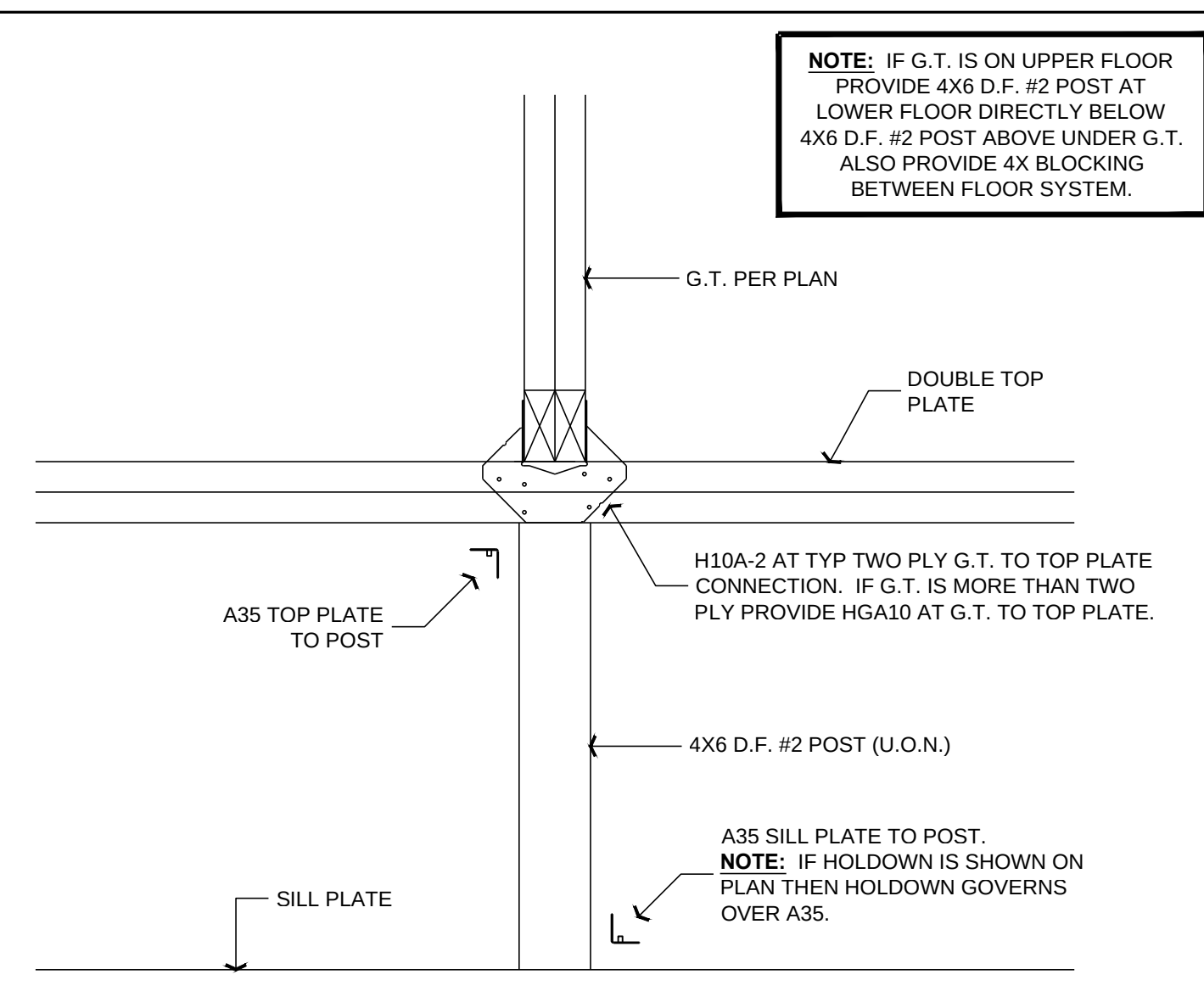
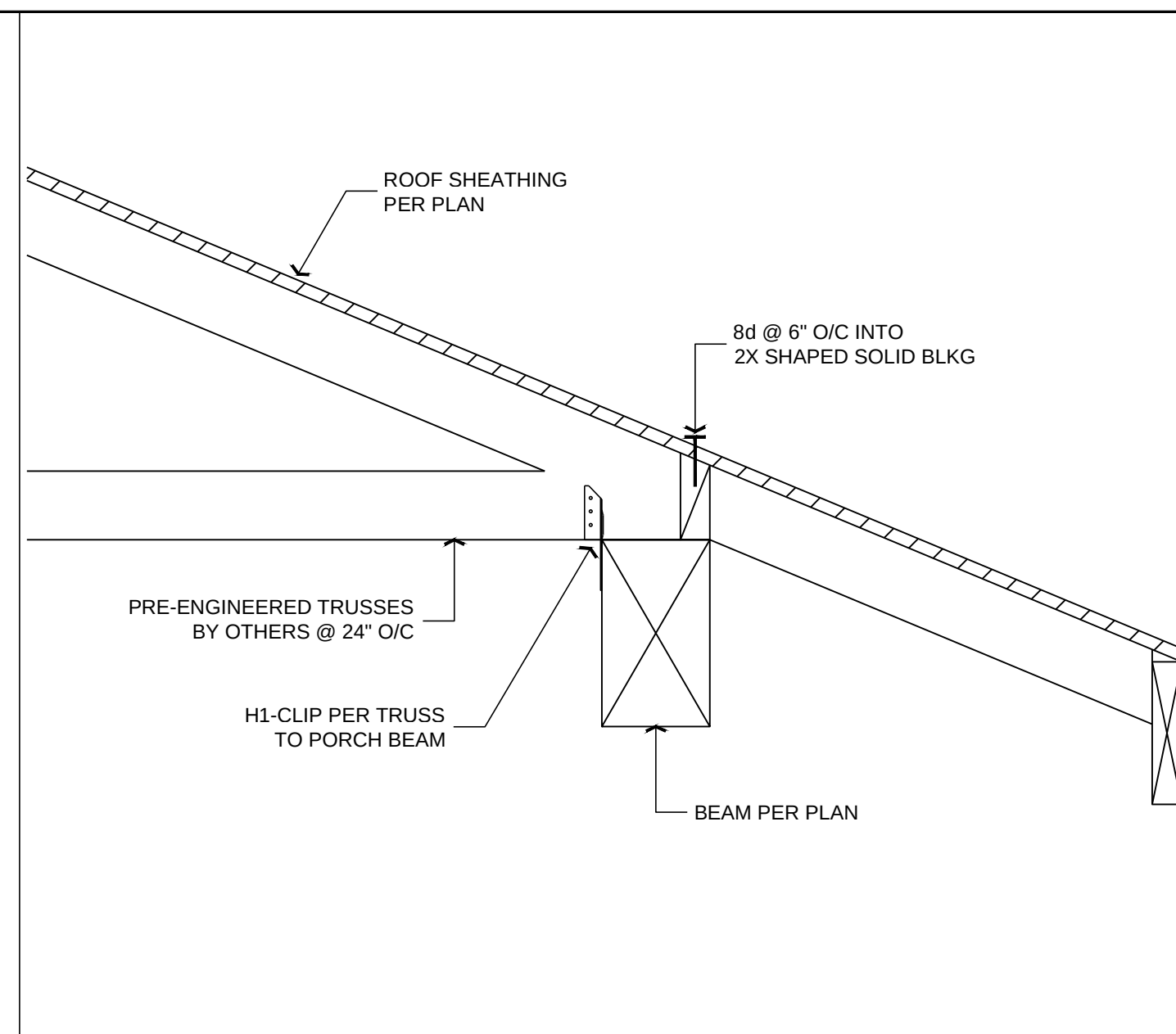
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PROJECT NO. ---
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 DRAWN BY JJK
 DATE 3/31/2017 8:00 AM
 SHEET TITLE:
 STRUCTURAL
 DETAILS

SHEET NUMBER:

D-1.1

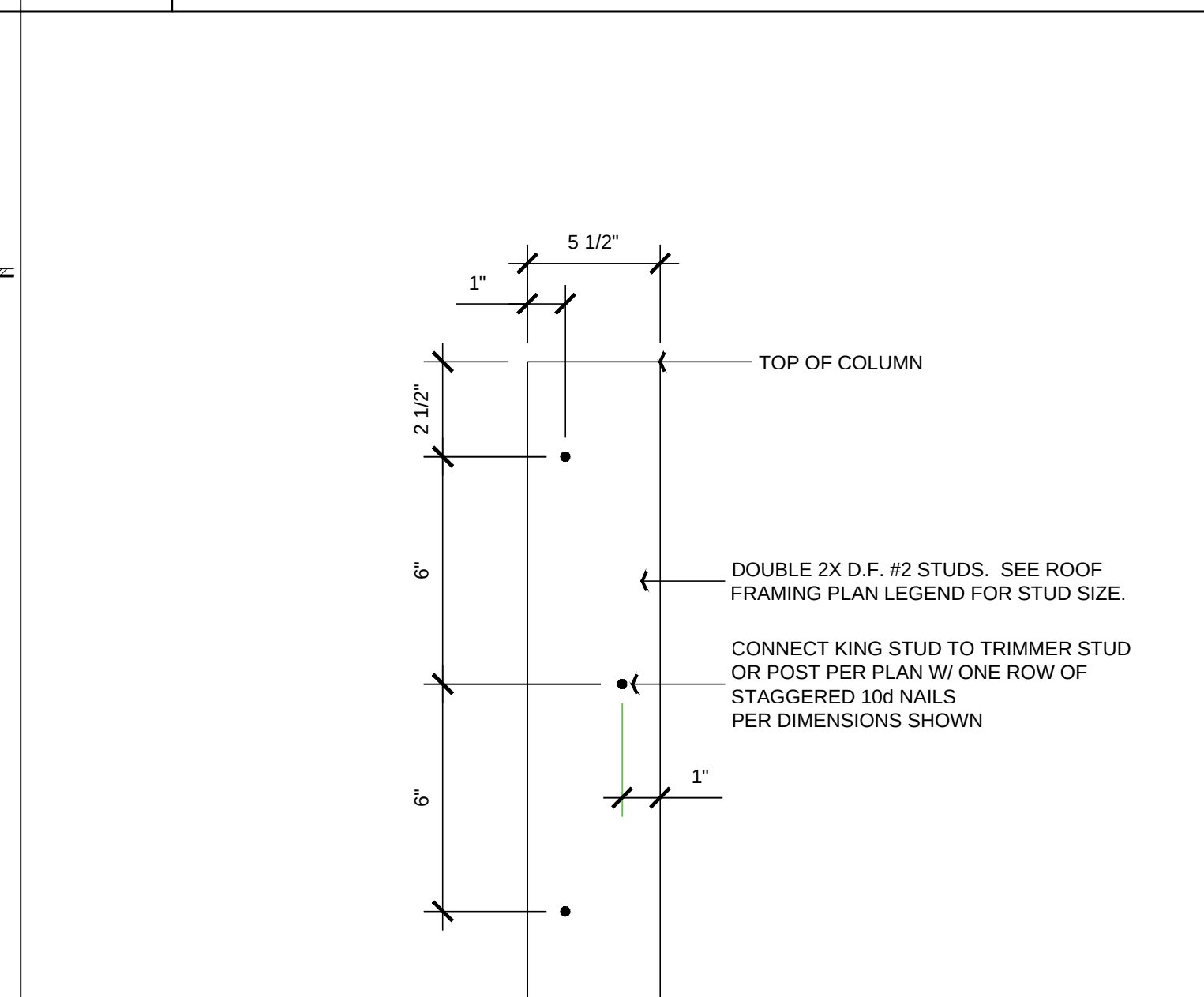
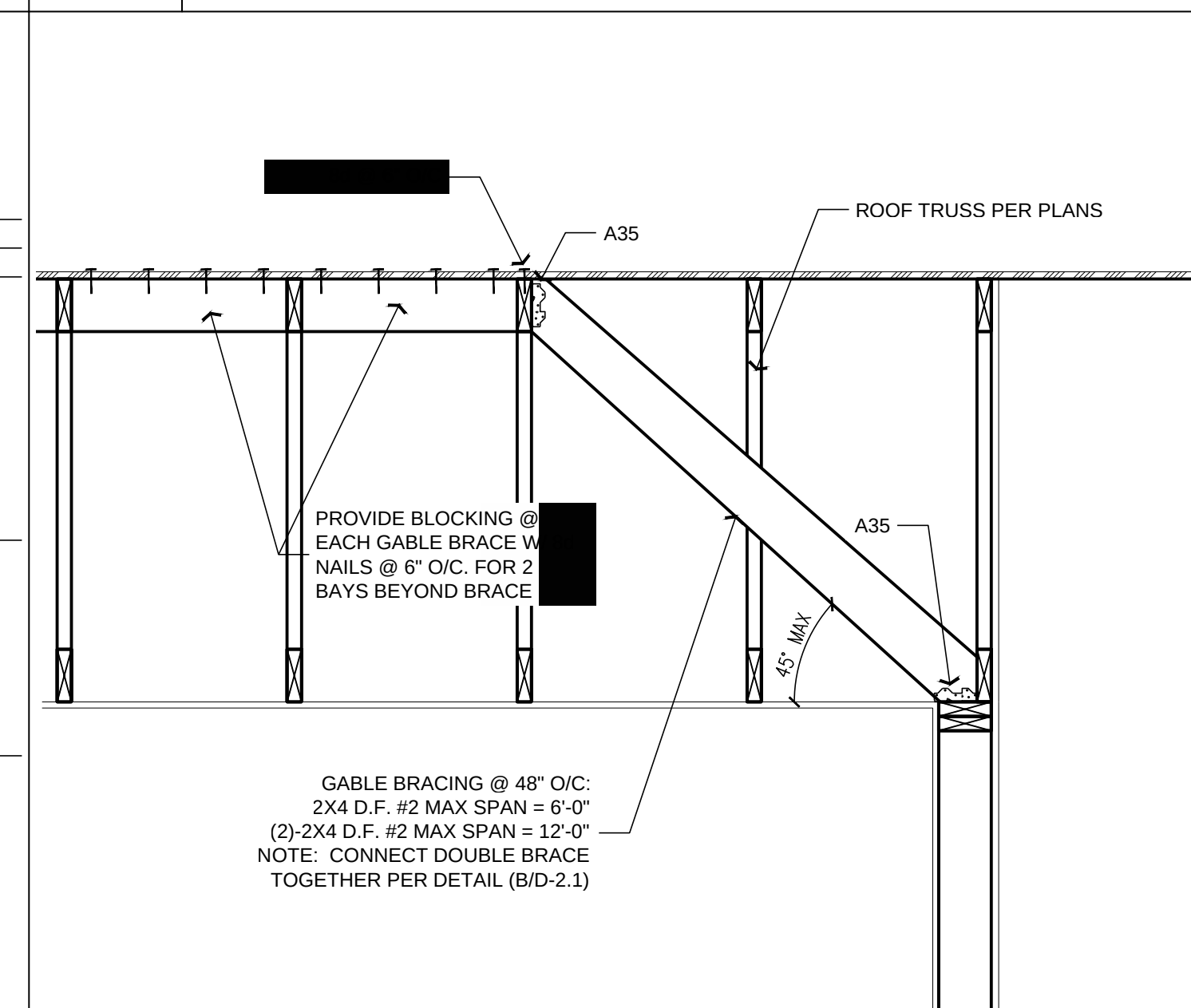
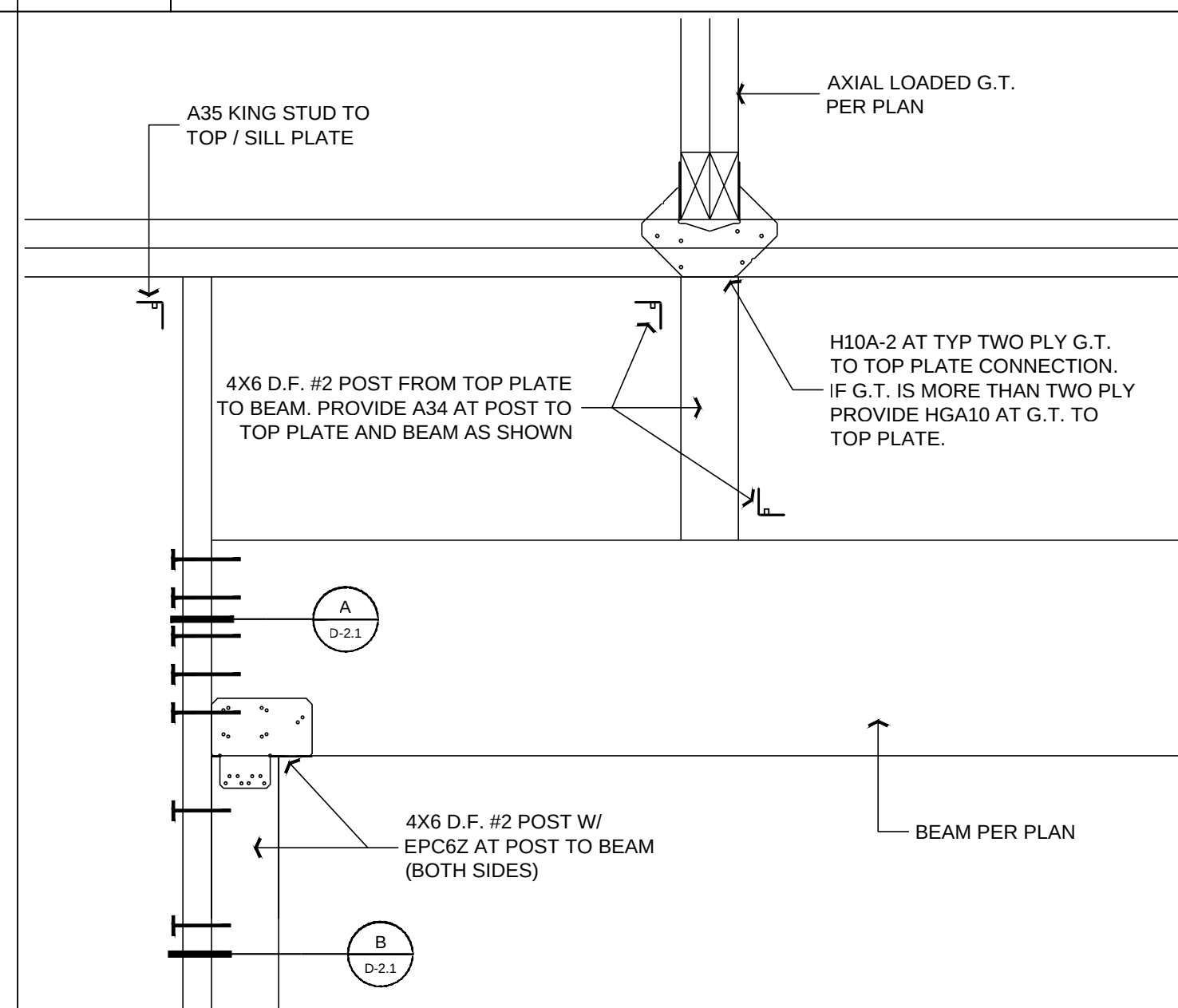
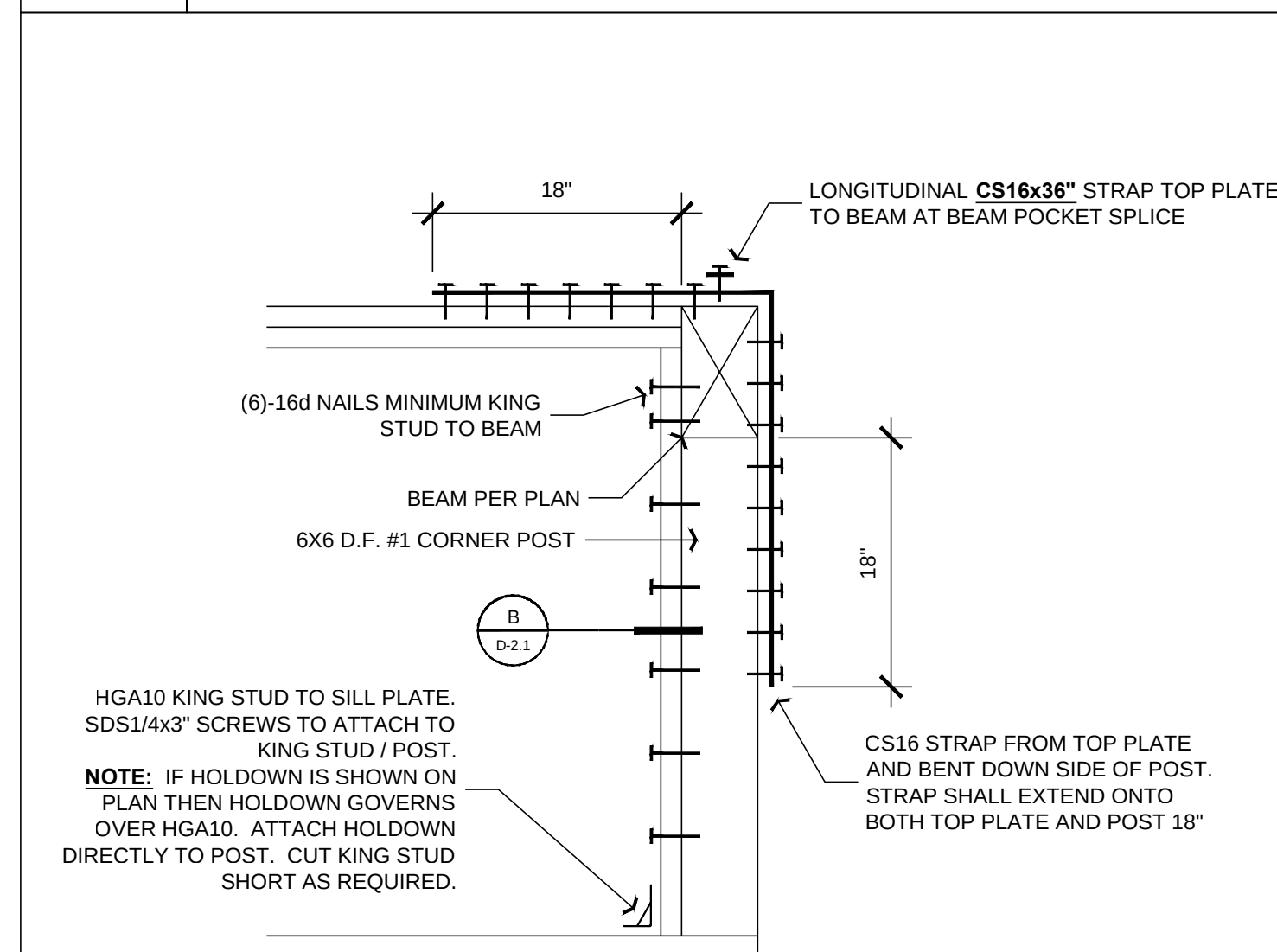


J | SHEAR TRANSFER - TRUSS EAVE TO BEAM

G	TYP GIRDER TRUSS CONNECTION
---	-----------------------------

D	30" OPENING @ TRUSS
---	---------------------

A	TYP FRM AT WINDOW / DOOR OPENING (U.O.N.)
---	---

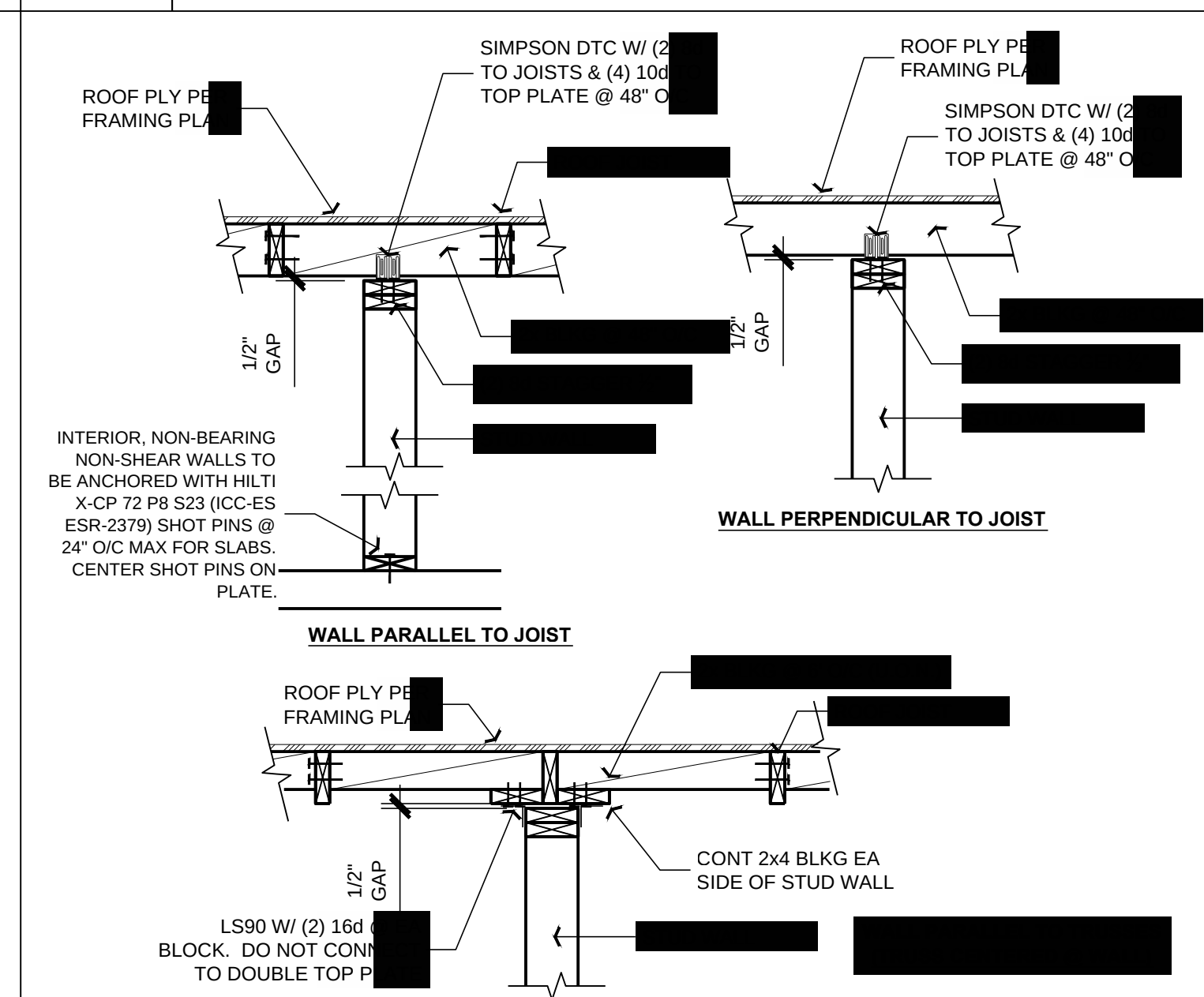
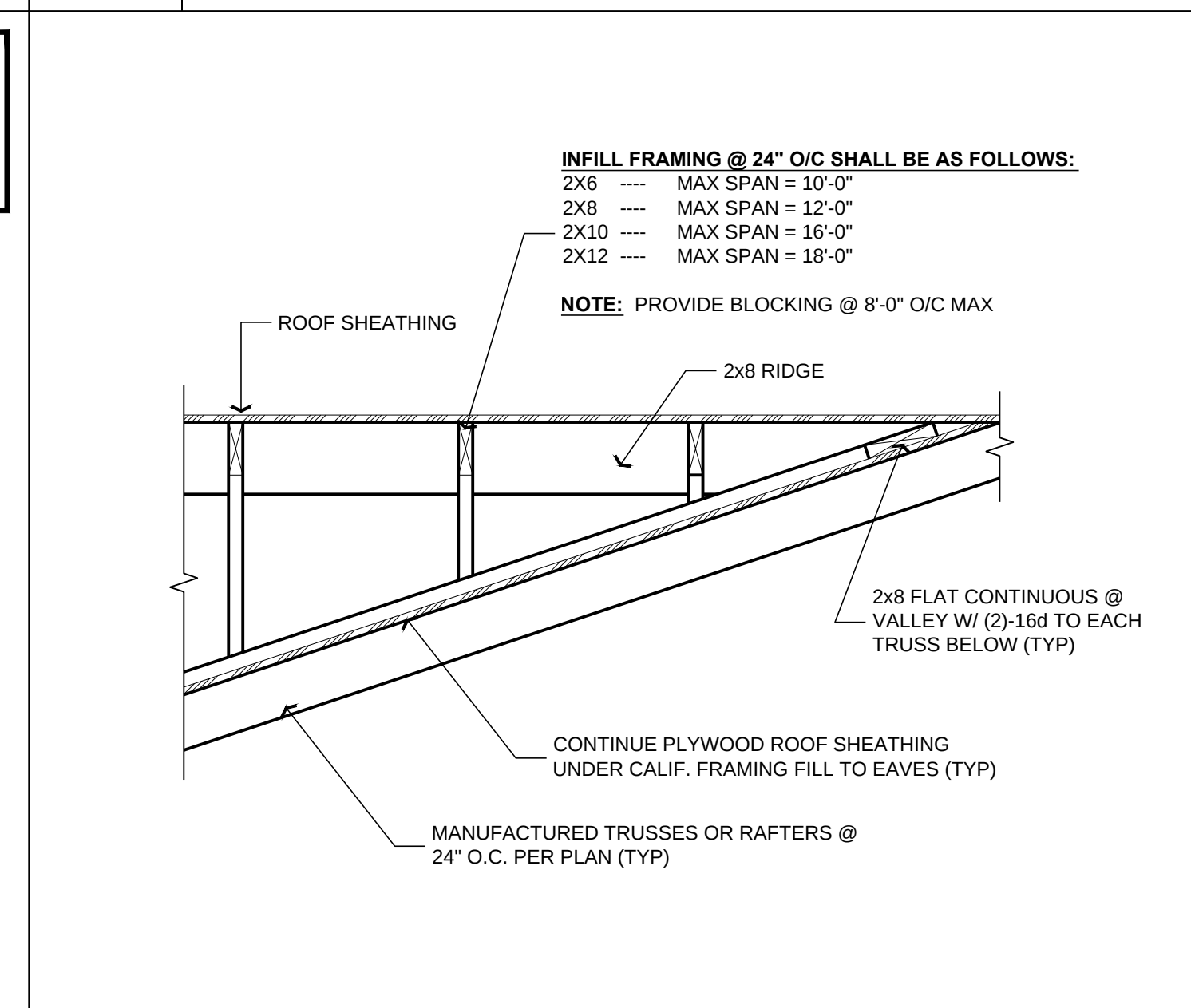
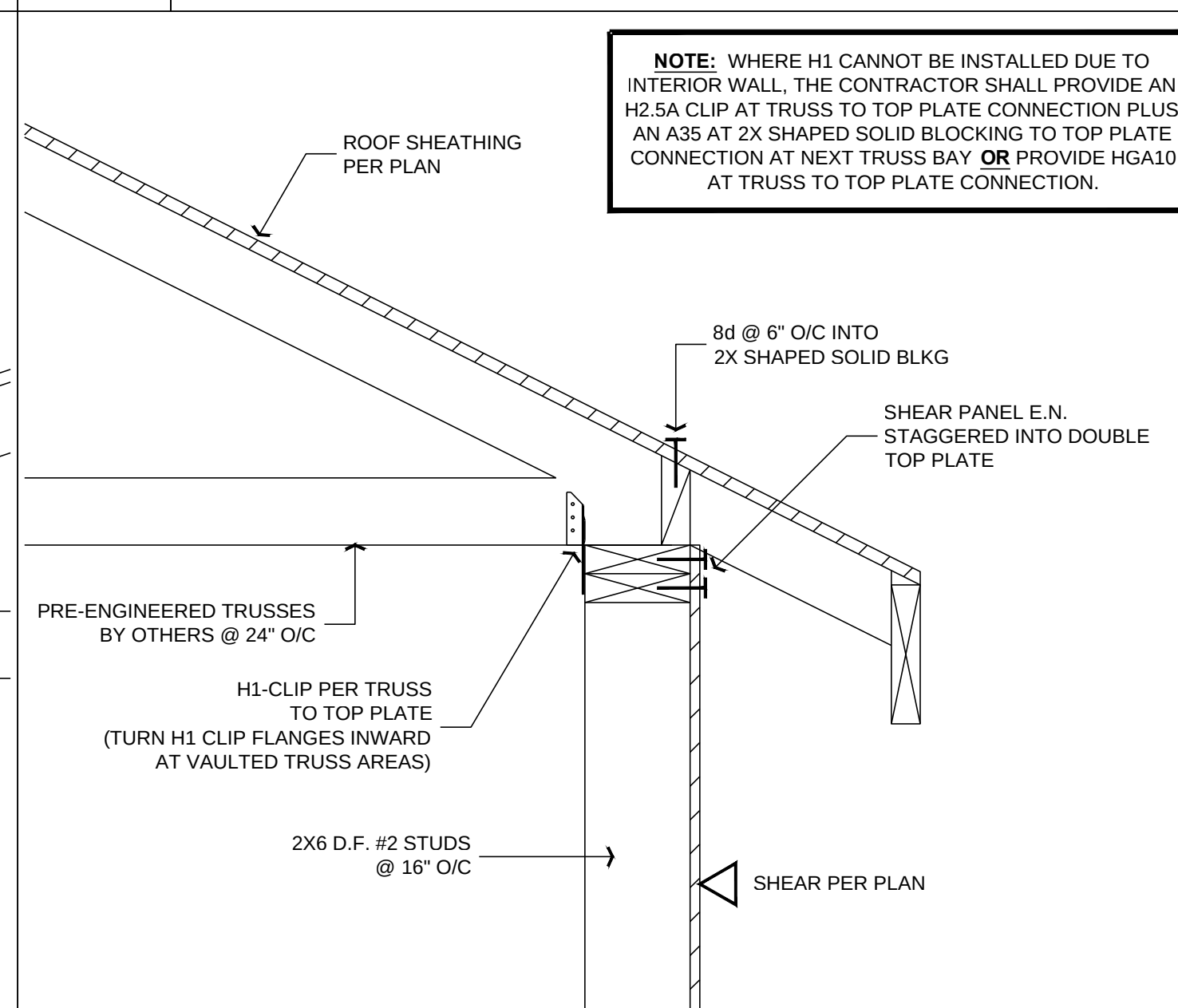
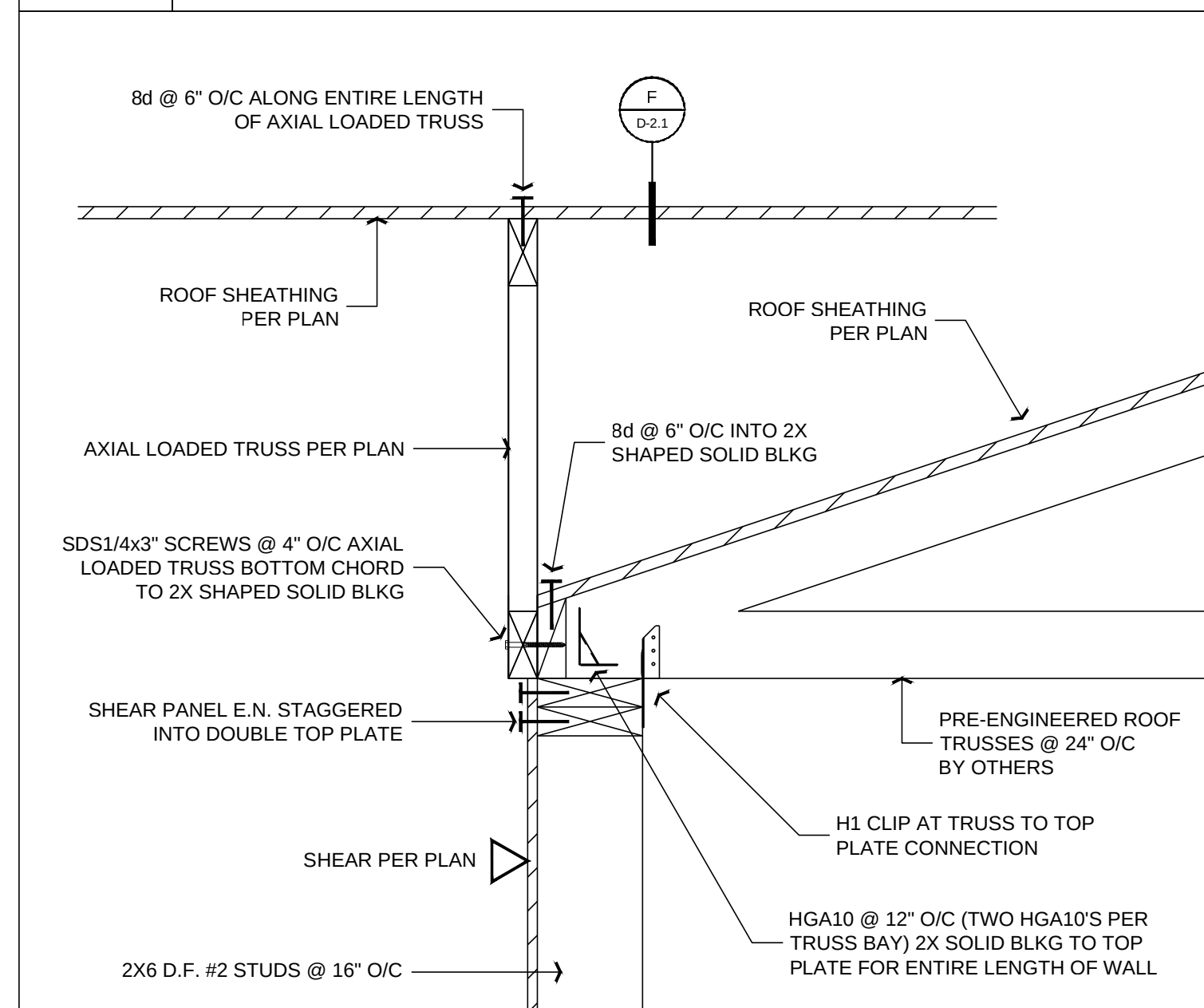


K	BEAM POCKET DETAIL
---	--------------------

H	TYP G.T. OVER HEADER CONNECTION
---	---------------------------------

E	OUT-OF-PLANE SHEAR TRANSFER
---	-----------------------------

B	BUILT UP DOUBLE STUD COLUMN
---	-----------------------------



L	SHEAR TRANSFER - LAYOUT TRANSITION
---	------------------------------------

I SHEAR TRANSFER - TRUSS EAVE CONNECTION

F	CALIFORNIA FRAMING
---	--------------------

C	INTERIOR NON-BRG WALL DETAIL
---	------------------------------

[illegible]

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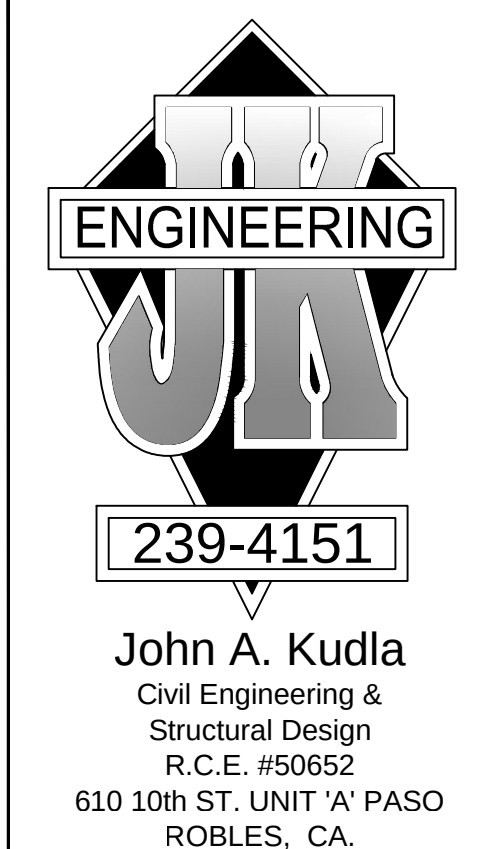
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DATE 3/31/2017 8:07 AM
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**STRUCTURAL
DETAILS**

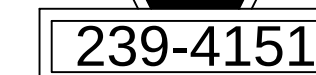
SHEET NUMBER

D-2.1

PLAN PREPARED FOR:

KIRK & CARRIE ALLEN
LOT 106 TRACT 1990-2
PASO ROBLES, CA 93446





John A. Kudla
Civil Engineering &
Structural Design
R.C.E. #50652
610 10th ST. UNIT 'A' PAS
ROBLES, CA.

PLAN PREPARED FOR:

KIRK & CARRIE ALLEN
LOT 106 TRACT 1990-2
PASO ROBLES, CA 93446

[illegible]

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PROJECT NO. ----
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DRAWN BY JUK

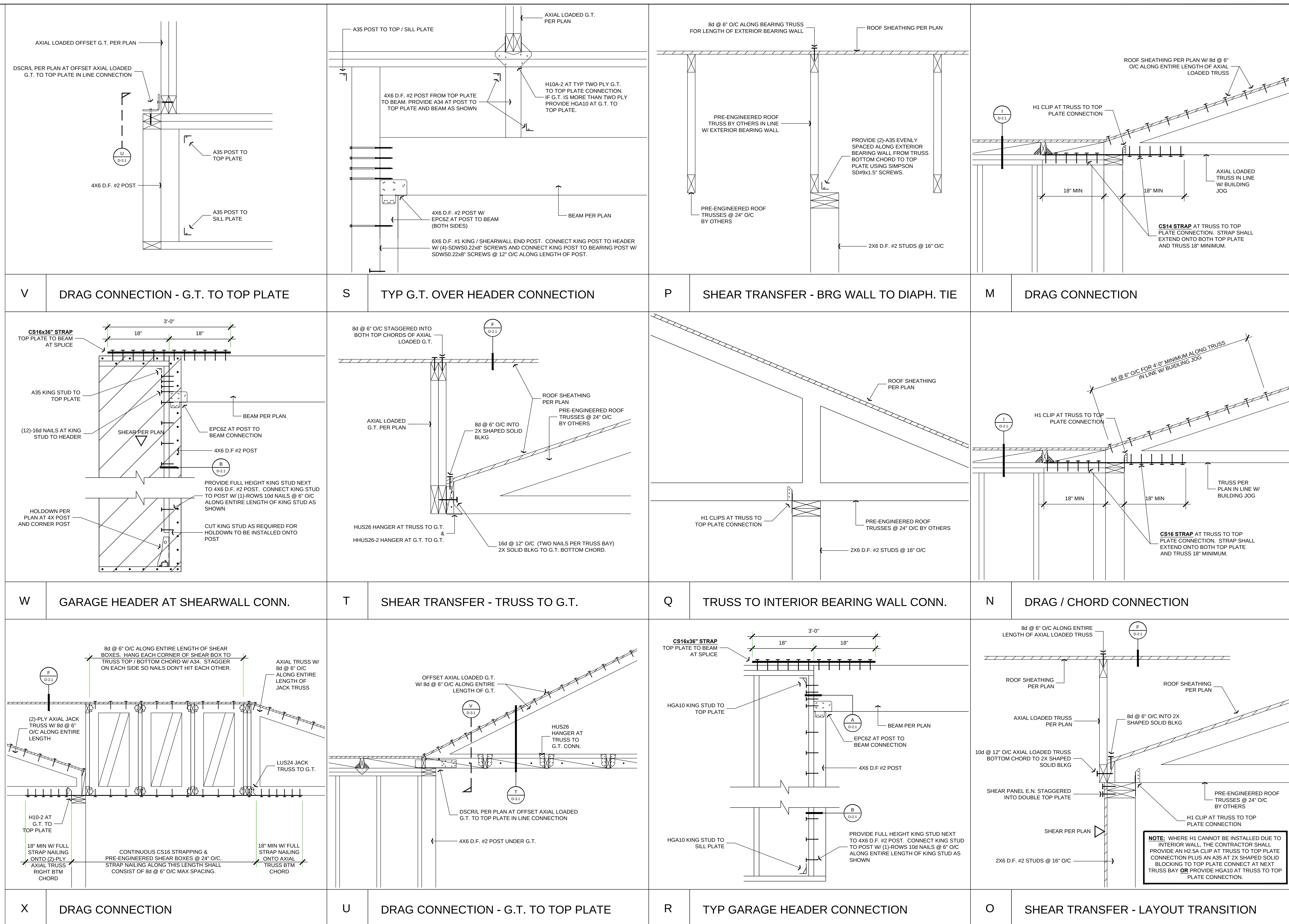
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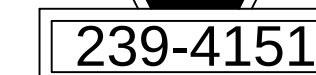
STRUCTURAL DETAILS

SHEET NUMBER

D-3.1



\\dratting\projects\cspgs_Devo\DOT=106\LOT 106\Sheets\O-3.1 STRUCTURAL DETAILS.dwg Johnny Kudva 2/14/2017 11:53:21 AM



John A. Kudla
Civil Engineering &
Structural Design
R.C.E. #50652
610 10th ST. UNIT 'A' PASO
ROBLES, CA.

PLAN PREPARED FOR:
KIRK & CARRIE ALLEN
LOT 106 TRACT 1990-2
PASO ROBLES, CA 93446

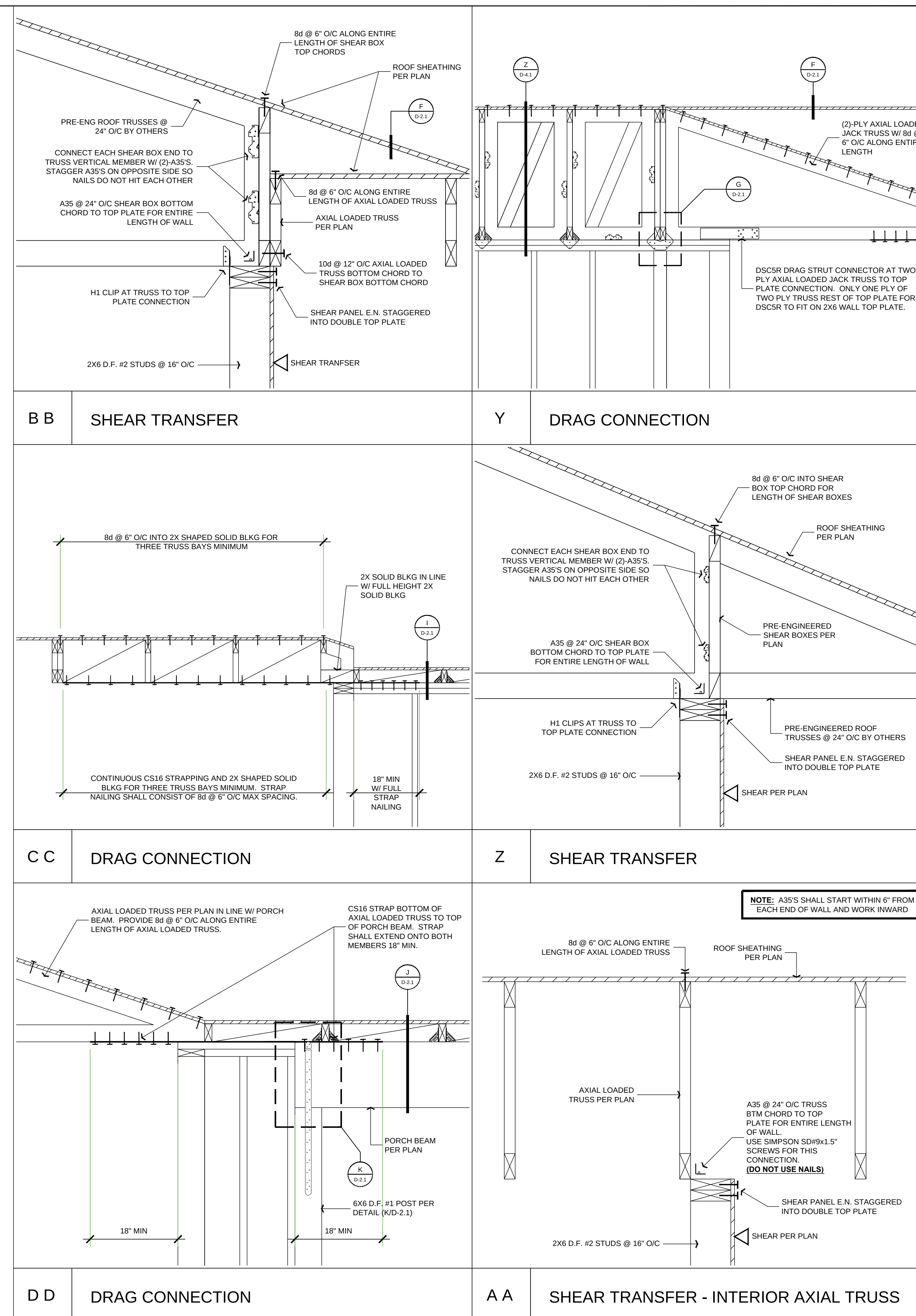
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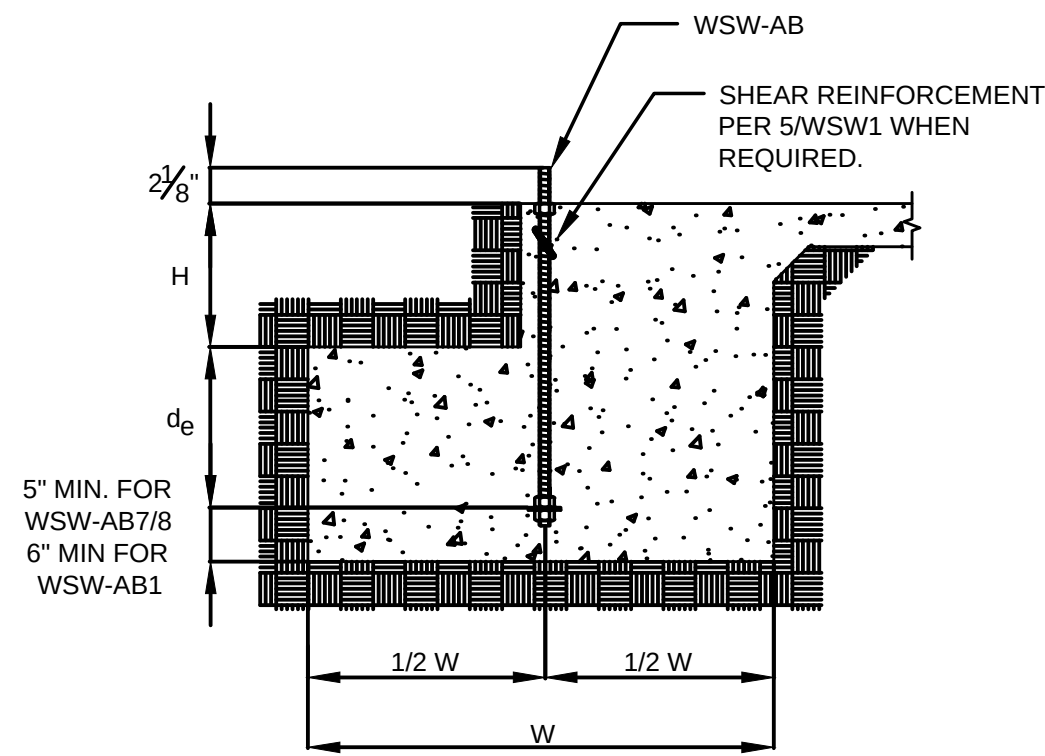
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DATE 3/31/2017 8:00 AM
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DETAILS**

SHEET NUMBER

D-4.1



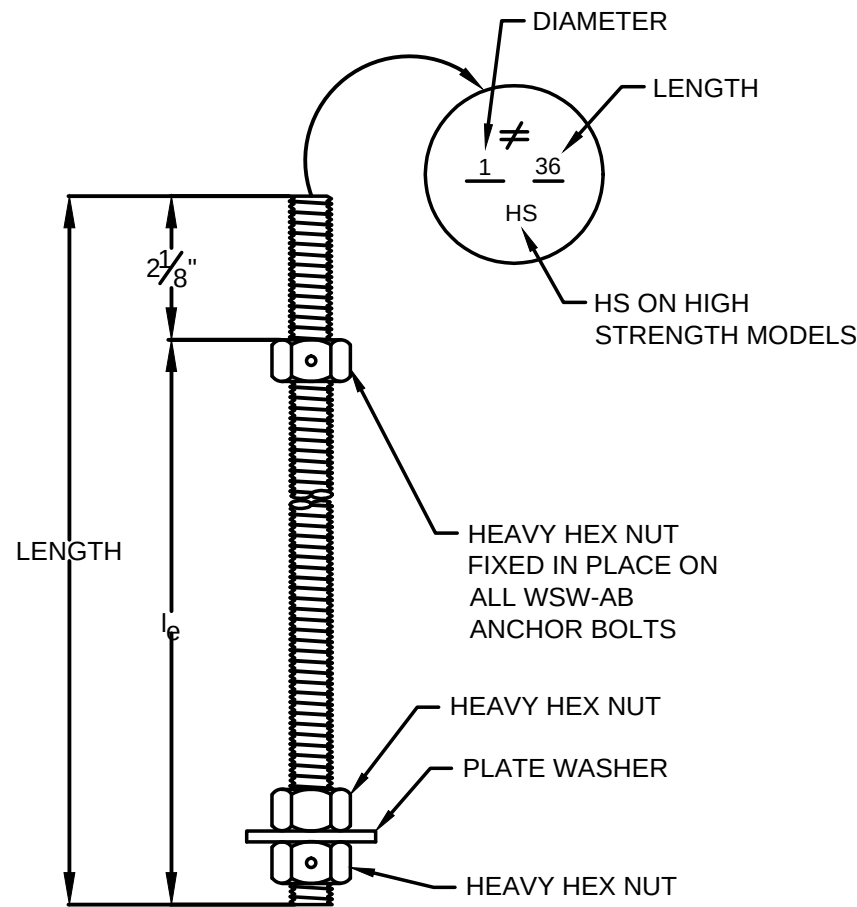
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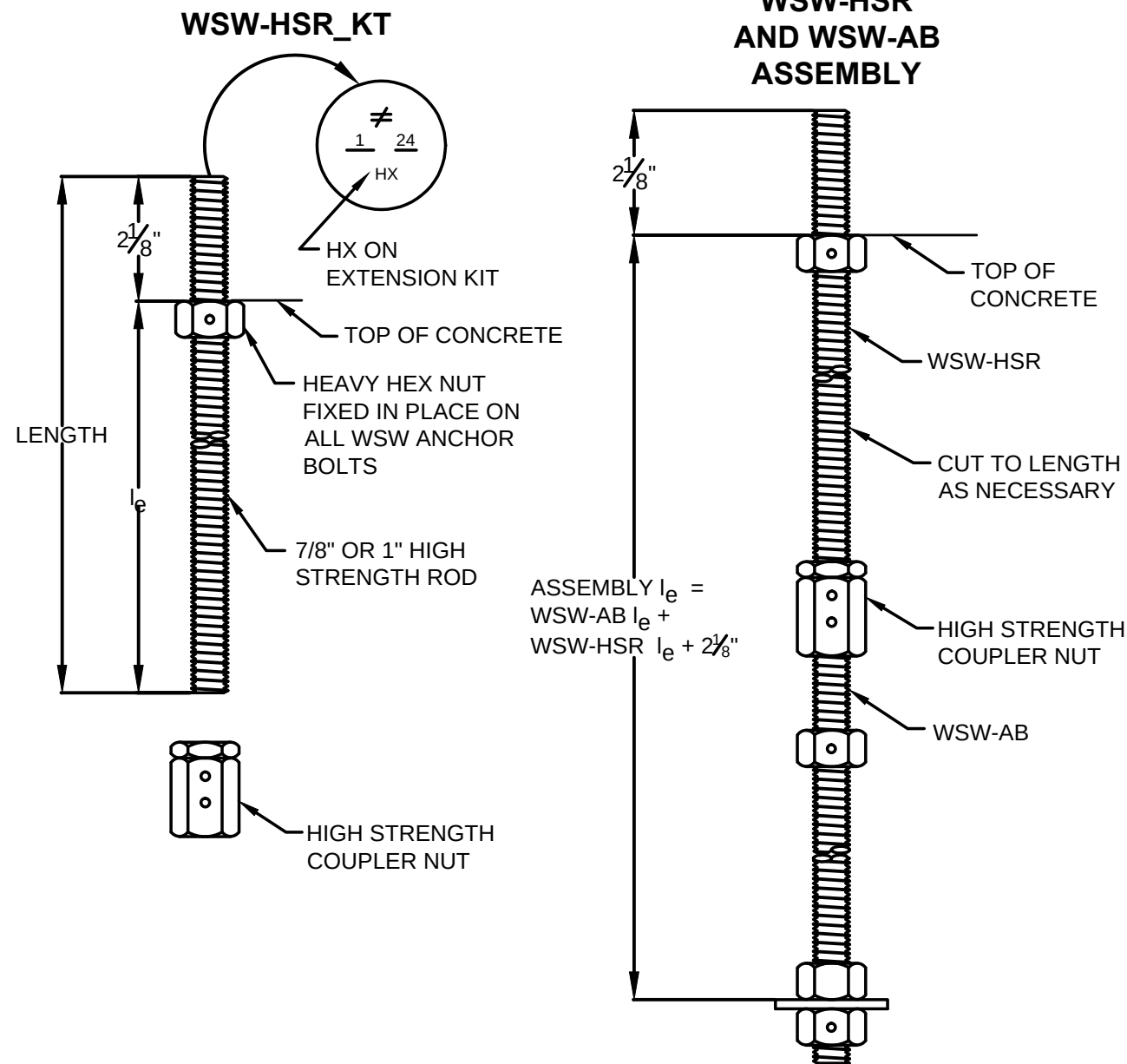
SLAB ON GRADE FOUNDATION

- NOTES:
- SEE 2/WSW1 FOR DIMENSIONS AND ADDITIONAL NOTES.
 - SEE 5/WSW1 FOR SHEAR REINFORCEMENT WHEN REQUIRED.
 - MAXIMUM H = $l_e - d_e$. SEE 3/WSW1 AND 4/WSW1 FOR l_e .

REGISTERED DESIGN PROFESSIONAL
IS PERMITTED TO MODIFY DETAILS
FOR SPECIFIC CONDITIONS.

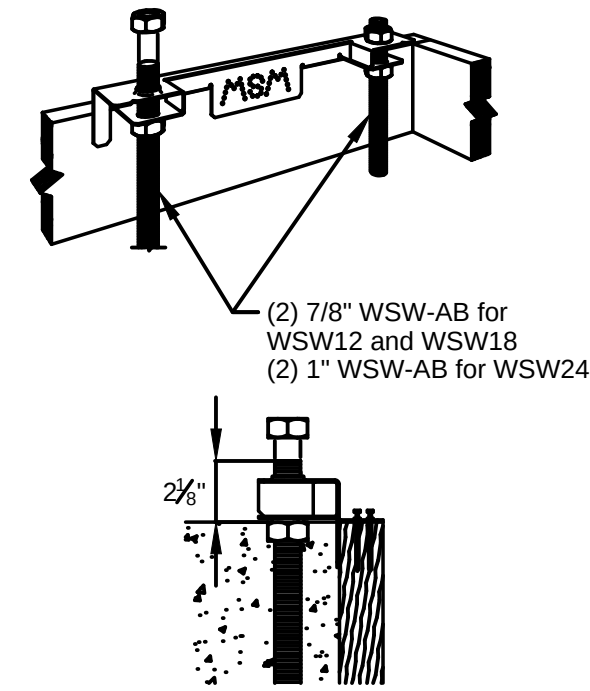


WSW PANEL MODEL	MODEL NO.	DIAMETER	LENGTH	l_e
WSW12 AND WSW18	WSW-AB7/8x24	7/8"	24"	20"
	WSW-AB7/8x24HS	7/8"	24"	20"
	WSW-AB7/8x30	7/8"	30"	26"
	WSW-AB7/8x30HS	7/8"	30"	26"
	WSW-AB7/8x36HS	7/8"	36"	32"
WSW24	WSW-AB1x24	1"	24"	20"
	WSW-AB1x24HS	1"	24"	20"
	WSW-AB1x30	1"	30"	26"
	WSW-AB1x30HS	1"	30"	26"
	WSW-AB1x36HS	1"	36"	32"

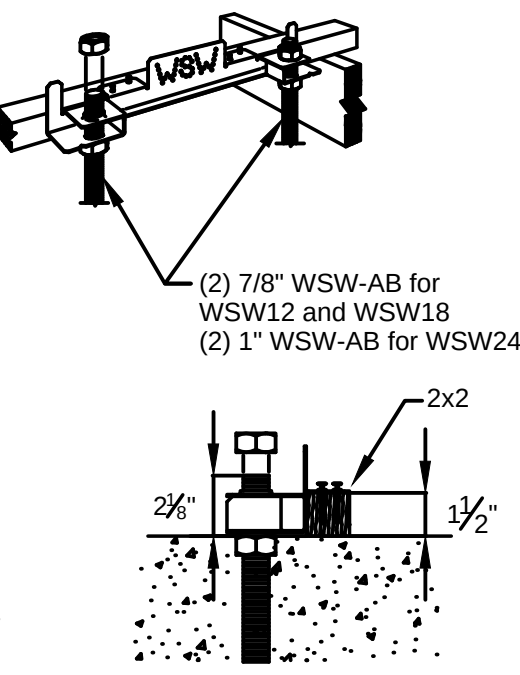


WSW PANEL MODEL	MODEL NO.	DIAMETER	LENGTH	l_e
WSW12 AND WSW18	WSW-HSR7/8x24KT	7/8"	24"	22"
	WSW-HSR7/8x36KT	7/8"	36"	34"
WSW24	WSW-HSR1x24KT	1"	24"	22"
	WSW-HSR1x36KT	1"	36"	34"

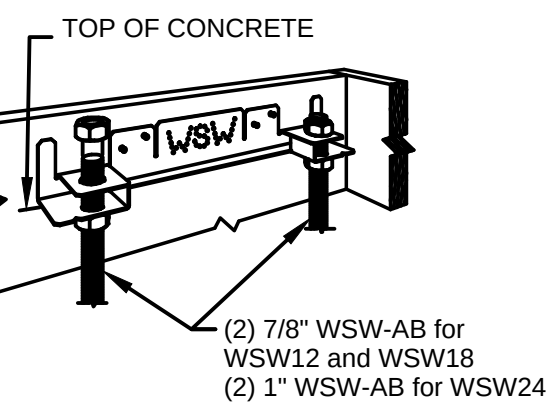
WSW-RT EXTERIOR INSTALLATION



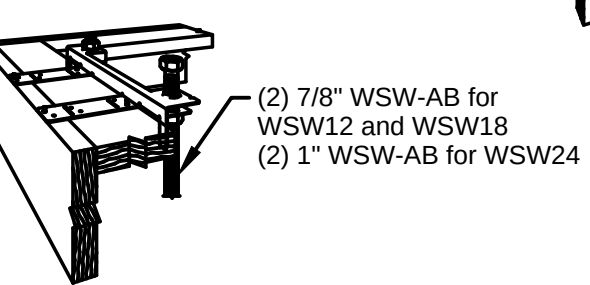
WSW-RT INTERIOR INSTALLATION



WSW-RTPF PANEL FORM INSTALLATION



WSW-RTBL BRICK LEDGE INSTALLATION

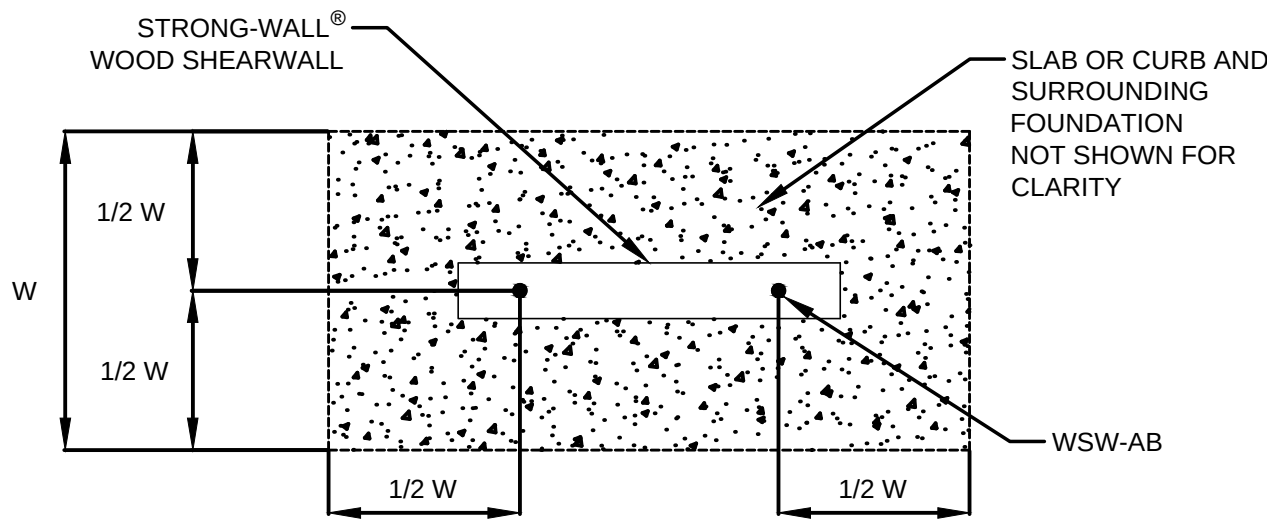


STRONG-WALL® WSW ANCHORAGE - TYPICAL SECTIONS

WSW ANCHOR BOLTS

WSW ANCHOR BOLT EXTENSION

WSW ANCHOR BOLT TEMPLATES



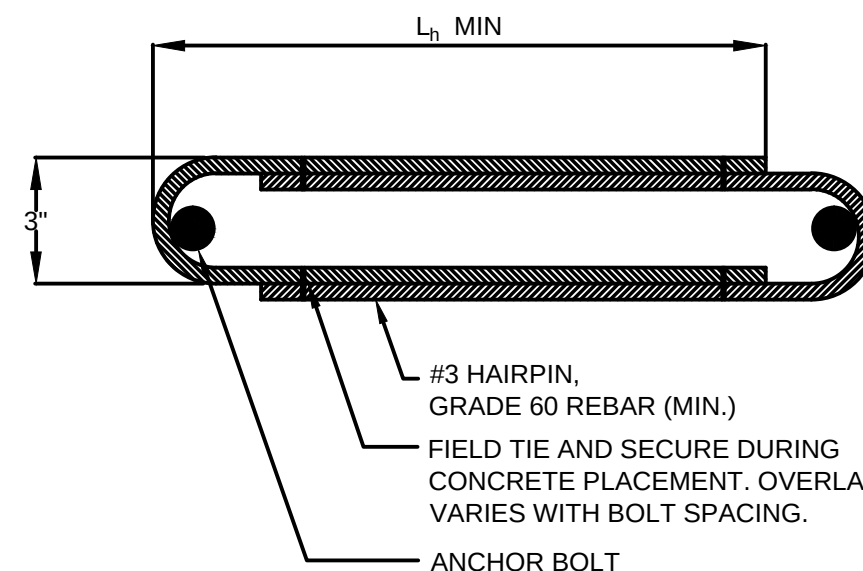
FOUNDATION PLAN VIEW

WSW ANCHORAGE SOLUTIONS FOR 2500 PSI CONCRETE								
DESIGN CRITERIA	CONCRETE CONDITION	ANCHOR STRENGTH	WSW-AB7/8 ANCHOR BOLT			WSW-AB1 ANCHOR BOLT		
			ASD ALLOWABLE TENSION (lb.)	W (in.)	d_e (in.)	ASD ALLOWABLE TENSION (lb.)	W (in.)	d_e (in.)
SEISMIC	CRACKED	STANDARD	11,900	27	9	16,100	33	11
			13,100	29	10	17,100	35	12
		HIGH STRENGTH	24,900	43	15	33,000	51	17
	UNCRACKED	STANDARD	27,100	46	16	35,300	54	18
			12,500	24	8	15,700	28	10
		HIGH STRENGTH	13,100	25	9	17,100	30	10
WIND	CRACKED	STANDARD	25,300	38	13	32,300	44	15
			27,100	40	14	35,300	47	16
			5,100	14	6	6,200	16	6
		HIGH STRENGTH	8,700	20	7	11,400	24	8
			13,100	27	9	17,100	32	11
			15,900	30	10	21,100	36	12
	UNCRACKED	STANDARD	18,400	33	11	27,300	42	14
			23,100	38	13	31,800	46	16
			27,100	42	14	35,300	50	17
		HIGH STRENGTH	5,000	12	6	6,400	14	6
			9,300	18	6	12,500	22	8
			13,100	23	8	17,100	28	10

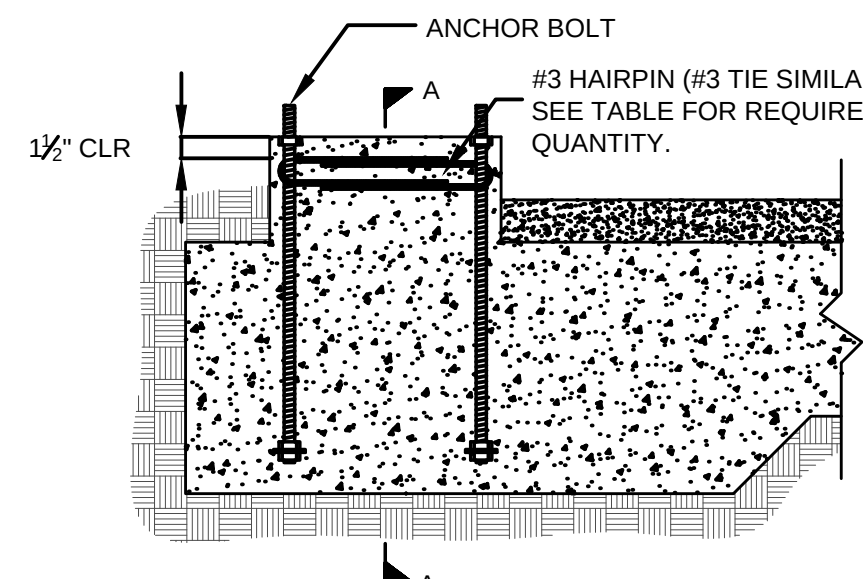
- NOTES:
- ANCHORAGE DESIGNS CONFORM TO ACI 318-11 APPENDIX D AND ACI 318-14 WITH NO SUPPLEMENTARY REINFORCEMENT FOR CRACKED OR UNCRACKED CONCRETE AS NOTED.
 - ANCHOR STRENGTH INDICATES REQUIRED GRADE OF WSW-AB ANCHOR BOLT. STANDARD (ASTM F1554 GRADE 36) OR HIGH STRENGTH (HS) (ASTM A449).
 - SEISMIC INDICATES SEISMIC DESIGN CATEGORY C - F. DETACHED 1 AND 2 FAMILY DWELLINGS IN SDC C MAY USE WIND ANCHORAGE SOLUTIONS. SEISMIC ANCHORAGE DESIGNS CONFORM TO ACI 318-11 SECTION D.3.3.4.3 AND ACI 318-14 SECTION 17.2.3.4.3.
 - WIND INCLUDES SEISMIC DESIGN CATEGORY A AND B AND DETACHED 1 AND 2 FAMILY DWELLINGS IN SDC C.
 - FOUNDATION DIMENSIONS ARE FOR ANCHORAGE ONLY. FOUNDATION DESIGN (SIZE AND REINFORCEMENT) BY OTHERS. THE REGISTERED DESIGN PROFESSIONAL MAY SPECIFY ALTERNATE EMBEDMENT, FOOTING SIZE OR ANCHOR BOLT.
 - REFER TO 1/WSW1 FOR d_e .

WSW ANCHORAGE SOLUTIONS FOR 3000 PSI CONCRETE								
DESIGN CRITERIA	CONCRETE CONDITION	ANCHOR STRENGTH	WSW-AB7/8 ANCHOR BOLT			WSW-AB1 ANCHOR BOLT		
			ASD ALLOWABLE TENSION (lb.)	W (in.)	d_e (in.)	ASD ALLOWABLE TENSION (lb.)	W (in.)	d_e (in.)
SEISMIC	CRACKED	STANDARD	12,300	26	9	16,000	31	11
			13,100	28	10	17,100	33	11
		HIGH STRENGTH	25,200	41	14	32,700	48	16
	UNCRACKED	STANDARD	27,100	43	15	35,300	51	17
			12,000	22	8	16,300	27	9
		HIGH STRENGTH	13,100	24	8	17,100	28	10
WIND	CRACKED	STANDARD	25,300	36	12	32,700	42	14
			27,100	38	13	35,300	44	15
			5,000	13	6	5,600	14	6
		HIGH STRENGTH	8,800	19	7	10,200	21	7
			13,100	25	9	17,100	30	10
			15,700	28	10	20,100	33	11
	UNCRACKED	STANDARD	19,200	32	11	25,300	38	13
			23,200	36	12	32,300	44	15
			27,100	40	14	35,300	47	16
		HIGH STRENGTH	5,500	12	6	6,200	13	6
			8,500	16	6	12,800	21	7
			13,100	22	8	17,100	26	9

WSW ANCHORAGE SOLUTIONS FOR 4500 PSI CONCRETE								
DESIGN CRITERIA	CONCRETE CONDITION	ANCHOR STRENGTH	WSW-AB7/8 ANCHOR BOLT			WSW-AB1 ANCHOR BOLT		
			ASD ALLOWABLE TENSION (lb.)	W (in.)	d_e (in.)	ASD ALLOWABLE TENSION (lb.)	W (in.)	d_e (in.)
SEISMIC	CRACKED	STANDARD	12,600	23	8	16,000	27	9
			13,100	24	8	17,100	29	10
		HIGH STRENGTH	24,800	36	12	32,100	42	14
	UNCRACKED	STANDARD	27,100	38	13	35,300	45	15
			12,700	20	7	15,700	23	8
		HIGH STRENGTH	13,100	21	7	17,100	25	9
WIND	CRACKED	STANDARD	24,600	31	11	32,500	37	13
			27,100	34	12	35,300	39	13
			5,400	12	6	6,800	14	6
		HIGH STRENGTH	8,300	16	6	11,600	20	7
			13,100	22	8	17,100	26	9
			15,300	24	8	21,400	30	10
	UNCRACKED	STANDARD	19,300	28	10	25,800	34	12
			23,600	32	11	31,000	38	13
			27,100	36	12	35,300	42	14
		HIGH STRENGTH	6,800	12	6	6,800	12	6
			9,400	15	6	12,400	18	6
			13,100	19	7	17,100	23	8

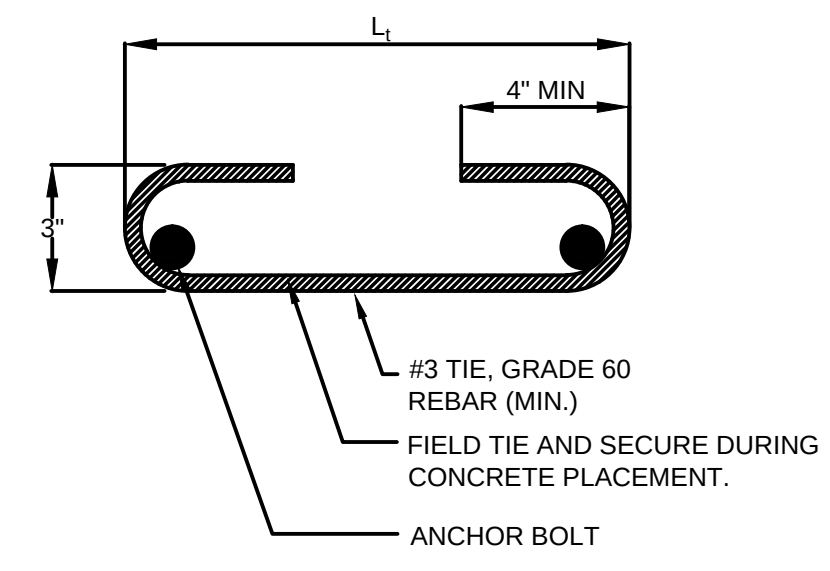


HAIRPIN SHEAR REINFORCEMENT

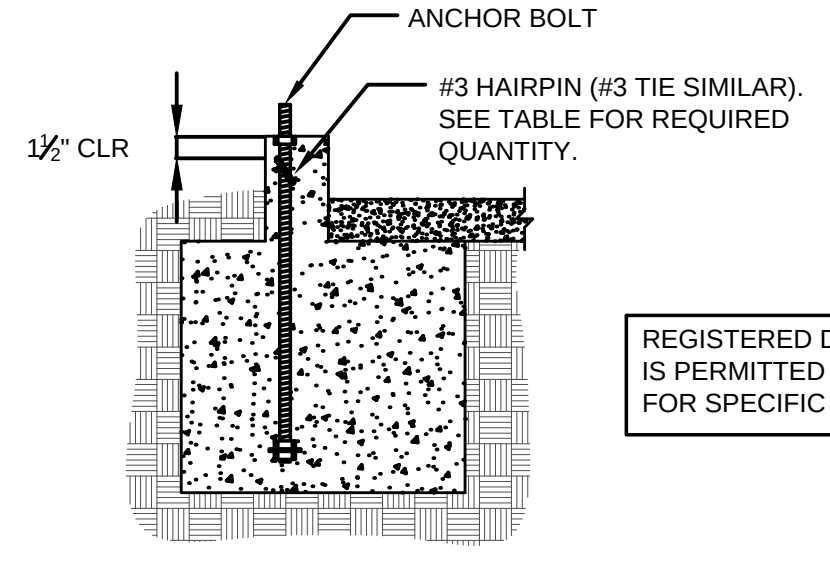


HAIRPIN INSTALLATION

(GARAGE CURB SHOWN. OTHER FOOTING TYPES SIMILAR.)



TIE SHEAR REINFORCEMENT



SECTION A-A

REGISTERED DESIGN PROFESSIONAL
IS PERMITTED TO MODIFY DETAILS
FOR SPECIFIC CONDITIONS.

STRONG-WALL® WOOD SHEARWALL SHEAR ANCHORAGE							
MODEL	L_1 OR L_2 (in.)	SEISMIC ³		WIND ⁴		ASD ALLOWABLE SHEAR LOAD, V (lb.) ⁶	
		SHEAR REINFORCEMENT	MINIMUM CURB/STEMWALL WIDTH (in.)	SHEAR REINFORCEMENT	MINIMUM CURB/STEMWALL WIDTH (in.)	UNCRACKED	CRACKED
WSW12	10 1/2	(1) #3 HAIRPIN	8 ⁵	SEE NOTE 6	6	1,035	740
WSW18	15	(1) #3 HAIRPIN	8 ⁵	(1) #3 HAIRPIN	6	HAIRPIN REINFORCEMENT ACHIEVES MAXIMUM ALLOWABLE SHEAR LOAD OF THE WSW	
WSW24	19	(2) #3 HAIRPINS	8 ⁵	(1) #3 HAIRPIN	6		

- NOTES:
- SHEAR ANCHORAGE DESIGNS CONFORM TO ACI 318-11 AND ACI 318-14 AND ASSUME MINIMUM 2,500 PSI CONCRETE.
 - SHEAR REINFORCEMENT IS NOT REQUIRED FOR INTERIOR FOUNDATION APPLICATIONS (PANEL INSTALLED AWAY FROM EDGE OF CONCRETE), OR BRACED WALL PANEL APPLICATIONS.
 - SEISMIC INDICATES SEISMIC DESIGN CATEGORY C THROUGH F. DETACHED 1 AND 2 FAMILY DWELLINGS IN SDC C MAY USE WIND ANCHORAGE SOLUTIONS.
 - WIND INCLUDES SEISMIC DESIGN CATEGORY A AND B AND DETACHED 1 AND 2 FAMILY DWELLINGS IN SDC C.
 - WHERE NOTED, MINIMUM CURB/STEMWALL WIDTH IS 6 INCHES WHEN STANDARD STRENGTH ANCHOR BOLT IS USED.
 - USE (1) #3 TIE FOR WSW12 WHEN PANEL DESIGN SHEAR FORCE EXCEEDS TABULATED ANCHORAGE ALLOWABLE SHEAR LOAD.
 - #4 GRADE 40 SHEAR REINFORCEMENT MAY BE SUBSTITUTED FOR WSW SHEAR ANCHORAGE SOLUTIONS.

STRONG-WALL® WOOD SHEARWALL TENSION ANCHORAGE SCHEDULE 2,500, 3,000 AND 4,500 PSI

STRONG-WALL® WSW SHEAR ANCHORAGE SCHEDULE AND DETAILS

NO.	DATE	REVISIONS
0	07/01/2016	FIRST RELEASE - 2015 B.C.

SIMPSON STRONG-TIE COMPANY, INC.
HOME OFFICE:
5956 W. LAS POSITAS BLVD.
PLEASANTON, CA 94588
TEL: (800) 999-5099



STRONG-WALL® WSW
ANCHORAGE DETAILS
ENGINEERED DESIGNS



NAME	DATE
07-01-2016	SCALE
N.T.S.	CHECKED
SHEET	WSW1
OF SHEETS	JOB NO.

STRONG-WALL® WOOD SHEARWALL MODELS		REGISTERED DESIGN PROFESSIONAL IS PERMITTED TO MODIFY DETAILS FOR SPECIFIC CONDITIONS. PLACE STRONG-WALL® WOOD SHEARWALL OVER THE ANCHOR BOLTS AND SECURE WITH WASHER AND HEX NUTS (PROVIDED). SNUG TIGHT FIT REQUIRED; DO NOT USE AN IMPACT WRENCH. • USE 1½" WRENCH FOR ¾" NUT • USE 1½" WRENCH FOR 1" NUT STRONG-WALL® WOOD SHEARWALL HEX NUT AND STRUCTURAL WASHER SEE SHEETS WSW1 AND WSW1.1 FOR ANCHORAGE SOLUTIONS																																																																																																																																																																	
<table><tr><th rowspan="2">MODEL NO.</th><th rowspan="2">W (in.)</th><th rowspan="2">H (in.)</th><th colspan="2">ANCHOR BOLTS</th><th rowspan="2">TOTAL WALL WEIGHT (lb.)</th></tr><tr><th>QUANTITY</th><th>DIA. (in.)</th></tr><tr><td>WSW12x7</td><td>12</td><td>78</td><td>2</td><td>7/8</td><td>100</td></tr><tr><td>WSW18x7</td><td>18</td><td>78</td><td>2</td><td>7/8</td><td>145</td></tr><tr><td>WSW12x7.5</td><td>12</td><td>85 1/2</td><td>2</td><td>7/8</td><td>110</td></tr><tr><td>WSW18x7.5</td><td>18</td><td>85 1/2</td><td>2</td><td>7/8</td><td>155</td></tr><tr><td>WSW12x8</td><td>12</td><td>93 1/4</td><td>2</td><td>7/8</td><td>115</td></tr><tr><td>WSW18x8</td><td>18</td><td>93 1/4</td><td>2</td><td>7/8</td><td>165</td></tr><tr><td>WSW24x8</td><td>24</td><td>93 1/4</td><td>2</td><td>1</td><td>225</td></tr><tr><td>WSW12x9</td><td>12</td><td>105 1/4</td><td>2</td><td>7/8</td><td>130</td></tr><tr><td>WSW18x9</td><td>18</td><td>105 1/4</td><td>2</td><td>7/8</td><td>185</td></tr><tr><td>WSW24x9</td><td>24</td><td>105 1/4</td><td>2</td><td>1</td><td>245</td></tr><tr><td>WSW12x10</td><td>12</td><td>117 1/4</td><td>2</td><td>7/8</td><td>140</td></tr><tr><td>WSW18x10</td><td>18</td><td>117 1/4</td><td>2</td><td>7/8</td><td>205</td></tr><tr><td>WSW24x10</td><td>24</td><td>117 1/4</td><td>2</td><td>1</td><td>270</td></tr><tr><td>WSW12x11</td><td>12</td><td>129 1/4</td><td>2</td><td>7/8</td><td>150</td></tr><tr><td>WSW18x11</td><td>18</td><td>129 1/4</td><td>2</td><td>7/8</td><td>220</td></tr><tr><td>WSW24x11</td><td>24</td><td>129 1/4</td><td>2</td><td>1</td><td>295</td></tr><tr><td>WSW12x12</td><td>12</td><td>141 1/4</td><td>2</td><td>7/8</td><td>165</td></tr><tr><td>WSW18x12</td><td>18</td><td>141 1/4</td><td>2</td><td>7/8</td><td>240</td></tr><tr><td>WSW24x12</td><td>24</td><td>141 1/4</td><td>2</td><td>1</td><td>320</td></tr><tr><td>WSW18x13</td><td>18</td><td>153 1/4</td><td>2</td><td>7/8</td><td>255</td></tr><tr><td>WSW24x13</td><td>24</td><td>153 1/4</td><td>2</td><td>1</td><td>345</td></tr><tr><td>WSW24x14</td><td>24</td><td>168</td><td>2</td><td>1</td><td>375</td></tr><tr><td>WSW24x16</td><td>24</td><td>192</td><td>2</td><td>1</td><td>425</td></tr><tr><td>WSW18x20</td><td>18</td><td>240</td><td>2</td><td>7/8</td><td>385</td></tr><tr><td>WSW24x20</td><td>24</td><td>240</td><td>2</td><td>1</td><td>520</td></tr></table> NOTES: 1. FOR HEIGHTS NOT LISTED, ORDER THE NEXT TALLEST PANEL AND TRIM TO FIT. MINIMUM TRIMMED HEIGHT FOR ALL PANELS IS 74½". 2. ALL PANELS COME WITH TWO PRE-ATTACHED HOLDDOWNS, TWO STANDARD HEX NUTS, TWO STRUCTURAL WASHERS, TWO WSW-TOW PLATES AND INSTALLATION INSTRUCTIONS. 3. ALL PANELS ARE 3½" THICK.				MODEL NO.	W (in.)	H (in.)	ANCHOR BOLTS		TOTAL WALL WEIGHT (lb.)	QUANTITY	DIA. (in.)	WSW12x7	12	78	2	7/8	100	WSW18x7	18	78	2	7/8	145	WSW12x7.5	12	85 1/2	2	7/8	110	WSW18x7.5	18	85 1/2	2	7/8	155	WSW12x8	12	93 1/4	2	7/8	115	WSW18x8	18	93 1/4	2	7/8	165	WSW24x8	24	93 1/4	2	1	225	WSW12x9	12	105 1/4	2	7/8	130	WSW18x9	18	105 1/4	2	7/8	185	WSW24x9	24	105 1/4	2	1	245	WSW12x10	12	117 1/4	2	7/8	140	WSW18x10	18	117 1/4	2	7/8	205	WSW24x10	24	117 1/4	2	1	270	WSW12x11	12	129 1/4	2	7/8	150	WSW18x11	18	129 1/4	2	7/8	220	WSW24x11	24	129 1/4	2	1	295	WSW12x12	12	141 1/4	2	7/8	165	WSW18x12	18	141 1/4	2	7/8	240	WSW24x12	24	141 1/4	2	1	320	WSW18x13	18	153 1/4	2	7/8	255	WSW24x13	24	153 1/4	2	1	345	WSW24x14	24	168	2	1	375	WSW24x16	24	192	2	1	425	WSW18x20	18	240	2	7/8	385	WSW24x20	24	240	2	1	520		
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STRONG-WALL® WSW MODELS		1	STANDARD INSTALLATION BASE CONNECTION		4	STANDARD TOP CONNECTION		6	TOP OF WALL HEIGHT ADJUSTMENTS		9																																																																																																																																																								
									EDGE DRILL ZONE MIDDLE ½ OF PANEL THICKNESS HOLES • MAX. THREE HOLES IN FACE AND THREE IN EDGE. • ¾"-DIAMETER HOLES, MAX. • 6" O.C., MIN. NO EDGE HOLES ALLOWED IN LOWER 26" OF PANEL FACE DRILL ZONE MAINTAIN 1½" MIN. EDGE DISTANCE FROM CHASE AND OUTSIDE EDGE, TYPICAL. NO FACE HOLES ALLOWED IN LOWER 40" OF PANEL NO HOLES ALLOWED IN TOP 8" OF PANEL FACE DRILL ZONE CENTER 4½" OF PANEL FACE AS SHOWN 12" ABOVE EXISTING HOLE, MIN. 16" 24½" ALLOWABLE SMALL HOLES FACE AND EDGE DRILL ZONES ALLOWABLE LARGE HOLES IN ADDITION TO ALLOWABLE SMALL HOLES																																																																																																																																																										
SINGLE STORY WSW ON CONCRETE		2	WOOD FLOOR SYSTEM BASE CONNECTION		5	ALTERNATE TOP CONNECTION		7	TRIM ZONE AND ALLOWABLE HOLES		10																																																																																																																																																								
									1. STRONG-WALL WOOD SHEARWALL IS MANUFACTURED AND TRADEMARKED BY "SIMPSON STRONG-TIE COMPANY INC."HOME OFFICE: 5956 W. LAS POSITAS BLVD., PLEASANTON, CA 94588 TEL: (800) 999-5099, FAX: (925) 847-1597. "SIMPSON STRONG-TIE COMPANY INC." IS AN ISO 9001-2008 REGISTERED COMPANY. 2. USE OF THIS PRODUCT IS SUBJECT TO THE APPROVAL OF THE LOCAL BUILDING DEPARTMENT. 3. THIS PRODUCT IS PART OF THE OVERALL LATERAL FORCE RESISTING SYSTEM OF THE STRUCTURE. DESIGN OF THE BUILDING'S LATERAL FORCE RESISTING SYSTEM, INCLUDING THE LOAD PATH TO TRANSFER LATERAL FORCES FROM THE STRUCTURE TO THE GROUND, IS THE RESPONSIBILITY OF THE DESIGNER. 4. ENGINEER OF RECORD IS PERMITTED TO MODIFY DETAILS FOR SPECIFIC CONDITIONS. 5. THE CONTRACTOR SHALL VERIFY ALL DIMENSIONS, CONDITIONS, ELEVATIONS, ETC. PRIOR TO INSTALLATION OF ANY COMPONENTS FOR THE STRONG-WALL SB SYSTEM. IF ANY DISCREPANCIES ARE FOUND, THEY SHALL BE BROUGHT TO THE ATTENTION OF THE DESIGNER FOR CLARIFICATION PRIOR TO CONSTRUCTION. 6. INSTALLATION OF PRODUCT SHALL BE DONE IN CONFORMANCE TO THESE DRAWINGS. THE PERFORMANCE OF MODIFIED PRODUCTS OR ALTERED INSTALLATION PROCEDURES ARE THE SOLE RESPONSIBILITY OF THE DESIGNER. 7. SIMPSON STRONG-TIE COMPANY INC. RESERVES THE RIGHT TO CHANGE SPECIFICATIONS, DESIGNS, AND MODELS WITHOUT NOTICE OR LIABILITY FOR SUCH CHANGES. 8. ALL HARDWARE CALLED OUT IS SIMPSON STRONG-TIE. 9. SEE ICC-ES ESR-2652 OR CITY OF LOS ANGELES RR25730 AS APPLICABLE FOR ADDITIONAL INFORMATION.																																																																																																																																																										
ALTERNATE WSW GARAGE FRONT OPTIONS			3	RAKE WALL		8	NOTES			11																																																																																																																																																									

REVISIONS

DATE

NO.

07/01/2016

0

STRONG-WALL WSW

FRAMING DETAILS

ENGINEERED DESIGNS

NAME

DATE

SCALE

CHECKED

SHEET

WSW2

OF SHEETS

JOB NO.

07-01-2016

N.T.S.

11

SIMPSON

Strong-Tie

THERE IS NO EQUAL

SIMPSON

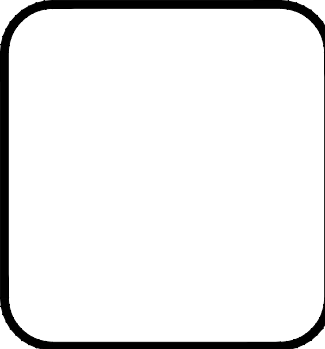
Strong-Tie

THERE IS NO EQUAL

SIMPSON STRONG-TIE COMPANY, INC.

HOME OFFICE: 5956 W. LAS POSITAS BLVD. PLEASANTON, CA 94588 TEL: (800) 999-5099

NO.	DATE	REVISIONS	
		NO.	DESCRIPTION
0	07/01/2016		FIRST RELEASE - 2015 EC



STRONG-TIE COMPANY, INC.

HOME OFFICE:
5956 W. LAS POSITAS BLVD.
PLEASANTON, CA 94588
TEL: (800) 999-5099

STRONG-WALL WSW

FRAMING DETAILS

ENGINEERED DESIGNS

NAME	
DATE	07-01-2016
SCALE	N.T.S.
CHECKED	
SHEET	WSW2
OF	SHEETS
JOB NO.	

GENERAL NOTES

- THE FOLLOWING NOTES, DETAILS, SCHEDULES & SPECIFICATIONS SHALL APPLY TO ALL PHASES OF THIS PROJECT UNLESS SPECIFICALLY OTHERWISE NOTED (UON), NOTES AND DETAILS ON THE STRUCTURAL PLANS SHALL TAKE PRECEDENCE OVER GENERAL NOTES AND TYPICAL DETAILS. WHERE NO DETAILS ARE GIVEN, CONSTRUCTION SHALL BE AS SHOWN FOR SIMILAR WORK.
- ALL DRAWINGS ARE CONSIDERED TO BE PART OF THE CONTRACT DOCUMENTS. THE CONTRACTOR SHALL BE RESPONSIBLE FOR THE REVIEW AND COORDINATION OF ALL DRAWINGS AND SPECIFICATIONS PRIOR TO THE START OF CONSTRUCTION, WITH ANY DISCREPANCIES BEING BROUGHT TO THE ATTENTION OF THE ENGINEER PRIOR TO THE START OF CONSTRUCTION SO THAT A CLARIFICATION CAN BE ISSUED WORK PERFORMED IN CONFLICT WITH THE CONTRACT DOCUMENTS OR ANY APPLICABLE CODE REQUIREMENTS SHALL BE CORRECTED BY THE CONTRACTOR AT NO EXPENSE TO THE OWNER OR ENGINEER.
- REFER TO THE ARCHITECTURAL PLANS FOR THE FOLLOWING:
 - DIMENSIONS
 - SIZE AND LOCATION OF ALL INTERIOR AND EXTERIOR WALL LOCATIONS.
 - SIZE AND LOCATION OF ALL FLOOR, ROOF AND WALL OPENINGS
 - SIZE AND LOCATION OF ALL DRAINS, SLOPES, DEPRESSIONS, STEPS, ETC.
 - SPECIFICATION OF ALL FINISHES & WATERPROOFING
 - ALL OTHER NON-STRUCTURAL ELEMENTS
- REFER TO THE MECHANICAL, ELECTRICAL AND PLUMBING PLANS FOR THE FOLLOWING:
 - SIZE AND LOCATION OF ALL EQUIPMENT
 - PIPE RUNS, SLEEVES, HANGERS AND TRENCHES
 - ALL OTHER MECHANICAL, ELECTRICAL OR PLUMBING RELATED ELEMENTS
- DO NOT SCALE STRUCTURAL PLANS. CONTRACTOR SHALL USE ALL WRITTEN DIMENSIONS ON ARCHITECTURAL PLANS.
- CONSTRUCTION MATERIALS SHALL BE UNIFORMLY SPREAD OUT & PLACED ON FLOOR OR ROOF SO AS TO NOT OVERLOAD THE FRAMING. LOAD SHALL NOT EXCEED THE DESIGN LIVE LOAD PER SQUARE FOOT. IT IS THE CONTRACTOR'S RESPONSIBILITY TO PROVIDE ADEQUATE SHORING AND BRACING AS REQUIRED.
- SPECIFICATIONS AND DETAILING OF ALL WATERPROOFING AND DRAINAGE ITEMS, WHILE SOMETIMES SHOWN ON THE STRUCTURAL PLANS FOR GENERAL INFORMATION PURPOSES ONLY, ARE SOLELY THE DESIGN RESPONSIBILITY OF OTHERS.
- THE ENGINEER WILL NOT BE RESPONSIBLE FOR AND WILL NOT HAVE CONTROL OR CHARGE OF CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE CONSTRUCTION UNLESSATED BY THESE PLANS. IT SHOULD BE UNDERSTOOD THAT THE CONTRACTOR OR HIS/HER AGENTS(SHALL SUPERVISE AND DIRECT ALL WORK AND BE COMPLETELY RESPONSIBLE FOR ALL CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES, PROCEDURES AND CONDITIONS ON THE JOB SITE, INCLUDING SAFETY OF ALL PERSONS AND PROPERTY DURING THE ENTIRE PERIOD OF CONSTRUCTION. PERIODIC OBSERVATIONS BY THE ENGINEER, HIS STAFF OR REPRESENTATIVES ARE NOT INTENDED TO INCLUDE VERIFICATION OF DIMENSIONS OR REVIEW THE ADEQUACY OF THE CONTRACTOR'S SAFETY MEASURES ON OR NEAR THE CONSTRUCTION SITE.
- MODIFICATIONS OF THE PLANS, NOTES, DETAILS AND SPECIFICATIONS SHALL NOT BE PERMITTED WITHOUT PRIOR APPROVAL FROM THE ENGINEER.
- ALL WORKSMANSHIP SHALL CONFORM TO THE BEST PRACTICE PREVAILING IN THE VARIOUS TRADES PERFORMING THE WORK. THE CONTRACTOR SHALL BE RESPONSIBLE FOR COORDINATING THE WORK OF ALL TRADES.
- IT IS THE CONTRACTOR'S RESPONSIBILITY TO ENSURE THAT ONLY APPROVED STRUCTURAL PLANS ARE USED DURING THE COURSE OF CONSTRUCTION. THE USE OF UNAPPROVED DOCUMENTS SHALL BE AT THE CONTRACTOR'S OWN RISK. CORRECTIONS OF ALL WORK BASED ON SUCH DOCUMENTS SHALL BE PERFORMED AT THE CONTRACTOR'S EXPENSE.
- THESE PLANS AND SPECIFICATIONS REPRESENT THE STRUCTURAL DESIGN ONLY. NO INFORMATION NOR WARRANTY IS PROVIDED FOR THE WORK OR ANY OTHER CONSULTANT (ARCHITECT, MECHANICAL, ELECTRICAL, ETC.) THIS INCLUDES, BUT IS NOT LIMITED TO, WATERPROOFING, DRAINAGE, VENTILATION, ACCESSIBILITY, OR DIMENSIONS.

FOUNDATIONS

- REFER TO STRUCTURAL DESIGN PARAMETERS SECTION ON SHEET S-1.1 FOR ALL SOIL DESIGN VALUES USED IN CALCULATIONS.
- SOILS VALUES PER GEOLOGICAL/GEOTECHNICAL REPORT REFERENCED ON FOUNDATION PLAN. THIS REPORT AND ALL RECOMMENDATIONS CONTAINED THEREIN ARE TO BE CONSIDERED A PART OF THESE PLANS.
- IT IS THE CONTRACTOR'S RESPONSIBILITY TO OBTAIN A COPY OF THE SOILS REPORT FROM THE OWNER. A COPY OF THE SOILS REPORT SHALL BE ON THE JOB SITE DURING THE COURSE OF CONSTRUCTION.
- UNEXPECTED SOIL CONDITIONS, ALLOWABLE VALUES AND SUBSEQUENT FOUNDATION DESIGNS ARE BASED ON SOIL CONDITIONS WHICH ARE SHOWN BY TEST BORINGS. ACTUAL SOIL CONDITIONS WHICH DEVIATE APPRECIABLY FROM THAT SHOWN IN THE TEST BORINGS SHALL BE REPORTED TO THE ENGINEER IMMEDIATELY.
- ALL COMPACTION, FILL, BACKFILLING AND SITE PREPARATION SHALL BE PERFORMED IN ACCORDANCE WITH PROJECT SOILS REPORT OR CBC APPENDIX CHAPTER J. ALL SUCH WORK SHALL BE PERFORMED UNDER THE SUPERVISION OF THE PROJECT SOILS ENGINEER.
- EXCAVATE TO REQUIRED DEPTHS AND DIMENSIONS (AS INDICATED IN THE DRAWINGS), CUT SQUARE AND SMOOTH WITH FIRM LEVEL BOTTOMS. CARE SHALL BE TAKEN NOT TO OVER-EXCAVATE FOUNDATION AT LOWER ELEVATION AND PREVENT DISTURBANCE OF SOILS AROUND HIGH ELEVATION.
- FOUNDATIONS SHALL BE POURED IN NEAT EXCAVATIONS.
- EXCAVATE ALL FOUNDATIONS TO REQUIRED DEPTHS UNTIL COMPACTED FILL (AS PER PLANS AND DETAILS) AND AS VERIFIED BY THE BUILDING OFFICIAL AND/OR SOILS ENGINEER.
- ALL FOUNDATIONS SHALL BE INSPECTED AND APPROVED BY THE APPROPRIATE BUILDING OFFICIAL AND/OR A REPRESENTATIVE OF THE SOILS ENGINEER PRIOR TO FORMING AND PLACEMENT OF REINFORCING OR CONCRETE.
- FOUNDATIONS SHALL NOT BE POURED UNTIL ALL REQUIRED REINFORCING STEEL, FRAMING HARDWARE, SLEEVES, PIPES, ETC. AND FORMWORK IS PROPERLY PLACED AND INSPECTED BY THE APPROPRIATE BUILDING OFFICIAL/INSPECTOR(S).
- IT IS THE RESPONSIBILITY OF THE CONTRACTOR IN CHARGE OF FRAMING TO PROPERLY POSITION ALL HOLLOWBOL BOLTS, ANCHOR BOLTS, COLUMN BARS AND ALL OTHER CAST-IN-PLACE HARDWARE. REFER TO TYPICAL DETAILS. ALL HARDWARE TO BE SECURED PRIOR TO FOUNDATION INSPECTIONS.
- THE SIDES AND BOTTOMS OF DEEP EXCAVATIONS MUST BE MOISTENED JUST PRIOR TO PLACING CONCRETE; CONVERSELY, DE WATER FOOTINGS AS REQUIRED TO REMOVE STANDING WATER AND TO MAINTAIN OPTIMUM WORKING CONDITIONS.
- THE CONTRACTOR SHALL BE SOLELY RESPONSIBLE FOR ALL EXCAVATION PROCEDURES INCLUDING LAGGING, SHORING, AND THE PROTECTION OF ADJACENT PROPERTY, STRUCTURES, STREETS, AND UTILITIES IN ACCORDANCE WITH ALL FEDERAL, STATE AND LOCAL SAFETY ORDINANCES. THE CONTRACTOR SHALL PROVIDE FOR THE DESIGN AND INSTALLATION OF ALL CRIBBING, BRACING AND SHORING REQUIRED.

CONCRETE

- ALL CONCRETE SHALL HAVE:
 - AN ULTIMATE COMPRESSIVE STRENGTH (F'c) OF 2500 PSI AT 28 DAYS (UON).
 - A MAXIMUM SLUMP OF 5" AT POINT OF PLACEMENT FOR SLABS AND FOOTINGS. CAISSONS SHALL HAVE A 4" TO 6" SLUMP AT "DRY" HOLES AND A 6" - 8" SLUMP AT "WET" HOLES.
 - A W/C RATIO OF 0.55 OR LESS FOR ALL SLABS, WALLS, AND COLUMNS, AND 0.60 OR LESS FOR ALL FOUNDATIONS.
 - A NORMAL DRY-WEIGHT DENSITY (UON)
- SPECIAL INSULATION IS NOT REQUIRED, EXCEPT WHERE SPECIFIED HEREIN, ON THE STRUCTURAL PLANS, OR BY THE BUILDING DEPARTMENT. AS A MINIMUM, SPECIAL INSULATION IS ALWAYS REQUIRED ON:
 - STRUCTURAL SLABS, FLAT PLATES
 - WALLS, COLUMNS, BEAMS
 - PILES, CAISSONS
 - WELDING OF REINFORCEMENT; INSTALLATION OF MECHANICAL BAR SPICE DEVICES, EPOXY APPLICATIONSWHEN REQUIRED OR SPECIFIED, SPECIAL INSPECTION SERVICES SHALL CONFORM TO CBC CHAPTER 17 AND SHALL BE PROVIDED BY AN ICC CERTIFIED INSPECTOR OR BUILDING DEPARTMENT APPROVED ENGINEER.
- THE BUILDING DEPARTMENT RESERVES THE RIGHT TO WAIVE OR REQUIRE THE SPECIAL INSPECTION REQUIREMENTS (CBC 1704.1 AND 1704.4). NOTHING IN THESE PLANS WAIVES THE BUILDING DEPARTMENT RIGHT TO REQUIRE SPECIAL INSPECTION ON AT ANY POINT AND ON ANY MATERIAL.
- TESTING OF MATERIALS USED IN CONCRETE CONSTRUCTION MUST BE PERFORMED AS NOTED ON STRUCTURAL PLANS OR AT THE REQUEST OF THE BUILDING DEPARTMENT TO DETERMINE IF MATERIALS ARE QUALITY SPECIFIED. TESTS OF MATERIALS AND OF CONCRETE SHALL BE MADE BY AN APPROVED AGENCY AND AT THE EXPENSE OF THE OWNER. SUCH TESTS SHALL BE MADE IN ACCORDANCE WITH THE STANDARDS LISTED IN CBC TABLE 1705.3.
- WHEN TESTING OF CONCRETE IS REQUIRED, FOUR (4) TEST CYLINDERS SHALL BE TAKEN FROM EACH 10 YARDS, OR FRACTION THEREOF, POURED IN ANY ONE DAY. ONE (1) CYLINDER SHALL BE TESTED AT SEVEN (7) DAYS, TWO (2) AT 28 DAYS, ONE (1) SHALL BE HELD IN RESERVE. IF CONTRACTOR ELECTS TO HAVE ADDITIONAL TESTS PERFORMED FOR "EARLY-BREAK" RESULTS, ADDITIONAL TEST CYLINDERS MUST BE TAKEN, AT NO TIME SHALL THE CONTRACTOR INSTRUCT THE TESTING AGENCY TO PERFORM TESTS ON A SCHEDULE DIFFERENT THAN ABOVE WITHOUT THE PRIOR AUTHORIZATION OF THE ENGINEER.
- CONTRACTOR IS RESPONSIBLE FOR COMPLYING WITH APPLICABLE TESTING REQUIREMENTS OF THE BUILDING DEPARTMENT. COPIES OF ALL TEST REPORTS SHALL BE PROVIDED TO ENGINEER AND BUILDING DEPARTMENT FOR REVIEW IN A TIMELY MANNER.
- THE CONTRACTOR SHALL REMOVE AND REPLACE ANY CONCRETE WHICH FAILS TO ATTAIN SPECIFIED 28 DAY COMPRESSIVE STRENGTH IF SO DIRECTED BY THE ENGINEER. ANY DEFECTS IN THE HARDENED CONCRETE SHALL BE REPAIRED TO THE SATISFACTION OF THE ENGINEER AND/OR

- ARCHITECT OR THE HARDENED CONCRETE SHALL BE REPLACED AT THE CONTRACTOR'S EXPENSE.
- ALL CONCRETE WORK SHALL CONFORM WITH CBC CHAPTER 19.
- ALL CEMENT SHALL BE PORTLAND CEMENT TYPE I OR II AND SHALL CONFORM TO ASTM C 150.
- ALL AGGREGATES SHALL CONFORM TO ASTM C33. MAXIMUM AGGREGATE SIZES:
 - FOOTINGS - 1-1/2"
 - SLABS, WALLS, COLUMNS - 3/4"
- WHERE NOT SPECIFICALLY DETAILED, THE MINIMUM CONCRETE COVER ON REINFORCING STEEL SHALL BE:
 - PERMANENTLY EXPOSED TO EARTH OR WEATHER
 - CAST AGAINST EARTH - 3"
 - CAST AGAINST FORMS - 2"
 - NOT EXPOSED TO EARTH OR WEATHER
 - SLAB WALLS, COLUMNS - 3/4"
 - BEAMS, GIRDERS, COLUMNS - 1-1/2"
- MINIMUM LAP SPICE LENGTH FOR ALL REINFORCING STEEL SHALL BE 48 BAR DIAMETER (UON) ON PLAN. STRUCTURAL PLANS AND DETAILS. ALL LAP SPICES TO BE STAGGERED.
- ALL ANCHOR BOLTS USED IN CONCRETE CONSTRUCTION SHALL HAVE A MINIMUM TOTAL EMBEDMENT AS FOLLOWS (UON):
 - 5/8" DIA. - 7"
 - 3/4" DIA. - 8"
 - 7/8" DIA. - 9"
 - 1" DIA. - 10"OVERALL LENGTH OF ANCHOR BOLTS SHALL BE COORDINATED WITH SLL PLATE REQUIREMENTS AS INDICATED ELSEWHERE IN THESE SPECIFICATIONS. ALL ANCHOR BOLTS IN CONTACT WITH PRESERVATIVE-TREATED WOOD SHALL BE HOT DIPPED ZINC GALVANIZED OR STAINLESS STEEL.
- ALL REINFORCING STEEL, ANCHOR BOLTS, DOWELS, INSERTS, AND ANY OTHER HARDWARE TO BE CAST IN CONCRETE SHALL BE WELL SECURED IN POSITION PRIOR TO FOUNDATION INSPECTION. ALL HARDWARE TO BE INSTALLED IN ACCORDANCE WITH RESPECTIVE MANUFACTURER'S SPECIFICATIONS. REFER TO ARCHITECTURAL AND STRUCTURAL PLANS FOR LOCATIONS OF EMBEDDED ITEMS.
- LOCATIONS OF ALL CONSTRUCTION JOINTS, OTHER THAN SPECIFIED ON THE STRUCTURAL PLANS, SHALL BE APPROVED BY THE ARCHITECT AND ENGINEER PRIOR TO FORMING. CONSTRUCTION JOINTS SHALL BE THOROUGHLY AIR AND WATER CLEANED AND HEAVILY ROUGHENED SO AS TO EXPOSE COARSE AGGREGATES. ALL SURFACES TO RECEIVE FRESH CONCRETE SHALL BE NOTED AND CONTINUOUSLY WET AT LEAST THREE (3) HOURS IN ADVANCE OF CONCRETE PLACEMENT.
- UNLESS SPECIFICALLY DETAILED OR OTHERWISE NOTED, CONSTRUCTION AND CONTROL JOINTS SHALL BE PROVIDED IN ALL CONCRETE SLABS ON-GRADE. JOINTS SHALL BE LOCATED SUCH THAT THE AREA DOES NOT EXCEED 400 SQ. FEET.
- THE ARCHITECT, ENGINEER AND APPROPRIATE INSPECTORS SHALL BE NOTIFIED IN A TIMELY MANNER FOR A REINFORCEMENT INSPECTION PRIOR TO THE PLACEMENT OF ANY CONCRETE.
- THE CONTRACTOR SHALL OBTAIN APPROVAL FROM THE ARCHITECT AND THE ENGINEER PRIOR TO PLACING SLEEVES, PIPES, DUCTS, CHASES, CORING AND OPENING ON OR THROUGH STRUCTURAL CONCRETE BEAMS, WALLS, FLOORS, AND ROOF SLABS UNLESS SPECIFICALLY DETAILED OR NOTED AND CONTINUOUSLY WET AT LEAST THREE (3) HOURS IN ADVANCE OF CONCRETE PLACEMENT.
- REMOVE FORM WORK IN ACCORDANCE WITH THE FOLLOWING SCHEDULE:
 - FORMS AT SLAB EDGE: 1 DAY
 - SIDE FORMS AT FOOTINGS: 2 DAYS
 - ALL OTHER VERTICAL FORMS: 3 DAYS
 - BEAMS, COLUMNS, GIRDERS: 15 DAYS
 - ELEVATED SLABS: 28 DAYSENGINEER RESERVES THE RIGHT TO MODIFY REMOVAL SCHEDULE ABOVE BASED ON FIELD OBSERVATIONS. CONCRETE SHALL BE 40 BAR DIAMETERS MINIMUM (UON). ADJACENT BAR LAPS SHALL BE STAGGERED 3'-0" MINIMUM. HOOKS SHALL BE 16 BAR DIAMETERS (UON).
- REINFORCING BARS TO HAVE GROUT COVERAGE OF AT LEAST ONE BAR DIAMETER (UON) MINIMUM FROM INSIDE FACE OF SHELL, HOWEVER THE CLEAR DISTANCE FROM OUTSIDE FACE OF MASONRY TO THE REINFORCING SHALL BE NOT LESS THAN 2" WHEN MASONRY IS EXPOSED TO SOIL OR 1" FOR OTHER CONDITIONS.
- THE CLEAR DISTANCE BETWEEN PARALLEL BARS IS " MINIMUM AND (AND) SHALL NOT BE LESS THAN 1 BAR DIAMETER, EXCEPT THAT THE TWO BARS IN A CONTACT SPICE SHALL BE IN CONTACT. THIS CLEAR DISTANCE REQUIREMENT ALSO APPLIES TO THE CLEAR DISTANCE BETWEEN A CONTACT SPICE AND ADJACENT SPICES OR BARS. EXCEPTION: THE MINIMUM CLEAR DISTANCE BETWEEN PARALLEL BARS IN COLUMNS AND PLASTERS IS 2 BAR DIAMETERS.
- REFER TO THE STRUCTURAL DETAILS FOR LAPS REINFORCING. AT A MINIMUM, BLOCK WALL REINFORCING SHALL BE #4 @ 16" O.C. AND HORIZONTAL REINFORCING SHALL BE #4 @ 16" O.C. AT THE TOP ONE CONTINUOUS HORIZONTAL #4 BAR OR LARGER SHALL BE PLACED IN BOTH THE BOTTOM AND THE LAST COURSE OF MASONRY WALL (UON).
- SEE STRUCTURAL SHEETS FOR TYPICAL WALL DETAILS. AT A MINIMUM, DOOR AND WINDOW JAMBS SHALL HAVE 2 - #8 BARS, AND HEADERS (OR "LINTELS") SHALL HAVE 2 - #8 BARS. UON ON THE PLANS. JAMB AND LINTEL BARS SHALL EXTEND A MINIMUM OF 48 BAR DIAMETERS PAST THE OPENING.
- JAMB REINFORCING STEEL SHALL EXTEND INTO THE FOUNDATION (OR DECK) BELOW WITH LAP BARS OF THE SAME DIAMETER BENT TO HORIZONTAL POSITION AND ANCHORED INTO THE FOUNDATION.
- JAMB STEEL SHALL CONTINUE TO THE TOP OF THE WALL, UNLESS DETAILED OTHERWISE ON THE PLANS, BUT SHALL NOT EXTEND LESS THAN 48 BAR DIAMETERS PAST THE OPENING.
- MASONRY COLUMNS & PLASTERS: REFER TO THE STRUCTURAL DETAILS FOR REINFORCEMENT REQUIREMENTS. PROVIDE AT LEAST 4 - #3 TIES IN THE TOP 5' OF THE COLUMN, AND ENGAGE AT LEAST FOUR VERTICAL BARS AND/OR ANCHOR BOLTS WITH THE TIES. THE UPPERMOST TIE SHALL BE WITHIN 2' OF THE TOP OF THE COLUMN. BARS SHALL BE PLACED NOT MORE THAN 12" AND NOT MORE THAN 5" FROM THE SURFACE OF THE COLUMN.
- ANCHOR BOLT INSTALLATION: SECURE IN PLACE PRIOR TO GROUTING. PROVIDE 1" MINIMUM GROUT COVER.
- CONDUIT SLEEVES SHALL NOT BE SPACED CLOSER THAN THREE (3) TIMES THE DIAMETER CENTER-TO-CENTER. CONDUIT AND OTHER OBSTRUCTIONS SHALL BE STRATEGICALLY LOCATED SO AS TO AVOID CONFLICT WITH WALL REINFORCING AND CELL GROUT SPACES AND THE REQUIRED CLEARANCES.
- WATERPROOFING SHALL BE TO BE PROVIDED ON THE FACE OF ALL MASONRY WALLS EXPOSED TO EARTH, PER THE ARCHITECTURAL PLANS AND SPECIFICATIONS.
- THE CONTRACTOR SHALL COORDINATE WITH ALL OTHER TRADES WHOSE WORK RELATES TO THE MASONRY INSTALLATION FOR PLACING OF ALL REQUIRED FRAMING. THIS INCLUDES, BUT IS NOT LIMITED TO, PLACING ANCHORS, BOLTS, PIPES, SLEEVES, NAILERS, BLOCKOUTS, REGLETS, FITTINGS, CONDUITS, ETC. PROVIDED BY OTHER TRADES WITHIN THE MASONRY CONSTRUCTION.
- RETAINING WALLS SHALL NOT BE BACKFILLED UNTIL GROUT HAS SET A MINIMUM OF 14 DAYS (28 DAYS PREFERRED). ALL WALLS ARE TO BE FULLY BACKFILLED PRIOR TO FRAMING BEING PLACED ON OR AGAINST THE WALL. PER THE SOILS REPORT, ALL BACKFILL IS TO BE INSPECTED BY THE SOILS-SELECTED TECHNICAL ENGINEER AT THE TIME OF PLACEMENT.
- HOT WEATHER CONSTRUCTION: MASONRY CONSTRUCTION IS NOT PERMITTED WHEN THE AMBIENT AIR TEMPERATURE EXCEEDS 100°F OR IF IT EXCEEDS 80°F WITH A WIND VELOCITY OF 8 MPH OR GREATER. EXCEPT IF PRECISE AND PROPER HOT WEATHER CONSTRUCTION AND PROTECTION REQUIREMENTS OF THE APPROPRIATE PRACTICE ARE IMPLEMENTED WHEN TEMPERATURES ARE FORECASTED TO REACH OR TO REACH THE LIMITS FOR NORMAL CONSTRUCTION. IF SUCH CONSTRUCTION IS NECESSARY, CONTACT THE ENGINEER FOR REQUIREMENTS.) CHECK LOCAL WEATHER REPORTS BEFORE THE START OF EACH DAY AND PERIODICALLY MEASURE AIR TEMPERATURE AND WIND SPEED DURING THE DAY. CONSTRAY ALL NEWLY CONSTRUCTED MASONRY UNTIL DAMP. AT LEAST THREE TIMES A DAY UNTIL THE MASONRY IS THREE DAYS OLD.
- COLD WEATHER CONSTRUCTION: COMPLY WITH CBC SECTION 2104.1.

REINFORCEMENT

- REINFORCING STEEL SHALL BE TO DEFORMED, CLEAN, FREE OF RUST, GREASE OR ANY OTHER SUBSTANCES THAT COULD IMPAIR CONCRETE BOND.
 - ALL BARS SHALL CONFORM TO ASTM A631, GRADE 60 MINIMUM (UON ON STRUCTURAL PLANS). EXCEPT THAT #3 & #4 BARS MAY BE GRADE 40. ALL WELD WIRE FABRIC (WWF) SHALL CONFORM TO ASTM A185.
 - REINFORCING STEEL THAT IS TO BE WELDED SHALL CONFORM TO ASTM A706. ALL WELDING OF REINFORCEMENT SHALL BE SUBJECT TO SPECIAL INSPECTION.
 - A CONTRACTOR SHALL TAKE NECESSARY STEPS (STANDARD TIES, ANCHORAGE DEVICES, ETC.) TO SECURE ALL REINFORCING STEEL IN THEIR TRUE POSITION AND PREVENT DISPLACEMENT DURING CONCRETE PLACEMENT.
 - FABRICATION, PLACEMENT AND INSTALLATION OF REINFORCING STEEL SHALL CONFORM TO:
 - CONCRETE REINFORCING STEEL INSTITUTE (CRSI) MANUAL OF STANDARD PRACTICE
 - SHOP DRAWINGS FOR FABRICATION OF REINFORCING STEEL SHALL BE APPROVED BY THE CONTRACTOR AND SUBMITTED TO THE ARCHITECT AND ENGINEER FOR REVIEW AND APPROVAL PRIOR TO FABRICATION. SHOP DRAWINGS ARE NOT REQUIRED FOR SLABS ON-GRADE OR FOUNDATIONS UON ON THE STRUCTURAL PLANS.
 - HEATING OF REINFORCING STEEL TO AID IN BENDING AND SHAPING OF BARS IS NOT PERMITTED. ALL BENDS IN REINFORCING STEEL ARE TO BE MADE COLD. ALL BEND RADI SHALL CONFORM TO CRSI MANUAL OF STANDARD PRACTICE.
 - REFER TO CONCRETE AND MASONRY NOTES FOR SPECIFIC MINIMUM SPICE LENGTH AND SPICE STAGGERING REQUIREMENTS. LAP WELDED WIRE FABRIC (WWF) REINFORCEMENT TWO (2) MODULES MINIMUM (UON). ALL SPICES ARE TO BE STAGGERED.
- STRUCTURAL STEEL**
- ALL STRUCTURAL STEEL AND CONNECTIONS SHALL BE FABRICATED AND ERECTED IN ACCORDANCE WITH AISC SPECIFICATIONS; SEISMIC PROVISIONS SUPPLEMENTS NO. 1 AND 2, AND CODE OF STANDARD PRACTICE AS AMENDED TO DATE.
 - STEEL FABRICATION SHOP DRAWINGS SHALL BE SUBMITTED FOR REVIEW BY THE ARCHITECT AND ENGINEER PRIOR TO FABRICATION.
 - MATERIALS:
 - TUBE SECTIONS (TS" OR "HSS) SHALL CONFORM TO ASTM A500 GR. B.
 - PIPE SECTIONS SHALL BE WELDED SEAMLESS PIPE CONFORMING TO ASTM A53 GR. B OR ASTM A501.
 - STD INDICATES STANDARD WALL
 - EXT INDICATES EXTRA STRONG
 - DBL INDICATES DOUBLE EXTRA STRONG
 - ALL OTHER MATERIAL (PLATE, BARS, ETC.) SHALL CONFORM TO ASTM A36 (UON)
 - BOLTS:
 - ALL BOLTS SHALL BE ASTM F1554 GRADE 36 (UON) ON THE STRUCTURAL PLANS.
 - HIGH STRENGTH BOLTS COMPLYING WITH ASTM A325 AND A490, WHEN SPECIFIED, SHALL BE DISTANCE TO PANEL OR FRAMING MEMBER.
 - THREADED ROD, WHERE SPECIFIED, SHALL CONFORM WITH ASTM F1554 GRADE 36 (UON) ON THE STRUCTURAL PLANS.
 - BOLT HOLES SHALL BE DRILLED 1/32" TO 1/16" LARGER THAN THE SPECIFIED BOLT DIAMETER.
 - WELDING:
 - ALL WELDING SHALL BE PERFORMED USING SMAW, GMAW OR FCAW PROCESSES.
 - ALL WELDED CONNECTIONS TO BE WELDED IN ACCORDANCE WITH THE LATEST EDITION OF THE AWS D1.1.
 - ALL WELDING SHALL BE PERFORMED BY CERTIFIED WELDERS.
 - ALL WELDING SHALL BE PERFORMED WITH E70XX ELECTRODES WITH A MINIMUM CWN TOUGHNESS OF 20 FTLB AT -200F.
 - WELD LENGTHS SPECIFIED ON THE PLANS ARE THE NET EFFECTIVE LENGTH REQUIRED, WHERE FILLET WELD SYMBOL IS PROVIDED WITHOUT INDICATION OF SIZE USE THE MINIMUM SIZE WELDS AS SPECIFIED IN SECTION 1.1.7.2 OF THE AISC MANUAL OF STEEL CONSTRUCTION 9TH EDITION.
 - NO FIELD WELDING SHALL BE PERMITTED UON ON THE PLANS OR DETAILS.
 - NO HOLES OTHER THAN THOSE SPECIFICALLY DETAILED SHALL BE ALLOWED THROUGH STRUCTURAL

- STEEL MEMBERS, BURNING OR TORCHING OF HOLES IS NOT PERMITTED UNDER ANY CIRCUMSTANCES.
- ALL STRUCTURAL STEEL SHALL BE PAINTED ONE SHOP COAT AND TOUCHED-UP IN THE FIELD WITH RED LEAD OR APPROVED ZINC CHROMATE PRIMER, AS NECESSARY.
- ANY STEEL MEMBER INTERFACING WITH WOOD FRAMING SHALL HAVE 1/2" DIAMETER STUDS WELDED AT 24" O.C. FOR ATTACHMENT OF WOOD NAILED. THRU-BOLTING OF NAILERS SHALL NOT BE PERMITTED UON ON THE PLANS OR DETAILS.
- PROVIDE HOT DIP GALVANIZING OR 3" MINIMUM CONCRETE COVER AROUND ALL STRUCTURAL STEEL BELOW GRADE.
- MASONRY**
 - SPECIAL INSPECTION IS REQUIRED FOR MASONRY WALLS PER CBC 1704.5
 - MASONRY UNITS: SHALL CONFORM TO ASTM C90, GRADE N, TYPE I, MEDIUM-WEIGHT, THE COMPRESSIVE STRENGTH OF THE MASONRY, F.M. SHALL BE 1500 PSI MINIMUM. REFER TO CBC 2103.
 - MORTAR SHALL BE TYPE S, WITH A STRENGTH OF 1800 PSI MINIMUM @ 28 DAYS. PROPORTIONED IN CONFORMANCE WITH CBC TABLE 21-4. WHEN THE SPECIFIED MASONRY STRENGTH, F.M. IS GREATER THAN 2000 PSI, THEN THE MORTAR SHALL BE TYPE M. MORTAR STRENGTH SHALL BE EQUAL TO OR GREATER THAN THE MASONRY STRENGTH, F.M. NO MORTARS SHALL BE USED THAT HAVE STOOD FOR MORE THAN ONE HOUR.
 - GROUT: STRENGTH SHALL BE NO LESS THAN 2500 PSIG 28 DAYS. CEMENT CONTENT OF THE GROUT SHALL BE INCREASED, AS NECESSARY, TO ACHIEVE THE SPECIFIED MASONRY ASSEMBLY STRENGTH, F.M. AND ADEQUATE WORKABILITY. GROUT COMPRESSIVE STRENGTH, WHEN TESTED PER USC STANDARD NO. 21-18 SHALL EQUAL OR EXCEED THE CONCRETE MASONRY UNIT STRENGTH. ALL GROUT ADDITIVES SHALL RECEIVE THE PRIOR APPROVAL OF THE ENGINEER AND THE BUILDING OFFICIAL.
 - ADMITTIVES: SHALL NOT BE PERMITTED IN MORTAR OR GROUT UNLESS SUSTAINING DATA HAS BEEN SUBMITTED TO AND APPROVED BY THE ENGINEER. FIRE CLAY, DIRT AND OTHER DELETERIOUS MATERIALS ARE PROHIBITED.
 - AGGREGATES: SAND FOR MORTAR SHALL CONFORM TO ASTM C144 EXCEPT THAT NOT LESS THAN 3% OF THE SAND SHALL PASS THE NUMBER 100 SIEVE. SAND AND PEA GRAVEL FOR GROUT SHALL CONFORM TO ASTM C644, TABLE 1. COARSE AGGREGATE, EXCEPT WHEN OTHER GRADINGS ARE SPECIFICALLY APPROVED BY THE ENGINEER.
 - WATER USED FOR MORTAR AND GROUT SHALL BE CLEAN AND FREE FROM DELETERIOUS AMOUNTS OF ACIDS, SALTS, ALKALI AND ORGANIC MATERIALS.
 - STEEL REINFORCING: SHALL CONFORM TO ASTM A615, GRADE 60, CLEAN AND FREE OF RUST. EXCEPT THAT #3 BARS MAY BE GRADE 40. REINFORCING STEEL THAT IS TO BE WELDED SHALL CONFORM TO ASTM A706, AND THE WELDING SHALL BE SPECIAL INSPECTED.
 - ANCHOR BOLTS: SEE THE "STRUCTURAL STEEL" SPECIFICATIONS SECTION HEREIN.
 - ALL CELLS SHALL BE SOLID GROUTED (OR "FULLY" GROUTED). MASONRY UNITS SHALL BE LAID IN RUNNING BOND. SURFACES TO BE CLEANED OF ALL LOOSE DEBRIS PRIOR TO SETTING BLOCK. CELLS TO BE IN VERTICAL ALIGNMENT. EACH THAT MINIMUM VERTICAL UNBROUGHTED CORE EXCLUDING HORIZONTAL BARS (S) 24" X 3" FOR GROUT POURS UP TO 4 FEET AND 3"X3" FOR GROUT POURS UP TO 6 FEET.
 - ALL BED JOINTS ARE TO BE FULL-BEDDED IN MORTAR. END WALLS AND CROSS WEBS FORMING CELLS TO BE FILLED SHALL BE FULL-BEDDED IN MORTAR TO PREVENT LEAKAGE OF GROUT. ALL HEAD JOINTS ARE TO BE SOLIDLY FILLED AT LEAST 1 1/2" BELOW TOP OF MASONRY. HORIZONTAL CONSTRUCTION JOINTS SHALL BE FORMED BY STOPPING THE GROUT FOUR 1 1/2" BELOW TOP OF MASONRY.
 - GROUT LETS SHALL NOT EXCEED 5 FEET A JOINTS. GROUT SHALL BE CONSOLIDATED BY MECHANICAL VIBRATION IMMEDIATELY AFTER PLACING TO HELP ENSURE FILLING OF ALL VOIDS. RECONSOLIDATION BY VIBRATION MUST BE DONE AFTER THE INITIAL WATER LOSS AND BEFORE INITIAL SET. FOR GROUT POURS EXCEEDING 5 FEET 4 INCHES, CLEANOUT OPENINGS SHALL BE 12" MINIMUM FROM INSIDE FACE OF SHELL, HOWEVER THE CLEAR DISTANCE FROM OUTSIDE FACE OF MASONRY TO THE REINFORCING SHALL BE NOT LESS THAN 2" WHEN MASONRY IS EXPOSED TO SOIL OR 1" FOR OTHER CONDITIONS.
 - THE CLEAR DISTANCE BETWEEN PARALLEL BARS IS " MINIMUM AND (AND) SHALL NOT BE LESS THAN 1 BAR DIAMETER, EXCEPT THAT THE TWO BARS IN A CONTACT SPICE SHALL BE IN CONTACT. THIS CLEAR DISTANCE REQUIREMENT ALSO APPLIES TO THE CLEAR DISTANCE BETWEEN A CONTACT SPICE AND ADJACENT SPICES OR BARS. EXCEPTION: THE MINIMUM CLEAR DISTANCE BETWEEN PARALLEL BARS IN COLUMNS AND PLASTERS IS 2 BAR DIAMETERS.

- REFER TO THE STRUCTURAL DETAILS FOR LAPS REINFORCING. AT A MINIMUM, BLOCK WALL REINFORCING SHALL BE #4 @ 16" O.C. AND HORIZONTAL REINFORCING SHALL BE #4 @ 16" O.C. AT THE TOP ONE CONTINUOUS HORIZONTAL #4 BAR OR LARGER SHALL BE PLACED IN BOTH THE BOTTOM AND THE LAST COURSE OF MASONRY WALL (UON).
- SEE STRUCTURAL SHEETS FOR TYPICAL WALL DETAILS. AT A MINIMUM, DOOR AND WINDOW JAMBS SHALL HAVE 2 - #8 BARS, AND HEADERS (OR "LINTELS") SHALL HAVE 2 - #8 BARS. UON ON THE PLANS. JAMB AND LINTEL BARS SHALL EXTEND A MINIMUM OF 48 BAR DIAMETERS PAST THE OPENING.
- JAMB REINFORCING STEEL SHALL EXTEND INTO THE FOUNDATION (OR DECK) BELOW WITH LAP BARS OF THE SAME DIAMETER BENT TO HORIZONTAL POSITION AND ANCHORED INTO THE FOUNDATION.
- JAMB STEEL SHALL CONTINUE TO THE TOP OF THE WALL, UNLESS DETAILED OTHERWISE ON THE PLANS, BUT SHALL NOT EXTEND LESS THAN 48 BAR DIAMETERS PAST THE OPENING.
- MASONRY COLUMNS & PLASTERS: REFER TO THE STRUCTURAL DETAILS FOR REINFORCEMENT REQUIREMENTS. PROVIDE AT LEAST 4 - #3 TIES IN THE TOP 5' OF THE COLUMN, AND ENGAGE AT LEAST FOUR VERTICAL BARS AND/OR ANCHOR BOLTS WITH THE TIES. THE UPPERMOST TIE SHALL BE WITHIN 2' OF THE TOP OF THE COLUMN. BARS SHALL BE PLACED NOT MORE THAN 12" AND NOT MORE THAN 5" FROM THE SURFACE OF THE COLUMN.
- ANCHOR BOLT INSTALLATION: SECURE IN PLACE PRIOR TO GROUTING. PROVIDE 1" MINIMUM GROUT COVER.
- CONDUIT SLEEVES SHALL NOT BE SPACED CLOSER THAN THREE (3) TIMES THE DIAMETER CENTER-TO-CENTER. CONDUIT AND OTHER OBSTRUCTIONS SHALL BE STRATEGICALLY LOCATED SO AS TO AVOID CONFLICT WITH WALL REINFORCING AND CELL GROUT SPACES AND THE REQUIRED CLEARANCES.
- WATERPROOFING SHALL BE TO BE PROVIDED ON THE FACE OF ALL MASONRY WALLS EXPOSED TO EARTH, PER THE ARCHITECTURAL PLANS AND SPECIFICATIONS.
- THE CONTRACTOR SHALL COORDINATE WITH ALL OTHER TRADES WHOSE WORK RELATES TO THE MASONRY INSTALLATION FOR PLACING OF ALL REQUIRED FRAMING. THIS INCLUDES, BUT IS NOT LIMITED TO, PLACING ANCHORS, BOLTS, PIPES, SLEEVES, NAILERS, BLOCKOUTS, REGLETS, FITTINGS, CONDUITS, ETC. PROVIDED BY OTHER TRADES WITHIN THE MASONRY CONSTRUCTION.
- RETAINING WALLS SHALL NOT BE BACKFILLED UNTIL GROUT HAS SET A MINIMUM OF 14 DAYS (28 DAYS PREFERRED). ALL WALLS ARE TO BE FULLY BACKFILLED PRIOR TO FRAMING BEING PLACED ON OR AGAINST THE WALL. PER THE SOILS REPORT, ALL BACKFILL IS TO BE INSPECTED BY THE SOILS-SELECTED TECHNICAL ENGINEER AT THE TIME OF PLACEMENT.
- HOT WEATHER CONSTRUCTION: MASONRY CONSTRUCTION IS NOT PERMITTED WHEN THE AMBIENT AIR TEMPERATURE EXCEEDS 100°F OR IF IT EXCEEDS 80°F WITH A WIND VELOCITY OF 8 MPH OR GREATER. EXCEPT IF PRECISE AND PROPER HOT WEATHER CONSTRUCTION AND PROTECTION REQUIREMENTS OF THE APPROPRIATE PRACTICE ARE IMPLEMENTED WHEN TEMPERATURES ARE FORECASTED TO REACH OR TO REACH THE LIMITS FOR NORMAL CONSTRUCTION. IF SUCH CONSTRUCTION IS NECESSARY, CONTACT THE ENGINEER FOR REQUIREMENTS.) CHECK LOCAL WEATHER REPORTS BEFORE THE START OF EACH DAY AND PERIODICALLY MEASURE AIR TEMPERATURE AND WIND SPEED DURING THE DAY. CONSTRAY ALL NEWLY CONSTRUCTED MASONRY UNTIL DAMP. AT LEAST THREE TIMES A DAY UNTIL THE MASONRY IS THREE DAYS OLD.
- COLD WEATHER CONSTRUCTION: COMPLY WITH CBC SECTION 2104.1.

TIMBER / LUMBER

- ALL STRUCTURAL LUMBER SHALL BE DOUGLAS FIR-LARCH, S4S AND SHALL CONFORM TO CBC SECTION 2303.1.
- THE MINIMUM LUMBER GRADE OF EACH MEMBER SHALL BE AS FOLLOWS UON ON PLANS AND DETAILS:
 - 2X STUDS, BLOCKING, PLATES: STUD
 - 2X JOISTS 12 OR BETTER
 - 4X4 BEAMS OR POSTS 12 OR BETTER
 - 4X6 OR LARGER BEAMS OR POSTS 14 OR BETTERIT IS RECOMMENDED (BUT NOT REQUIRED) THAT ALL EXPOSED MEMBERS BE SELECT STRUCTURAL OR BETTER AND FREE OF HEART CENTER DUE TO VISUAL CHARACTERISTICS.
- ALL LUMBER IN CONTACT WITH CONCRETE OR MASONRY SHALL BE REDWOOD OR PRESSURE TREATED DOUGLAS FIR. CONTRACTOR SHALL COORDINATE WITH EOR IF PRESSURE TREATED MATERIAL UTILIZES A CORROSIVE TREATMENT GREATER THAN "DOT" PRIOR TO INSTALLATION. WHENEVER IT IS NECESSARY TO CUT, NOTCH, BORE OR SPICE PRESSURE TREATED MATERIAL, ALL NEWLY CUT SURFACES SHALL BE THOROUGHLY PAINTED WITH THE SAME PRESERVATIVE.
- MAXIMUM MOISTURE CONTENT FOR ALL STRUCTURAL MEMBERS SHALL NOT EXCEED 19%.

- ALL PLYWOOD SHEATHING SHALL BE CDX GRADE (OR BETTER) DOUGLAS FIR WITH EXTERIOR GLUE. ALL SHEATHING SHALL CONFORM TO CBC STANDARD 23-2 AND GRADE-MARKED BY THE AMERICAN PLYWOOD ASSOCIATION (APA). PANEL INDEX TO BE 4020 FOR FLOORS AND 240 FOR ROOFS (UON) ON THE PLANS AND DETAILS.

FASTENERS

- NAILS:
 - SHALL BE WITH "COMMON" NAILS (UON).
 - SHALL NOT BE DRIVEN CLOSER THAN 3" THEIR LENGTH NOR CLOSER THAN 1/2" OF THEIR LENGTH TO THE EDGE OR END OF A MEMBER, EXCEPT FOR SHEATHING.
 - SHALL BE INSTALLED IN THE JOINTS AND AT SPACES NOT TO EXCEED 48" O.C.
 - IN CONTACT WITH PRESERVATIVE-TREATED WOOD SHALL BE HOT DIPPED ZINC GALVANIZED OR STAINLESS STEEL.
 - ALL NAILING CONFORM TO 2016 CBC TABLE 2304.10.1.
- LAG SCREWS:
 - SHALL BE INSTALLED INTO PRE-DRILLED LEAD HOLES. LUBRICANT (OR SOAP) SHALL BE USED TO FACILITATE INSTALLATION AND PREVENT DAMAGE TO THE SCREWS.
 - IN CONTACT WITH PRESERVATIVE-TREATED WOOD SHALL BE HOT DIPPED ZINC GALVANIZED OR STAINLESS STEEL.
- BOLTS:
 - SHALL CONFORM TO ASTM F1554 GRADE 36 (UON) ON PLANS AND DETAILS.
 - SHALL BE INSTALLED IN PRE-DRILLED HOLES A MAXIMUM OF 1/16" LARGER THAN THE SPECIFIED BOLT DIAMETER.
 - WHEN INSTALLED AGAINST WOOD SURFACES, SHALL HAVE STANDARD WASHERS UNDER THE HEADS AND NUTS.
 - IN CONTACT WITH PRESERVATIVE-TREATED WOOD SHALL BE HOT DIPPED ZINC GALVANIZED OR STAINLESS STEEL.
- ANCHOR BOLTS:
 - SHALL BE 5/8" DIAMETER WITH 33X20 2" STEEL PLATE WASHERS AT SHEARWALLS.
 - SHALL HAVE " MINIMUM EMBEDMENT" (CONTRACTOR TO COORDINATE LENGTH OF BOLTS WITH SLL PLATE THICKNESSES)
 - SHALL CONFORM TO ASTM F1554 GRADE 36
 - SHALL BE HOT DIPPED ZINC GALVANIZED OR STAINLESS STEEL
 - SHALL NOT BE SPACED GREATER THAN 27" O.C. REFER TO SHEARWALL SCHEDULE FOR SPECIFIC ANCHOR BOLT SPACING REQUIREMENTS.
 - SHALL BE PLACED A MAXIMUM OF 12" FROM WALL CORNERS, WALL ENDS, AND SLL PLATE SPICES (BUT NOT LESS THAN 7" DIAMETERS), AND A MINIMUM OF TWO BOLTS PER FACE OF SLL PLATE IS REQUIRED.
 - SHALL BE SECURED IN PLACE PRIOR TO FOUNDATION INSPECTION

CARPENTRY

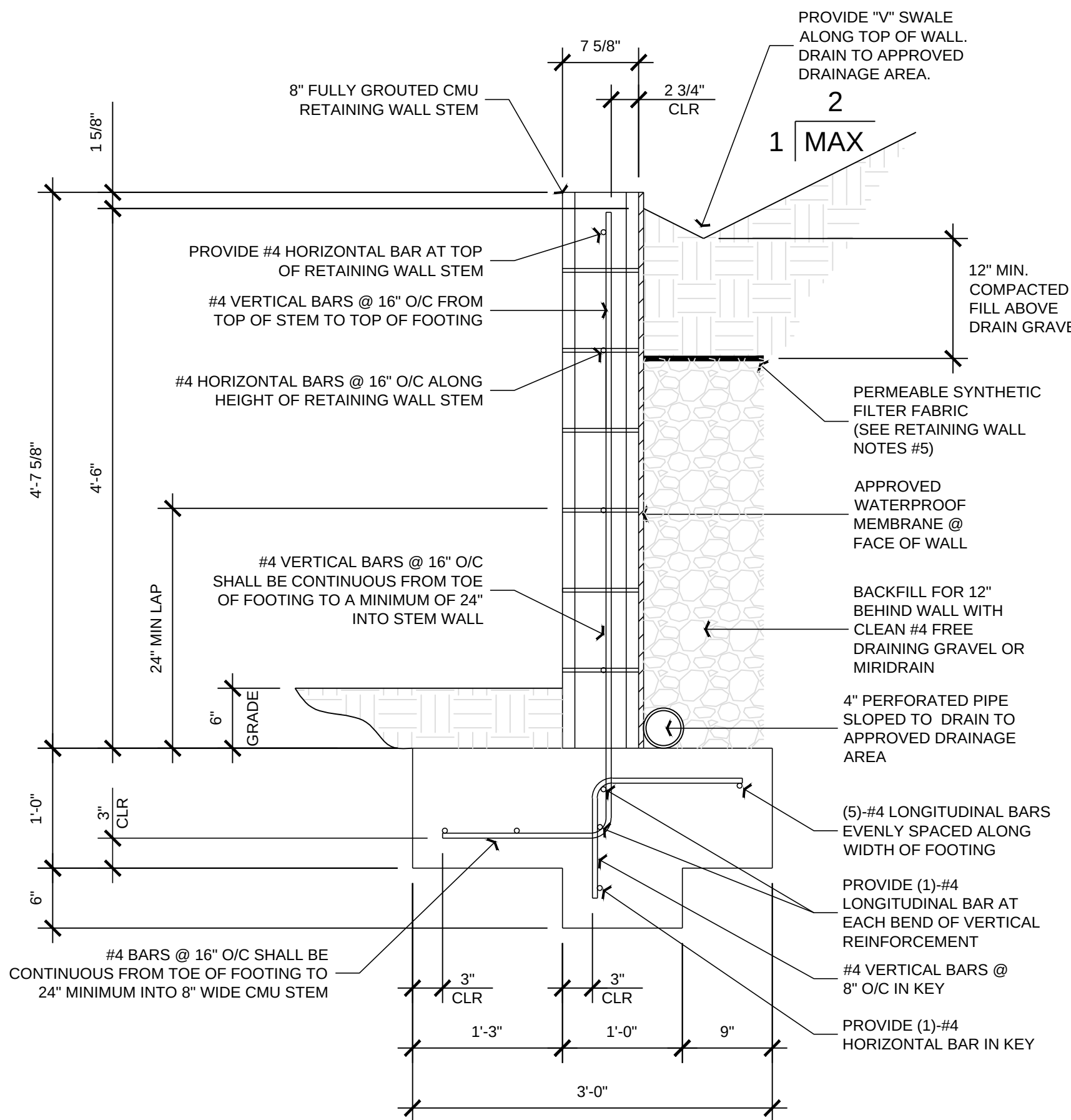
- REFER TO 2019 CBC TABLE 2304.10.1 FOR ALL MINIMUM NAILING REQUIREMENTS.
- REFER TO INDIVIDUAL SECTIONS FOR APPLICABLE MATERIAL SPECIFICATIONS.
- FABRICATE, SIZE, INSTALL, CONNECT, FASTEN, BORE, NOTCH, AND CUT WOOD AND PLYWOOD WITH JOINTS TRUE, TIGHT, AND WELL-NAILED, SCREWED OR BOLTED AS REQUIRED. ALL MEMBERS TO BE SOLID BEARING WITHOUT BEING SHIMMED (UON). SET HORIZONTAL MEMBERS SUBJECT TO BENDING AND CURVING. INSTALL FRAMING PLUMB, SQUARE, TRUE AND CUT FOR FULL BEARING. SPICES ARE NOT PERMITTED BETWEEN BEARINGS. USE FULL LENGTHS (UON).
- METAL FRAMING ANGLES, ANCHOR, CLIPS, STRAPS, TIES, HOLLOWDOWNS, ETC. SHALL BE MANUFACTURED BY SIMPSON STRONG-TIE CO. NO SUBSTITUTIONS SHALL BE PERMITTED WITHOUT PRIOR APPROVAL OF THE ENGINEER.
- ALL WALLS ARE TO HAVE CONTINUOUS DOUBLE 2X TOP PLATES SPICED AS FOLLOWS (UON) ON THE PLANS AND DETAILS.
- WALL STUDS SPECIFIED:
 - (UON) USE THE FOLLOWING GUIDELINES FOR WALL FRAMING:
 - USE 2X4 STUDS AT 16" O.C. FOR WALLS LESS THAN 9'-0" TALL.
 - WALLS 9'-0" TO 10'-0" TALL SHALL BE CONSTRUCTED OF 2X6 STUDS AT 16" O.C.
 - REQUEST SPECIFICALLY ENGINEERED WALL DETAILS FOR WALLS GREATER THAN 16'-0" TALL.
- BLOCKING:
 - PROVIDE MIN. ONE ROW OF NOMINAL 2" THICK BLOCKING OF SAME WIDTH AS STUD, FITTED SNUGLY AND SPOKE INTO STUDS AT MID-HEIGHT OF PARTITIONS OR WALLS OVER EIGHT FEET HIGH.
 - DO NOT CRIPPLE WALLS (OR "POKEY WALLS") LESS THAN 14" IN HEIGHT SHALL BE SOLID BLOCKING.
 - REFER TO SHEARWALL SECTION FOR ADDITIONAL BLOCKING REQUIREMENTS.
 - NOTHING:
 - IS NOT PERMITTED OF ANY STRUCTURAL MEMBER WITHOUT PRIOR APPROVAL
 - IN EXTERIOR AND BEARING WALLS, NOTCHES SHALL NOT EXCEED 25% OF THE STUD DEPTH.
 - NON-BEARING PARTITIONS OR PARTITIONS SHALL NOT EXCEED 40% OF THE STUD DEPTH.
 - A SUCCESSIVE NOTCHES IN THE SAME MEMBER SHALL BE SPACED A MINIMUM OF 16" APART.
- BORING:
 - IS NOT PERMITTED OF ANY STRUCTURAL MEMBER WITHOUT PRIOR APPROVAL
 - IN EXTERIOR AND BEARING WALLS, HOLES SHALL NOT EXCEED 40% OF THE STUD DEPTH.
 - NON-BEARING PARTITION WALLS, SHALL MAY BE DRILLED NOT GREATER THAN 60% OF THE STUD DEPTH.
 - A SUCCESSIVE HOLES IN THE SAME MEMBER SHALL BE SPACED A MINIMUM OF 16" APART.
- BEARING:
 - PROVIDE A MINIMUM OF 1 1/2" OF BEARING FOR ALL 2X JOISTS AND ALL 4X10 / 6X8 HEADERS & SMALLER.
 - PROVIDE A MINIMUM OF 3" OF BEARING FOR ALL BEAMS AND HEADERS 4X12 / 6X10 & LARGER (UON).
 - MEMBER BEARING ON PREFABRICATED HANGERS ARE TO HAVE FULL BEARING AND NAILING PER MANUFACTURER'S SPECIFICATIONS.
- POSTS:
 - POSTS INSIDE WALLS SHALL BEAR ON SILL PLATES AND SHALL BE CONTINUOUS BETWEEN TOP AND BOTTOM PLATES (UON).
 - PROVIDE POSTS UNDER ALL BEAMS, GIRDERS OR DOUBLE JOISTS EQUAL TO THE WIDTH OF THE SUPPORTED MEMBER.
 - POSTS ON UPPER LEVELS ARE TO BE STACKED ON POSTS OF EQUAL SIZE AT LEVELS BELOW, UNLESS A LARGER POST IS SPECIFIED ON THE PLANS.
 - VERTICAL BLOCKING ("SQUASH BLOCKS") SHALL BE USED TO FULLY TRANSFER THE POST AREA THROUGH FLOORS TO FOUNDATION. VERTICAL BLOCKING SHALL BE EQUAL TO FLOOR THICKNESS PLUS 1/16"
 - HEADERS FRAMING BEARING ON CONTINUOUS POSTS WITHOUT TRIMMER STUDS SHALL BE SUPPORTED IN SIMPSON HUC HANGERS (UON).
 - POSTS WHEN ISOLATED, SHALL BE SEATED IN SIMPSON POST OR COLUMN BASES (UON)
- FLOOR FRAMING:
 - PROVIDE WOOD JOISTS, AS SPECIFIED, LAID WITH THE CROWN UP AND SPACES AS INDICATED.
 - PROVIDE A MINIMUM OF 1 1/2" END BEARING UNLESS OTHERWISE SHOWN.
 - PROVIDE FULL DEPTH SOLID 2X BLOCKING OR CROSS-BRIDGING BETWEEN THE JOISTS AT 6'-0" O.C. MAX. FOR FLOORS FRAMED WITH JOISTS. REFER TO THE MANUFACTURER'S SPECIFICATIONS FOR BLOCKING REQUIREMENTS.
 - PROVIDE FULL DEPTH SOLID 2X BLOCKING BETWEEN THE JOISTS UNDER ALL WALLS AND PARTITIONS WHERE THE WALL OR PARTITION IS PERPENDICULAR TO THE FLOOR FRAMING (INCLUDING FLOORS FRAMED WITH JOISTS)
 - STAGGERED AND THE EDGES OF SHEETS CENTERED OVER SUPPORTS. IF 1x6 PLYWOOD IS NOT USED, PROVIDE BLOCKING AT ALL PLYWOOD EDGES. GLUE TO JOISTS AND FULLY NAIL WITH COMMON NAILS PER THE PLANS.
- ROOF FRAMING:
 - PROVIDE WOOD JOISTS, AS SPECIFIED, LAID WITH THE CROWN UP AND SPACES AS INDICATED.
 - PROVIDE A MINIMUM OF 1 1/2" END BEARING (UON).
 - PROVIDE FULL DEPTH SOLID 2X BLOCKING OR CROSS-BRIDGING BETWEEN THE JOISTS AT 6'-0" O.C. MAX.
 - PROVIDE ALL CRACKED FRAMING REQUIRED TO ACHIEVE POSITIVE DRAINAGE PER ARCHITECTURAL DRAWINGS.
 - INSTALL PLYWOOD PANELS WITH THE FACE GRAN ACROSS THE FRAMING AND CLOSE JOINTS AND NAIL AT EACH SUPPORT. FULLY NAIL WITH COMMON NAILS PER THE PLANS.
 - PROVIDE SIMPSON "PSC" CLIPS AT ALL PLYWOOD JOINTS PERPENDICULAR TO FRAMING. PROVIDE CLIPS MIDWAY BETWEEN FRAMING MEMBERS AT THE UNSUPPORTED EDGES OF PLYWOOD WHEN MEMB VISUAL OR OTHERWISE VERIFIED. IF CLIPS ARE NOT USED, PROVIDE SOLID BLOCKING FOR JOINTS PERPENDICULAR TO FRAMING.
- SHEARWALLS:
 - REFER TO PLANS FOR ALL SHEARWALL LOCATIONS, LENGTH TYPE AND NAILING.
 - REFER TO SHEARWALL SCHEDULE ON TITLE SHEET FOR ADDITIONAL INFORMATION.
 - SHEARWALL LENGTHS SPECIFIED ON PLANS ARE MINIMUM REQUIRED.
 - SHEARWALLS TO BE NAILED WITH COMMON NAILS. ALL NAILS TO HAVE MINIMUM 38" EDGE DISTANCE TO PANEL OR FRAMING MEMBER.
 - IF 3X FRAMING IS REQUIRED, STAGGER EDGE NAILING. 3X FRAMING IS REQUIRED AT:
 - ALL PANEL JOINTS
 - ALL SILL PLATES ON CONCRETE OR MASONRY
 - ALL SILL PLATES AT DOUBLE-SIDED SHEARWALLS
 - OSB MAY BE USED IN LIEU OF PLYWOOD.

ENGINEERED LUMBER

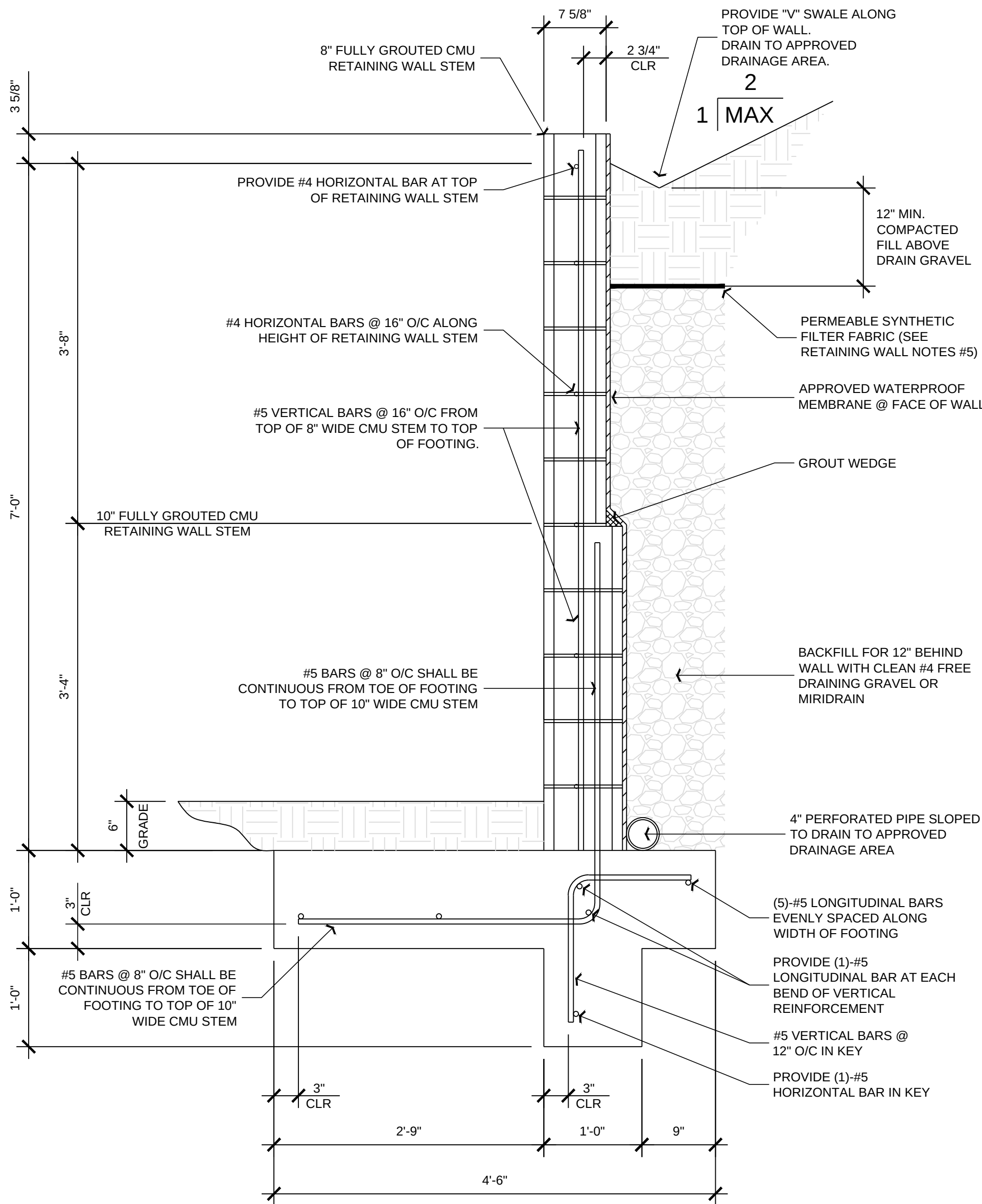
- GLULAMINATED BEAMS:
 - SHALL BE 24F-V8 FOR SIMPLE SPANS AND 24F-V8 FOR BEAMS WITH CANTILEVERS WITH THE FOLLOWING MINIMUM PROPERTIES:
 - FB = 2400 KSI
 - FE = 145 FSI
 - FC = 450 FSI
 - E = 1800 PSI
 - SHALL NOT BE NOTCHED, CUT OR DRILLED WITHOUT PRIOR APPROVAL FROM THE ENGINEER

- SHALL HAVE EXTERIOR GLUE AND WEATHER-TREATMENT PRIOR TO INSTALLATION
- SHALL BE FABRICATED BY AN APPROVED MANUFACTURER. AN A1.1.C. CERTIFICATE OF COMPLIANCE SHALL BE OBTAINED BY THE INSPECTOR PRIOR TO INSTALLATION
- SHALL HAVE FACTORY STANDARD NUMBER, EXCEPT WHERE NOTED OTHERWISE ON THE PLANS
- LAMINATED V

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SCALE: 1" = 1'-0"



SCALE: 1" = 1'-0"

MASONRY NOTES & REQUIRMENTS:

CONCRETE BLOCK	ASTM C90, Grade N, Type I, Normal Wt, fm=1500psi.
MORTAR	Minimum Compressive Strength of 2500 psi at 28 days. Slump to be maintained at 2 1/2" to 3". ASTM C270, Type M, with Type II Portland Cement. Proportions: (By volume) 1 part Portland Cement (ASTM C150). 1/4 part Hydrated Lime (ASTM C150). 2 1/4 to 3 parts sand (ASTM C144).
GROUT	Minimum Compressive strength of 2000 psi at 28 days. Slump to be minimum 8" to 10" with 3/8" Pea Gravel. ASTM C476, with Type I or II Portland Cement. Proportions: (By volume) 1 part Portland Cement (ASTM C150). 1 to 2 parts Pea Gravel. 2 1/4 to 3 parts sand. Placement of all grout to be per the requirements set forth in CBC. Solid grout all cells (with or without rebar). Low Lift Grouting: Blocking to be grouted shall not exceed 4'-0" in height, and shall be grouted in one continuous operation. All grout shall be vibrated when placed, and a second time approx. 1/4 hr after placing. When grouting is stopped for more than one hour, keep grout cold joint minimum 1 1/2" below the top of the blocks.
WATER	All water must be potable, clean and free of deleterious amounts of acid, alkalis or organic materials.
CONSTRUCTION	All construction methodology shall conform to the requirements of the 2010 CBC. Bond shall be provided by lapping units in successive vertical courses (Running Bond). All masonry walls in excess of 10'-0" in height shall be braced to withstand a wind load of 10 psf, applied perpendicular to wall in either direction during construction. Bracing shall remain in place until the supporting element (roof diaphragm, etc.) is completed and attached.
REBAR	All rebar to conform to ASTM A615. #4 bars & smaller: Grade 40. #5 bars & larger: Grade 60. Adjacent rebar laps to be staggered minimum of 24" Lap lengths: #4: 24" #5: 32" All bars shall be free of loose and flaky rust and scale, grease, or other material, which might affect or impair bond.
INSPECTIONS	Special Inspection, when required on plans or details, shall conform to the 2010 CBC, using the "Unit Strength Method" for determining unit compressive strength.

REQ'D VERIFICATION & INSPECTION OF SOILS

SPECIAL INSPECTIONS SHOULD BE PERFORMED IN ACCORDANCE WITH TABLE 1705.6 BELOW:

VERIFICATION AND INSPECTION TASK	CONTINUOUS DURING TASK LISTED	PERIODICALLY DURING TASK LISTED
VERIFY MATERIAL BELOW FOOTINGS ARE ADEQUATE TO ACHIEVE THE DESIGN BEARING CAPACITY	NO	YES
VERIFY EXCAVATIONS ARE EXTENDED TO PROPER DEPTH AND HAVE REACHED PROPER MATERIAL	NO	YES
PERFORM CLASSIFICATION AND TESTING OF CONTROLLED FILLED MATERIALS	NO	YES
VERIFY USE OF PROPER MATERIALS DENSITIES AND LIFT THICKNESSES DURING PLACEMENT AND COMPACTION OF CONTROLLED FILL	YES	NO
PRIOR TO PLACEMENT OF CONTROLLED FILL, OBSERVE SUBGRADE AND VERIFY THAT THE SITE HAS BEEN PREPARED PROPERLY	NO	YES

CONSULTANTS

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STRUCTURAL DESIGN & CIVIL ENGINEERING JK ENGINEERING 610 10TH ST, STE A PASO ROBLES, CA 93446 805.239.4151	

GENERAL SPECIFICATIONS FOR CONCRETE :

- All concrete shall have 2500 psi minimum compressive strength at 28 days and shall be normal weight. (U.O.N.)
- All concrete work shall comply with ACI building code (ACI 318)
- The maximum concrete slump shall be : Slabs 3" (plus or minus 1")
All other work ... 4" (plus or minus 1")
- The minimum cement content shall be 5 sacks per cubic yard and shall be Portland Cement, Type I or II, low alkali, per ASTM C-150.
- Any water reducing agents added shall be used to reduce the water/cement ratio. Admixtures shall be approved by the Engineer.
- Aggregate shall conform to ASTM C-33. Maximum aggregate size shall be 1" (U.O.N.)
Use 3/4" aggregate for slabs on grade.
- Concrete Placement :
 - Concrete shall not free-fall more than five (5) feet. Use tremie pump or other approved methods.
 - Vibrate all concrete (including slabs) as it is placed with a mechanical vibrator operated by experienced personnel. Reinforcing and forms shall not be vibrated.
- Curing : Freshly deposited concrete shall be protected from premature drying and excessively hot or cold temperatures, and shall be maintained with minimal moisture loss at a relatively constant temperature for the period of time necessary for the hydration of the cement. (typically 7 days)
- Unless specifically detailed or noted otherwise, construction and control joints shall be provided on all concrete slabs, and shall be located such that the area within joints does not exceed 400 square feet, and is roughly square without interior corners.

GENRAL SPECIFICATIONS FOR REINFORCING :

- Reinforcing steel shall be clean of rust, grease, or other material likely to impair bond.
- All reinforcing steel to be continuous and lapped (with staggered splices at adjacent bars) minimum 24" at splices (#4 bars), 42" at splices (#5 bars). Reinforcing bars shall have minimum bend radius of 4 times the bar diameter. Bars shall not be heated to facilitate bending. Once bent, steel shall not be straightened.
- Reinforcing bars to be deformed bars conforming ASTM A-615:
3, # 4 Grade 40 # 5 & larger Grade 60
- All reinforcing steel, anchor bolts, and foundation hardware shall be located in the formwork and held firmly in place prior to and during concrete placement by means of wire supports.
- Concrete cover is required as follows over reinforcing :
3" where concrete is exposed to and cast against earth
2" where concrete is exposed to earth but cast against formwork
1-1/2" where concrete is not exposed to earth or weather
- Reinforcing steel shall not be welded, unless specifically notes on the structural drawings. If allowed, welding shall conform to CBC Standards.

RETAINING WALL DESIGN VALUES

THE LISTED DESIGN VALUES BELOW FROM THE PROJECT SOILS REPORT WERE USED FOR THE STRUCTURAL DESIGN OF THE RETAINING WALLS FOR THIS PROJECT:

LATERAL PRESSURE	EQUIVALENT FLUID PRESSURE
ACTIVE CASE	49 pcf
AT REST CASE	50 pcf
PASSIVE CASE	400 pcf
MAXIMUM TOE PRESSURE	2400 pcf
COEFFICIENT OF SLIDING FRICTION	0.50

SOIL NOTE

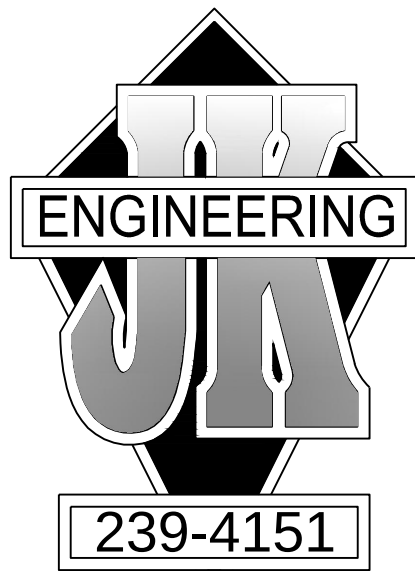
SOILS EXPANSION INDEX : VERY LOW
ORIGINAL REPORT: SLO2830-3
BY: GEOSOLUTIONS, INC.
DATED: OCTOBER 9, 2002
SOILS UPDATE LETTER: H-16996
BY: HALLIN GEOTECHNICAL SERVICES, LLC.
DATED: DECEMBER 5, 2016

THE SOILS REPORT REFERENCED IS PART OF THESE PLANS AND ALL RECOMMENDATIONS THERE IN SHALL BE COMPLIED WITH.

RETAINING WALL NOTES PER SOILS REPORT

- THE PRESSURES LISTED WERE BASED ON THE ASSUMPTION THAT BACKFILL SOILS WILL BE COMPACTED TO 90% OF MAXIMUM DRY DENSITY AS DETERMINED BY THE ASTM D 1557 TEST METHOD.
- A BACK DRAIN OR AN EQUIVALENT SYSTEM OF BACKFILL DRAINAGE SHOULD BE INCORPORATED INTO THE RETAINING WALL DESIGN. BACKFILL IMMEDIATELY BEHIND THE RETAINING STRUCTURE SHOULD BE A FREE-DRAINING GRANULAR MATERIAL. ALTERNATIVELY, THE BACK OF THE WALL COULD BE LINED WITH A GEODRAIN SYSTEM.
- COMPACTION ON THE UPHILL SIDE OF THE WALL WITHIN A HORIZONTAL DISTANCE EQUAL TO ONE WALL HEIGHT SHOULD BE PERFORMED BY HAND-OPERATED OR OTHER LIGHTWEIGHT COMPACTION EQUIPMENT. THIS IS INTENDED TO REDUCE POTENTIAL "LOCKED-IN" LATERAL PRESSURES CAUSED BY COMPACTION WITH HEAVY GRADING EQUIPMENT.
- WATER SHOULD NOT BE ALLOWED TO POND NEAR THE TOP OF THE WALL. TO ACCOMPLISH THIS, THE FINAL BACKFILL SITE GRADE SHOULD BE SUCH THAT ALL WATER IS DIVERTED AWAY FROM THE RETAINING WALL.
- TO REDUCE INFILTRATION OF THE SOIL INTO THE DRAIN GRAVEL, THE GRAVEL SHOULD BE ENCAPSULATED IN A PERMEABLE GEOTEXTILE FABRIC. A SUITABLE PERMEABLE GEOTEXTILE FABRIC, SUCH AS NON-WOVEN NEEDLE-PUNCHED MIRAFI 140N OR EQUAL, MAY BE UTILIZED AND SHOULD CONFORM TO CALTRANS STANDARD SPECIFICATION 88-1.03 FOR UNDERDRAINS.
- CONTRACTOR SHALL PROVIDE 6" OF FILL OVER TOE OF RETAINING WALL FOOTING. ALL FILL SHALL BE BROUGHT TO A MOISTURE CONTENT NEAR OPTIMUM MOISTURE CONTENT, AND COMPACTED TO A MINIMUM OF 90 PERCENT OF THE MAXIMUM DENSITY (ASTM D1557).

NOTE: FOR ALL OTHER INFORMATION NOT STATED HERE THE OWNER / CONTRACTOR SHALL READ THE PROJECT SOILS REPORT PRIOR TO START OF CONSTRUCTION. ANY QUESTIONS REGARDING THE SOILS REPORT SHALL BE BROUGHT TO THE ATTENTION OF HALLIN GEOTECHNICAL SERVICES, LLC.



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PLAN PREPARED FOR:

KIRK & CARRIE ALLEN
LOT 106 TRACT 1990-2
PASO ROBLES, CA 93446

REVISION LOG

REV.	DESCRIPTION	DATE

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PROJECT NO. ----
FILE NAME CAPPS-LOT-106 STRUCTURAL.DWG
DRAWN BY JJK
DATE 3/31/2017 8:01 AM
SHEET TITLE:

RETAINING WALLS

SHEET NUMBER:

R-1.1



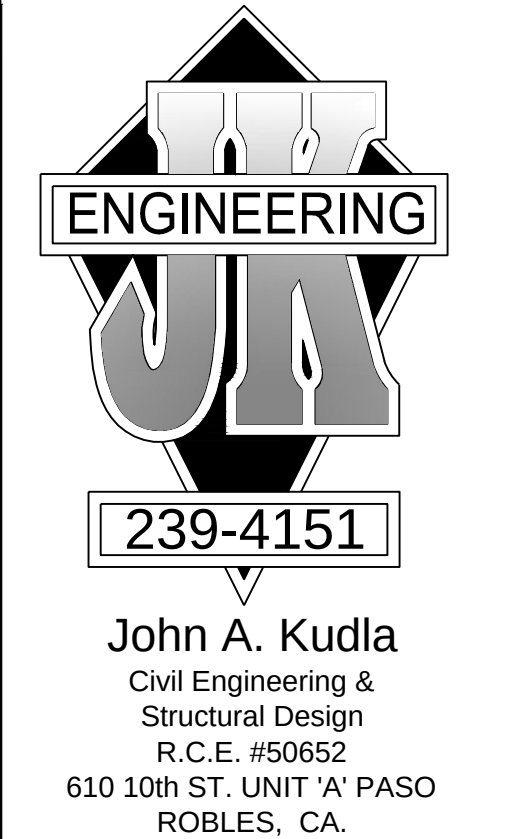
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PROJECT NO.	----
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DRAWN BY	JKK
DATE	3/31/2017 8:01 AM
SHEET TITLE:	

RETAINING WALL

SHEET NUMBER:



PLAN PREPARED FOR:

KIRK & CARRIE ALLEN
LOT 106 TRACT 1990-2
PASO ROBLES, CA 93446

R-2.1